Algebraic K-Theory

— ***EYNTKA*** **Series** —

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Contents

0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7⋆; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Introduction

Algebraic K-theory attaches to a ring R a sequence of abelian groups $K_0(R), K_1(R), K_2(R), \ldots$ that measure, in successively subtler ways, how far linear algebra over R departs from linear algebra over a field. Over a field every module is free and every invertible matrix is a product of elementary ones. Over a general ring neither holds, and the K-groups are precisely the record of the obstructions. The group $K_0(R)$ is the Grothendieck group of finitely generated projective modules, measuring how far projectives are from free; $K_1(R) = GL(R)/E(R)$ is the Whitehead group, measuring how far invertible matrices are from being products of elementary ones (the determinant, and more, when R is commutative); and $K_2(R)$ records the relations among the elementary matrices.

In each degree the construction springs from one impulse, the same that builds the integers from the natural numbers: complete a monoid to a group by adjoining inverses. This is transparent for K_0, where the monoid is projective modules under direct sum. Quillen's insight was that the whole tower $K_0, K_1, K_2, \ldots$ is governed by a single homotopy-theoretic construction: the higher groups are the homotopy groups of a space built from R, whether by the plus construction applied to $BGL(R)$ or by the Q-construction on the category of projective modules. The two agree, and they reproduce the classical K_0, K_1, K_2 in the bottom three degrees.

Unlike its topological sibling ([22]), algebraic K-theory is not periodic: the groups $K_n(R)$ form an infinite tower of distinct invariants whose computation, even for the ring of integers, runs to the frontier of mathematics. Their arithmetic connects directly to number theory, the K-groups of rings of integers encoding special values of L-functions and étale cohomology. The bridge to the topological side is the Serre-Swan theorem, under which $K_0(C(X)) = K^0(X)$ ([6]), so that the algebraic and topological theories coincide exactly in degree zero before diverging above it.

This book builds the subject from the ground up. It assumes comfort with rings and modules ([8], [15]) and with homological algebra and exact categories ([13]). The higher theory additionally draws on the homotopy groups, classifying spaces and fibration sequences of *Algebraic Topology* [20], and on the nerve and geometric realization of a small category, developed in *Higher Category Theory* [12]. Quillen's own theorems, by contrast, are proved here: Theorem A, Theorem B and the plus construction are results about K-theory that happen to be stated in homotopy-theoretic language, and nothing is gained by sourcing them elsewhere. Throughout, rings are associative with $1 \neq 0$, and modules are right modules unless stated otherwise, so that a homomorphism $R^n \to R^m$ is an $m \times n$ matrix acting on column vectors.

The route runs from projective modules and the Picard group through K_0, with its ties to homology and Morita equivalence, up to K_1 and K_2. Milnor's symbolic K_*^M then reproduces those three groups by generators and relations, which is the last point at which they can be written down by hand. Quillen's two constructions take over from there, the fundamental theorems make them computable, schemes widen the input, and the closing chapters compute: $K_*(\mathbb{F}_q)$, K_* of a Dedekind domain, and the arithmetic of $K_*(\mathcal{O}_K)$.

By the end, a reader should be able to:

1. compute K_0, K_1 and K_2 of a given ring, and recognize which structural feature of the ring each group is detecting;
2. read a research paper that forms $K_*(\mathcal{E})$ of an exact category and follow what the Q-construction contributes;
3. apply the fundamental theorems (resolution, dévissage, localization, Bass) to reduce an unknown K-group to known ones;
4. state what is and is not known about $K_*(\mathbb{Z})$, and say why the question is hard.

Omitted Material

Several topics fall under the title and are deliberately absent.

Topological and operator K-theory. Bott periodicity, $K^0(X) = [X, \mathbb{Z} \times BU]$, and the K-theory of C^*-algebras are the periodic half of the subject and carry their own volumes: *Vector Bundles and K-Theory* [22] and *C^*-Algebras and K-Theory* [6]. The two theories meet in degree zero, through Serre-Swan, and diverge above it. That meeting is treated here, the periodic theory is not.

Waldhausen K-theory and the K-theory of ring spectra. The $S_\bullet$-construction and A-theory generalize the Q-construction to categories with weak equivalences, but require stable homotopy theory that is not a prerequisite here. *Higher Category Theory* [12] builds the setting; Waldhausen's "Algebraic K-theory of spaces" remains the source.

Motivic cohomology and the Bloch-Kato conjecture. The proof of Bloch-Kato, and the motivic spectral sequence relating motivic cohomology to K-theory, need motivic homotopy theory and a book of their own. **??** surveys the statements. Mazza, Voevodsky and Weibel's *Lecture Notes on Motivic Cohomology* is the standard entry point.

Trace methods in full. Topological Hochschild and cyclic homology compute K-theory in practice, and the cyclotomic trace is the modern computational tool, but the equivariant stable homotopy theory behind them is a separate subject. Surveyed in **??**; Nikolaus and Scholze's "On topological cyclic homology" is the current reference.

Hermitian and quadratic K-theory, and L-theory. These replace projective modules by modules with a form, and their applications are to surgery and the classification of manifolds rather than to the arithmetic pursued here. Ranicki's *Algebraic L-theory and Topological Manifolds* is the standard account.

The étale and Galois cohomology behind Quillen-Lichtenbaum. **??** states the comparison between K-theory and étale cohomology and uses it, but the cohomology itself belongs to *Galois Cohomology* [11] and *Étale Cohomology* [10].

Riemann-Roch in its general form. Chapter 6 proves the version without denominators. Chow groups, the full Grothendieck-Riemann-Roch theorem, and the intersection theory they rest on are *Intersection Theory* [14]'s.

1

Projective modules

If R is a field or a divisor ring, every R-module is a vector-space which is algebraically characterized up to dimension. More generally, a free R-module is a module that is R-module isomorphic to $R^{\oplus I}$ for same indexing set I. Any R-module homomorphism between two free modules is based on where chosen basis of the domain is represented by the chosen basis of the codomain; namely it can be represented by a matrix (or the generalization thereof for infinite dimension). These are the simplest modules that can be put over a ring. A natural question would be to see what are the possible "next weakest" conditions we can put, and what rings R "force" this condition. We shall explore a few of these conditions, one important one shall be for a module to be *projective* which we shall show how this could be thought of as being *locally free*. Measuring how this structure differs from being free will be the beginnings of the study of the K_0-group in the next chapter.

1.1 IBP, Stably Free, and Projective Modules

If R is a field or a divisor ring, then $R^n \cong R^m$ implies $n = m$. If R is commutative this is still the case (consider $R/\mathfrak{m}$ by a maximal ideal), group rings (for finite groups), and any (left)-Noetherian ring satisfies this property as well (a version of Nakayama's lemma proves this, see [8, Nakayama's Lemma I]). This property is *not* true for general rings:

Example 1.1.1: Failure of Invariant Basis Number
let R be a commutative ring and let M be the $\mathbb{N} \times \mathbb{N}$ matrices where each column has finitely many nonzero entries, this is sometimes denoted $CFM_{\mathbb{N}}(R)$. Then $CFM_{\mathbb{N}}(R) \cong CFM_{\mathbb{N}}(R)^2$ via the map:

$$\varphi : CFM_{\mathbb{N}}(R) \to CFM_{\mathbb{N}}(R)^2 \qquad M \mapsto (\text{odd columns of } M,\ \text{even columns of } M)$$

Another example the reader could work through is $\mathrm{End}_k(k^{\oplus \mathbb{N}})$.

Definition 1.1.2: Invariant Basis Property

Let R be a ring. Then R satisfies the (right) *invariant basis property* (IBP) if

$$R^n \cong R^m \qquad \Rightarrow \qquad n = m$$

If R satisfies IBP, then the rank of a free R-module M is an invariant, independent of the choice of basis.

Proposition 1.1.3: Sufficient Condition for IBP

Let R be a ring. Then R satisfies IBP if there exists a ring map $f \colon R \to F$ from R to a field (or division ring) F.

***Proof*:**
Suppose $R^n \cong R^m$ as R-modules. Applying $- \otimes_R F$, which sends R to F and commutes with finite direct sums, gives $F^n \cong F^m$ as F-vector spaces. Dimension over a division ring is well-defined, so $n = m$, as we sought to show.

Kernels of a surjective R-module homomorphism $\varphi : R^n \to R^m$ need not be free, however they must be stably free:

Definition 1.1.4: Stably Free

Let P be an R-module. Then P is called *stably free* of rank $n - m$ if there exists n, m such that

$$P \oplus R^m \cong R^n$$

If R satisfies IBP, then the rank of a stably free module is independent of the choice of n and m.

The kernel is stably free: R^m is free, hence projective, so the sequence splits ([15, Equivalencies for Projective Modules](3)). Not all R-modules are stably free (or more interestingly, not all stably free modules are free). The notion of stably free is close to *projective*, but no condition is imposed on the second summand, which makes it a slightly weaker notion. Projective modules follow shortly.

Consider next the case of a module P with $P \oplus R \cong R^n$, so that P is stably free of rank $n - 1$ and the complement being cancelled is a single copy of R. When can the R be cancelled outright, leaving P free? The question is the first genuinely K-theoretic one in the book, and it is controlled by a row vector.

Definition 1.1.5: Unimodular Row

Let R be a ring. A row vector $\sigma = (r_1, \ldots, r_n) \in R^n$ is *unimodular* if its entries generate the unit ideal, that is, if $r_1 R + \cdots + r_n R = R$.

Proposition 1.1.6: Unimodular Rows and Splittings

Let $\sigma = (r_1, \dots, r_n) \in R^n$ and let $\pi \colon R^n \to R$ be the map it defines. The following are equivalent.

1. σ is unimodular.
2. There are $s_1, \dots, s_n \in R$ with $r_1 s_1 + \cdots + r_n s_n = 1$.
3. π is surjective.
4. π splits, so that $R^n \cong \ker(\pi) \oplus R$.

***Proof*:**
(1) and (2) are the same statement: the ideal $r_1 R + \cdots + r_n R$ contains 1 if and only if some R-combination of the r_i equals 1. (2) and (3) are equivalent because the image of π is exactly that ideal, and a submodule of R is all of R precisely when it contains 1. Finally (3) implies (4): R is free, hence projective, so a surjection onto it splits by [15, Equivalencies for Projective Modules](3), and (4) implies (3) since a split epimorphism is an epimorphism, completing the proof.

Writing $P = \ker(\pi)$, so that $R^n \cong P \oplus R$, suppose P were free. A basis of P together with the splitting would give a new basis of R^n, and the change-of-basis matrix g has first row the unimodular row σ. This is the whole content of the next proposition.

Proposition 1.1.7: Free and Unimodular

Let $\sigma \in R^n$ be unimodular and let $P = \ker(\pi)$, so $P \oplus R \cong R^n$. Then P is free if and only if σ can be completed to an invertible matrix.

***Proof*:**
Write $p_1 \colon R^n \to R$ for the first-coordinate projection, and note that a matrix g has first row σ exactly when $p_1 \circ g = \pi$, since $(gx)_1$ is the dot product of the first row of g with x.

Suppose σ completes to $g \in \mathrm{GL}_n(R)$. Then $\pi = p_1 \circ g$, so

$$P = \ker(\pi) = g^{-1}\big(\ker p_1\big) = g^{-1}\big(0 \oplus R^{n-1}\big)$$

and g^{-1} is an isomorphism, so $P \cong R^{n-1}$ is free.

Conversely, suppose P is free. Its rank is then $n-1$: from $P \oplus R \cong R^n$ and $P \cong R^m$ we get $R^{m+1} \cong R^n$, and R satisfies IBP by hypothesis, so $m = n-1$. Choose a splitting $s \colon R \to R^n$ of π, which exists by proposition 1.1.6(4), and a basis $b_1, \dots, b_{n-1}$ of P. Then $s(1), b_1, \dots, b_{n-1}$ is a basis of R^n, so the matrix h having these as columns is invertible. Checking on the standard basis, $he_1 = s(1)$ and $\pi(s(1)) = 1 = p_1(e_1)$, while $he_{j+1} = b_j$ and $\pi(b_j) = 0 = p_1(e_{j+1})$. So $\pi \circ h = p_1$, hence $\pi = p_1 \circ h^{-1}$, and by the opening remark h^{-1} is an invertible matrix whose first row is σ, as we sought to show.

If R is commutative, then every unimodular row of length 2 may be completed: if $r_1 s_1 + r_2 s_2 = 1$,

then the invertible matrix is:
$$\begin{pmatrix} r_1 & r_2 \\ -s_2 & s_1 \end{pmatrix}$$
And so, if $R^2 \cong R \oplus P$, then $P \cong R$. Stronger versions of this appear later (note here though that commutativity was important, this fails for a unimodular row of length 2 over a non-commutative ring).

In dimension 3 there is already a counterexample:

Example 1.1.8: Stably Free But Not Free

Take
$$R = \frac{\mathbb{R}[x,y,z]}{(x^2+y^2+z^2-1)}$$
Note that every element $(f,g,h) \in R^3$ gives a vector-field in $\mathbb{R}^3$. Let σ be the unimodular row $\sigma = (x,y,z)$. Then σ is the vector field pointing radially outward. If P were free, a basis of P would yield two tangent vector fields on S^2 which are linearly independent at every point of S^2 (since together with σ, they span the tangent space of 3-space at each point). But this contradicts the hairy ball theorem ([20, Hairy ball theorem]), which says S^2 carries no nowhere-vanishing continuous tangent vector field.

Let us now return to the discussion of projective modules.

Definition 1.1.9: Projective Module

Let P be an R-module. Then P is *projective* if there exists an R-module Q such that $P \oplus Q$ is a free R-module.

See [15, Equivalencies for Projective Modules] for the equivalent characterizations. One of them is that there is always a lift in the following commutative diagram:

$$\begin{tikzcd} & P \arrow[dl, dashed, "\exists!"'] \arrow[d, "g"] & \\ M \arrow[r, "f"] & N \arrow[r, "0"] & 0 \end{tikzcd}$$

where f is surjective. Naturally, free and stably free module are projective. A more nuanced example is that all $M_n(k)$-modules (with $n > 1$) are projective but not free (which can be deduced using Artin-Wederburn). A more canonical set of examples comes from vector bundles. Take $R = C^0(X)$ to be the set of continuous real valued function on a compact topological space X, and let $\Psi : E \to X$ be a vector-bundle. Then the set $\Gamma(E)$ of continuous sections naturally forms a projective R-module. Let us show this (see [4] for the algebraic geometry perspective of the equivalence between vector bundle and locally free sheaves). The trivial bundle is $\mathbb{R}^n \times X \to X$, with $\Gamma(\mathbb{R}^n \times X) = R^n$. Now assume $\Psi : E \to X$ is a non-trivial vector-bundles (up to vector-bundle isomorphism). Then if $\Gamma(E)$ were free, there would be sections $\{s_1, \ldots, s_n\}$ forming a basis $f \colon T^n \to E$ which would give an isomorphism $\Gamma(T^n) = \Gamma(E)$. As the kernel and cokernel of bundles of f have no nonzero sections, they must vanish, and so f would be an isomorphism, and so $\Psi : E \to X$ would in fact be trivial.

If X is compact, then the famous *Swan's Theorem* gives an equivalence between the category of projective modules over $C^0(X)$ and the category of vector bundles over X, showing that vector bundles are the geometric interpretation of projective modules, which motivates similar discussions over sheaves of module for a scheme.

Proposition 1.1.10: Finitely Generated Projective Modules and Idempotents

Let R be a ring. An R-module P is finitely generated projective if and only if there is an n and an idempotent $e = e^2 \in M_n(R)$ with $P \cong eR^n$.

***Proof*:**
Suppose P is finitely generated projective, so $P \oplus Q \cong R^n$ for some Q and some n. Let $\pi \colon R^n \to P$ be the projection onto the first summand and $\iota \colon P \to R^n$ the inclusion, and set

$$e := \iota\pi \colon R^n \to R^n$$

Then $e^2 = \iota(\pi\iota)\pi = \iota\pi = e$, since $\pi\iota = \mathrm{id}_P$. Writing e as a matrix in the standard basis of R^n exhibits it as an idempotent of $M_n(R)$, and its image is $\iota(P)$, so $eR^n \cong P$.

Conversely, given an idempotent $e \in M_n(R)$, the decomposition $R^n = eR^n \oplus (1-e)R^n$ presents eR^n as a direct summand of a finitely generated free module, hence as a finitely generated projective module, completing the proof.

If P is a finitely generated projective R-module over a PID, then it is in fact free, by the structure theorem for finitely generated modules over a PID[1]. Another famous class of modules where all finitely generated projectives are free comes from Quillen and Suslin stating that finitely generated projective modules over a polynomial ring (or a Laurent polynomial ring) over a PID are free[2]. This settles the group ring $\mathbb{Z}[G]$ for G finitely generated free abelian, since for such a G of rank r the ring $\mathbb{Z}[G]$ is the Laurent polynomial ring $\mathbb{Z}[x_1, x_1^{-1}, \ldots, x_r, x_r^{-1}]$: every finitely generated projective $\mathbb{Z}[G]$-module is then free. On the other hand, if G is *not* abelian (but is torsion-free and nilpotent), then any projective $\mathbb{Z}[G]$-module P is stably free but not free, in particular that:

$$P \oplus \mathbb{Z}[G] \cong (\mathbb{Z}[G])^2$$

As of writing this book, it is unknown of the condition "nilpotent" can be dropped and produce a similar (or the same) result.

Another important class of finitely generated projective R-module that are always free are those over *local rings*. For informative purposes, the result can in fact be extended to infinitely generated projective R-modules over local rings (Kaplansky gave such a proof). The following is given as a reminder:

Proposition 1.1.11: Projective Over Local Ring, then Free

Let R be a local ring. Then every finitely generated projective Rmodule is free, where

$$P \cong R^n \qquad n = \dim_{R/\mathfrak{m}}(P/\mathfrak{m}P)$$

[1]The finiteness hypothesis can be dropped: Kaplansky proved that every projective module over a PID is free, at any rank. That argument is not the structure theorem and is not developed anywhere in this series.

[2]This result appears in [4] where it is used to show that all locally free $\mathcal{O}_X$-modules on affine space must be free

***Proof*:**
This is [8, Projective Module over Local Ring], which additionally identifies the rank: lifting a basis of $P/\mathfrak{m}P$ over the residue field $R/\mathfrak{m}$ gives a basis of P over R.

Proposition 1.1.12: Free At Stalks

Let R be a commutative ring, $\mathfrak{p}$ a prime ideal, and P a finitely generated projective R-module. Then the localization $P_{\mathfrak{p}}$ is isomorphic to $(R_{\mathfrak{p}})^n$ for some $n \geq 0$. Furthermore, there exists some $s \in R \setminus \mathfrak{p}$ where the localization away from s is free:

$$P\left[\frac{1}{s}\right] \cong R\left[\frac{1}{s}\right]^n$$

Moreover $P_{\mathfrak{p}'} \cong (R_{\mathfrak{p}'})^n$ for every prime ideal $\mathfrak{p}'$ of R not containing S

***Proof*:**
As P is projective, let $P \oplus Q = R^m$. Then $P_{\mathfrak{p}} \oplus Q_{\mathfrak{p}} = (R_{\mathfrak{p}})^m$, hence $P_{\mathfrak{p}}$ is a finitely generated projective $R_{\mathfrak{p}}$-module. As $R_{\mathfrak{p}}$ is local, $P_{\mathfrak{p}}$ is in fact free. Now, every element of $P_{\mathfrak{p}}$ is of the form p/s, for $p \in P$ and $s \in R \setminus \mathfrak{p}$. By choosing a basis for $P_{\mathfrak{p}}$, we may clear the denominators to get an R-module homomorphism $\varphi : R^n \to P$ that become an isomorphism when localizing at $\mathfrak{p}$.

Now, as $\operatorname{coker}(\varphi)$ is a finitely generated R-module which vanishes upon localization, it must be annihilated by some $s \in R \setminus \mathfrak{p}$. Given this s, we get a surjective map:

$$\varphi\left[\frac{1}{s}\right] : \left(R\left[\frac{1}{s}\right]\right)^n \to P\left[\frac{1}{s}\right]$$

As $P\left[\frac{1}{s}\right]$ is projective (as localization preserves projectiveness), $\left(R\left[\frac{1}{s}\right]\right)^n$ is isomorphic to the direct sum of $P\left[\frac{1}{s}\right]$ and a finitely generated $R\left[\frac{1}{s}\right]$-module M (where $M_{\mathfrak{p}} = 0$). As M is annihilated by some $t \in R \setminus \mathfrak{p}$, we get:

$$\varphi\left[\frac{1}{st}\right] : \left(R\left[\frac{1}{st}\right]\right)^n \xrightarrow{\cong} P\left[\frac{1}{st}\right]$$

completing the proof.

Modules carry a "vector bundle" structure, and defining it takes a little algebraic geometry, nothing more than spectra (see [4]). Let $\operatorname{Spec} R$ be the collection of prime ideals of R, and define the Zariski topology with basic open sets given by:

$$D(s) = \{\mathfrak{p} \in \operatorname{Spec} R \mid s \notin \mathfrak{p}\} \cong \operatorname{Spec} R\left[\frac{1}{s}\right]$$

Definition 1.1.13: Rank At Prime Ideal

Let R be a commutative ring and M an R-module. Then the *rank* at $\mathfrak{p}$ is:

$$\operatorname{rank}_{\mathfrak{p}}(M) = \dim_{\kappa(\mathfrak{p})} M \otimes_R \kappa(\mathfrak{p})$$

a cardinal, taking the value ∞ when $M \otimes_R \kappa(\mathfrak{p})$ is not finite-dimensional. The module M has *constant rank* n if $\operatorname{rank}_{\mathfrak{p}}(M) = n$ is the same finite value at every prime $\mathfrak{p}$.

Note that $M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}} \cong \kappa(\mathfrak{p})^{\operatorname{rank}_{\mathfrak{p}}(M)}$, hence $\operatorname{rank}_{\mathfrak{p}}(M)$ is the minimal number of generators for $M_{\mathfrak{p}}$. If P is a finitely generated projective R-module, then $\operatorname{rank}(P) : \mathfrak{p} \mapsto \operatorname{rank}_{\mathfrak{p}}(P)$ is a continuous function between the $\operatorname{Spec} R$ with the Zariski topology and $\mathbb{N}$ with the discrete topology.

If $\operatorname{Spec} R$ is connected, then $\operatorname{rank}(P) : \operatorname{Spec} R \to \mathbb{N}$ must always be a constant function, hence every finitely generated projective R-module has constant rank. In the simplest case, R is an integral domain, then the rank is constant with:

$$\operatorname{rank}(P) = \dim_{\operatorname{Frac}(R)}(P \otimes_R \operatorname{Frac}(R))$$

The converse direction also holds, and is worth recording now that rank is defined for every module: a *projective* R-module of constant rank is finitely generated[3]. Projectivity is what carries the argument, so the hypothesis is not decoration. If M is not projective, then $\operatorname{rank}(M)$ need not even be continuous: consider $R = \mathbb{Z}$ with $M = \mathbb{Z}/p\mathbb{Z}$.

The section closes with an observation about how little higher dimensions add. For the following, recall that M, M' are stably isomorphic if $M \oplus R^m \cong M' \oplus R^m$ for some $m \geq 0$.

Theorem 1.1.14: Base Serre Cancellation

Let R be a commutative Noetherian ring with $\operatorname{kdim}(R) = d$, and let P be a projective R-module of constant rank $n > d$. Then:

1. $P \cong P_0 \oplus R^{n-d}$ for some projective R-module P_0 of constant rank d
2. If P is stably isomorphic to P', then $P \cong P'$
3. For all M, M', if $P \oplus M$ is stably isomorphic to M', then

$$P \oplus M \cong M'$$

The idea is that any extra structure when considering higher dimension vector bundles only adds "trivial" information, making the information from vector-bundles ties to the local information of the base space.

> ***Proof Key ideas*:**
> The argument is Bass's, and it runs by induction on the number of generators, using a general-position argument over $\operatorname{Spec} R$ to replace a generating set of size n by one of size $n-1$ whenever n exceeds the dimension. That step is where the Noetherian and dimension hypotheses are spent, and it requires prime avoidance over the whole spectrum rather than at a single prime. The machinery is developed nowhere else in this book and is not used again, so the details are cited rather than reproduced. See [3, Bass's cancellation theorem].

[3] Prove it for the case of rank 1, then for the rank r case show that $\Lambda^r(P)$ is finitely generated.

1.1.1 *Milnor Square

The following material is going to be more useful in section 2.5 when looking over Mayer-Vietoris sequences, however it appears when studying the Picard group of a ring. The idea is that the Milnor square gives us a way of gluing together different parts of a ring akin to the Mayer-Vietoris sequence from algebraic topology (see [20]). Recall from algebraic topology that this construction was express done to be computationally rather simple. This section algebraizes that technique by looking at a particularly nice way to "patch" two rings in such a way that will allow for some simple computations of various invariance (ex. Picard group, K-groups,etc)

Let R be a ring, $I \subseteq R$ an ideal, $\varphi : R \to S$ a ring homomorphism that maps I isomorphically to an ideal in S, which we shall also label I. Then R is the pullback of S and R/I in the following diagram:

$$\begin{tikzcd} R \arrow[r] \arrow[d] & S \arrow[d] \\ R/I \arrow[r] & S/I \end{tikzcd}$$

This square is called the *Milnor square*. In algebraic geometry, this represents R as the fibered product of S and R/I over S/I.

We can patch modules given a Milnor square. Let M_1 be an S-module and M_2 be an R/I-module. The Milnor square defines the natural S/I-module isomorphism:

$$g : M_2 \otimes_{R/I} S/I \cong M_1/IM_1$$

Then M can be thought of as a triple (M_1, g, M_2) and is a subset of $M_1 \times M_2$ given by the kernel of the R-module homomorphism:

$$M_1 \times M_2 \to M_1/IM_1 \qquad (m_1, m_2) \mapsto \overline{m}_1 - g(\overline{f}(m_2))$$

Example 1.1.15: Milnor Patching A Module

Take S^n and $(R/I)^n$ with $1 \in \mathrm{GL}_n(S/I)$. Consider $\overline{f} : R/I \to S/I$ Then:

$$\left\{ (\vec{s}, r \vec{+} I) \;\middle|\; \overline{\vec{s}} = \overline{f}(r + I) \right\}$$

This patches two vector spaces of the same dimension over different rings, glued over R/I. Geometrically, two trivial bundles over R/I and over S are glued into a bundle over R.

The properties of Milnor patching used later are these:

Theorem 1.1.16: Milnor patching

Let

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R/I & \longrightarrow & S/I \end{array}$$

be a commutative diagram where $I \subseteq R$ is identified with its image in S and also represented as I (i.e. a Milnor square). Then:

1. If P is obtained by patching together a finitely generated projective S-module P_1 and a finitely generated projective R/I-module P_2, then P is a finitely generated projective R-module.
2. $P \otimes_R S \cong P_1$ and $P/IP \cong P_2$.
3. Every finitely generated projective R-module arises in this way.
4. If P is obtained by patching free modules along $g \in \mathrm{GL}_n(S/I)$ and Q is obtained by patching free modules along g^{-1}, then $P \oplus Q \cong R^{2n}$.
5. Write $P(g)$ for the module patched from free modules along $g \in \mathrm{GL}_n(S/I)$. Then $P(1) \cong R^n$; $P(g) \oplus P(h) \cong P(g \oplus h)$; and

$$P\big(\overline{\alpha}\, g\, \overline{f}(\beta)\big) \cong P(g) \qquad \text{for } \alpha \in \mathrm{GL}_n(S),\ \beta \in \mathrm{GL}_n(R/I)$$

so the isomorphism class of $P(g)$ depends only on the double coset of g in $\mathrm{GL}_n(S) \backslash \mathrm{GL}_n(S/I) / \mathrm{GL}_n(R/I)$.

***Proof*:**

1. Let's start with a Milnor square, which is a pullback diagram:

$$\begin{array}{ccc} R & \to & S \\ \downarrow & & \downarrow \\ R/I & \to & S/I \end{array}$$

Suppose we have a finitely generated projective S-module P_1 and a finitely generated projective R/I-module P_2, along with an isomorphism $g : P_1 \otimes_S S/I \cong P_2/IP_2$. Let

$$P = \{(x, y) \in P_1 \times P_2 \big| \overline{x} = g(\overline{f}(y))\}$$

To show P is finitely generated: since P_1 and P_2 are finitely generated, there exists a surjection $S^n \to P_1$ for some n and a surjection $(R/I)^m \to P_2$ for some m. Then it is certainly the case that we can construct a surjection $R^{m+n} \to P$, showing that P is finitely generated.

For showing it is projective, since P_1 is projective over S, there exists an S-module Q_1 such that $P_1 \oplus Q_1 \cong S^n$ for some n. Similarly, there exists an R/I-module Q_2 such that $P_2 \oplus Q_2 \cong (R/I)^m$ for some m.

The isomorphism $g : P_1 \otimes_S S/I \cong P_2/IP_2$ extends $\tilde{g} : S^n \otimes_S S/I \cong (R/I)^m/I(R/I)^m$, namely it is the isomorphism induced by taking $Q_1 \otimes_S S/I$ and Q_2/IQ_2. Now, define F by:

$$F = \{(x,y) \in S^n \times (R/I)^m \big| \overline{x} = g(\overline{f}(y))\}$$

By the properties of the Milnor square, F is a free R-module. The inclusion maps $P_1 \hookrightarrow S^n$ and $P_2 \hookrightarrow (R/I)^m$ induce an inclusion $P \hookrightarrow F$. Moreover, the projections $S^n \to P_1$ and $(R/I)^m \to P_2$ induce a projection $F \to P$ such that the composition $P \to F \to P$ is the identity on P. But then P is projective,

2. Both isomorphisms are read off the construction in (1). Tensoring the defining inclusion $P \subseteq P_1 \times P_2$ with S kills the second factor, since P_2 is an R/I-module and I maps isomorphically into S, leaving $P \otimes_R S \cong P_1$; reducing mod I kills the first factor for the same reason, leaving $P/IP \cong P_2$.

3. If M is any R-module, the Milnor square gives a natural map from M to the R-module M' by patching $M_1 = M \otimes_R S$ and $M_2 = M \otimes_R (R/I) = M/IM$ along the natural isomorphism

$$(M/IM) \otimes_{R/I} (S/I) \cong M \otimes_R (S/I) \cong (M \otimes_R S)/I(M \otimes_R S)$$

Now, tensor M with $0 \to R \to (R/I) \oplus S \to S/I \to 0$. This gives an exact sequence (recall that tensoring is right-exact, but not necessarily left-exact).

$$\mathrm{Tor}_1^R(M, S/I) \to M \to M' \to 0$$

so in general M' is just a quotient of M. However, if M is projective, the Tor term is zero and $M \cong M'$. Thus every projective R-module may be obtained by patching.

4. Write $P(g)$ for the R-module patched from S^n and $(R/I)^n$ along $g \in \mathrm{GL}_n(S/I)$, that is

$$P(g) = \big\{(x,y) \in S^n \times (R/I)^n \;\big|\; \overline{x} = g(\overline{f}(y))\big\}$$

so that $P = P(g)$ and $Q = P(g^{-1})$. Three properties follow directly from this description.

Step 1: $P(1) \cong R^n$. Patching along the identity imposes $\overline{x} = \overline{f}(y)$, which is precisely the condition defining R as the pullback of S and R/I over S/I, applied coordinatewise.

Step 2: $P(g) \oplus P(h) \cong P(g \oplus h)$. Regrouping the coordinates identifies $P(g) \oplus P(h)$ with the set of pairs $((x,x'),(y,y'))$ satisfying $\overline{x} = g\overline{f}(y)$ and $\overline{x'} = h\overline{f}(y')$, and those two conditions together say exactly $(\overline{x}, \overline{x'}) = (g \oplus h)\overline{f}(y,y')$.

Step 3: invariance under the double coset. For $\alpha \in \mathrm{GL}_n(S)$ the map $(x,y) \mapsto (\alpha x, y)$ is R-linear with inverse $(x,y) \mapsto (\alpha^{-1}x, y)$, and it lands in $P(\overline{\alpha}g)$ because $\overline{\alpha x} = \overline{\alpha}\,\overline{x} = \overline{\alpha}g\overline{f}(y)$. For $\beta \in \mathrm{GL}_n(R/I)$ the map $(x,y) \mapsto (x, \beta y)$ carries $P(g\,\overline{f}(\beta))$ to $P(g)$, since the defining condition $\overline{x} = g\,\overline{f}(\beta)\overline{f}(y) = g\,\overline{f}(\beta y)$ is exactly membership of $(x, \beta y)$ in $P(g)$; it is bijective because $y \mapsto \beta y$ permutes $(R/I)^n$. Composing the two gives $P(\overline{\alpha}\,g\,\overline{f}(\beta)) \cong P(g)$, which is (5).

By Step 2, $P \oplus Q \cong P(g \oplus g^{-1})$. Now $g \oplus g^{-1}$ lifts to $\mathrm{GL}_{2n}(S)$. Writing $E(X)$ and $F(X)$ for the $2n \times 2n$ block matrices with identity diagonal and X in the upper-right, respectively lower-left, block, a direct multiplication gives

$$\begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} = E(g)\,F(-g^{-1})\,E(g)\,E(-1)\,F(1)\,E(-1)$$

where 1 denotes I_n. Each $E(X)$ is the product of the commuting elementary matrices $e_{i,n+j}(X_{ij})$, and each $F(X)$ the product of the $e_{n+i,j}(X_{ij})$, so $g \oplus g^{-1}$ lies in $E_{2n}(S/I)$. Since $S \to S/I$ is surjective, each generator $e_{k\ell}(\bar{s})$ is the image of $e_{k\ell}(s)$, so $g \oplus g^{-1} = \overline{\alpha}$ for some $\alpha \in E_{2n}(S) \subseteq \mathrm{GL}_{2n}(S)$. Steps 3 and 1 then give

$$P \oplus Q \cong P(\overline{\alpha}) \cong P(1) \cong R^{2n}$$

completing the proof.

1.2 Picard Group of a Commutative Ring

Let us now tackle the connection between projective modules and vector bundles more closely. To fully understand the interplay between the algebraic and the geometric, the following theorem must be quoted:

Theorem 1.2.1: Serre-Swan's Theorem

1. **(Serre's Theorem)** Let R be a commutative Noetherian ring. Then the category of finitely generated projective R-modules is equivalent to the category of algebraic vector bundles[a] on $\operatorname{Spec} R$
2. **(Swan's Theorem)** Let X be a compact Hausdorff space. Then the category of finitely generated projective R-modules over the continuous functions $C(X)$ is equivalent to the category of finite-rank vector bundles on X, the equivalence being established by sending vector bundles its module of continuous section.

[a] locally free sheaves of structure sheaf-modules of constant finite rank

***Proof Key ideas*:**
Both halves construct the equivalence the same way, by sending a bundle to its module of sections and a module to the bundle whose fibre at a point is the module's reduction there. What has to be checked is that the two constructions are mutually inverse, and in Swan's case that argument rests on partitions of unity and Tietze extension on a compact Hausdorff space, machinery this book does not develop. The proofs are therefore cited: Serre's is in [23] and Swan's in [25].

These connections motivate the following definition:

Definition 1.2.2: Algebraic Line Bundle

Let R be a commutative ring. Then an *algebraic line bundle* L over R is a finitely generated projective R-module of constant rank 1.

This section says just *line bundle*. Line bundles are locally 1-dimensional, so the tensor product of two of them is again one, since $\dim(V \otimes W) = \dim(V) \cdot \dim(W)$ locally; and $L \otimes_R R \cong L$ for any module. So $\otimes_R$ makes the isomorphism classes a commutative monoid. An inverse makes it a group, and one exists:

Lemma 1.2.3: Inverse Line Bundle

Let L be a line bundle and $L^\vee = \mathrm{Hom}_R(L, R)$. Then

$$L \otimes_R L^\vee \cong R$$

***Proof*:**
First, $L^\vee$ is line bundle as $\mathrm{rank}(L) = \mathrm{rank}(L^\vee)$. Next, take:

$$f \otimes_R \ell \mapsto f(\ell)$$

If $L = R$, this is certainly an isomorphism. Hence in the general case, localizing at $\mathfrak{p}$ we get:

$$(L^\vee \otimes_R L)_\mathfrak{p} \cong L^\vee_\mathfrak{p} \otimes_{R_\mathfrak{p}} L_\mathfrak{p} \to R_\mathfrak{p}$$

which is then certainly an isomorphism. As being an isomorphism is a local property, we get the global isomorphism, completing the proof.

Definition 1.2.4: Picard Group

Let R be a commutative ring. Then the *picard group* of R is the set of isomorphism classes of line bundles on R with the operation being the tensor product $\otimes_R$ and inverses $L^{-1} = L^\vee$.

Proposition 1.2.5: Picard Group Homomorphism

Let $R \to S$ be a ring homomorphism between commutative rings. Then $\mathrm{Pic}\,(R) \to \mathrm{Pic}\,(S)$ is a group homomorphism mapping $L \mapsto L \otimes_R S$

***Proof*:**
Certainly $L \otimes_R S$ is a line bundle over S, since base change preserves both finite generation and local freeness of rank 1. For multiplicativity, associativity of the tensor product together with $S \otimes_S S \cong S$ gives

$$(L \otimes_R M) \otimes_R S \cong (L \otimes_R S) \otimes_S (M \otimes_R S)$$

the outer product on the right being taken over S, which is where both factors live. Hence $- \otimes_R S$ is a group homomorphism, as we sought to show.

One representation of line bundles will be needed later:

Lemma 1.2.6: Representation of Line Bundle

Let L be a line bundle over R. Then

$$\mathrm{End}_R(L) \cong R$$

***Proof*:**
Multiplication by elements of r give a map $R \to \mathrm{End}_R(L)$. This map is locally an isomorphism, and hence an isomorphism, completing the proof.

Example 1.2.7: Picard Groups of Commutative Rings

1. Let $R = k$. Then as k is a field, every projective k-module is free. But then any projective k module of rank 1 is a dimension 1 vector-space, namely it is isomorphic to k. Hence:
$$\mathrm{Pic}\,(k) = 0$$

2. Similarly, as every module over a PID is free, if $R = \mathbb{Z}$, then:
$$\mathrm{Pic}\,(\mathbb{Z}) = 0$$

3. Let R be a Dedekind domain (usually a number ring). The structure theory used here is [5, Classifying Torsion-Free Modules Over Dedekind Domain]. First, recall that each ideal $I \subseteq R$ is a projective R-module, and it is certainly locally of rank 1 (every ideal is generated by at most 2 elements, and localizing at the appropriate prime makes the ideal principal). Next, recall that any torsion-free R-module of rank n is isomorphic to:
$$M \cong R^{n-1} \oplus I$$
for an ideal I that can be replaced by $J = (a)I$ for $a \in R$. As all projective modules are torsion free, and the modules here have rank 1, the line bundles on R correspond to the ideals of R. Principal ideals are isomorphic to R, so:
$$\mathrm{Pic}\,(R) \cong \mathrm{Cl}\,(R)$$

4. (If you know some divisor theory) Building off the work from the last point, If X is a nonsingular affine variety over an algebraically closed field $\overline{k}$ then $\mathrm{Pic}\,(k[X])$ (the Picard group of its coordinate ring) is isomorphic to the divisor class group of X.

5. Quillen and Suslin proved that every finitely generated projective module over a polynomial ring in finitely many variables over a field is free. Thus the Picard group of $k[x_1, \ldots, x_n]$ is always trivial:
$$\mathrm{Pic}\,(k[x_1, ..., x_n]) = 0$$

6. (foreshadowing for elliptic curve theory). An elliptic curve is a smooth algebraic curve (over an algebraically closed field $\overline{k}$) whose genus is 1[a]. There is a natural subgroup of the Picard group, $\mathrm{Pic}^{\,0}$, and that if E is an elliptic curve associated to $R = \overline{k}[x, y, z]/I$, then:
$$\mathrm{Pic}^{\,0}(R) \cong E$$
so the curve recovers itself as a group. See [9] for more details.

[a]Every nonsingular curve maps onto a compact Riemann surface of genus g

The *determinant line bundle* is defined next. Its feature of interest is that rank and determinant together determine a finitely generated projective module. First, recall the following properties of the exterior product:

Proposition 1.2.8: Properties of The Exterior Product

Let R be a commutative ring and M an R-module. Then:

1. $\Lambda^k(R^n)$ is a free R-module of rank $\binom{n}{k}$ with basis the $e_{i_1} \wedge \cdots \wedge e_{i_k}$ for $i_1 < \cdots < i_k$. In particular $\Lambda^n(R^n) \cong R$, and $\Lambda^k(R^n) = 0$ for $k > n$

2. If $R \to S$ is a ring map, there is an isomorphism:

$$\left(\Lambda^k M\right) \otimes_R S \cong \Lambda^k(M \otimes_R S)$$

3. There is a natural isomorphism:

$$\bigwedge^k(P \oplus Q) \cong \bigoplus_{i=0}^{k}(\bigwedge^i P) \otimes (\bigwedge^{k-i} Q)$$

4. If P is projective, $\Lambda^k(P)$ is projective (as it is locally free).

***Proof*:**

1. This is [15, Bases of the Exterior Powers], applied to the free module R^n with its standard basis.

2. Λ^k is the quotient of the k-fold tensor power $M^{\otimes k}$ by the submodule generated by tensors with a repeated entry. Base change is right exact and commutes with tensor powers, so applying $- \otimes_R S$ to that presentation gives the presentation of $\Lambda^k_S(M \otimes_R S)$, whence the isomorphism.

3. Wedging gives an R-bilinear map $\Lambda^i P \times \Lambda^{k-i} Q \to \Lambda^k(P \oplus Q)$, hence a map

$$\theta \colon \bigoplus_{i=0}^{k} \left(\Lambda^i P\right) \otimes \left(\Lambda^{k-i} Q\right) \longrightarrow \Lambda^k(P \oplus Q)$$

natural in P and Q. Both sides commute with localization by (2), and being an isomorphism is a local property, so it suffices to check θ after localizing at each prime. There P and Q are free, by proposition 1.1.12, and θ carries the basis of the left side described in (1) bijectively onto that of the right side, sorting a basis wedge of $P_\mathfrak{p} \oplus Q_\mathfrak{p}$ into its P-factors and Q-factors. Hence θ is an isomorphism.

4. If P is projective, choose Q with $P \oplus Q \cong R^n$. By (3) the module $\Lambda^k(P)$, the $i = k$ summand, is a direct summand of $\Lambda^k(R^n)$, which is free by (1). A direct summand of a free module is projective, completing the proof.

Recall that $\Lambda^n(M)$ for a n-rank R-module is generated by a single element, and by moving all the coefficients to the front leaves the determinant. This motivates the following terminology:

Definition 1.2.9: Determinant Bundle

Let P be a finitely generated projective R-module of constant rank n. Then $\Lambda^n(P)$ is called the *determinant line bundle* of P, and is denoted $\det(P)$.

For P of non-constant rank the definition is applied component-wise. For an endomorphism $g\colon P \to P$, functoriality of det gives $\det(g)\colon \det(P) \to \det(P)$, which by lemma 1.2.6 is an element of R.

Rank and determinant together are a complete invariant in dimension one:

Proposition 1.2.10: Classification of Projective Module Over 1-dimensional Ring

Let R be a commutative Noetherian 1-dimensional ring and P a finitely generated projective R-module of *constant* rank $n \geq 1$. Then $P \cong L \oplus R^{n-1}$ with $L = \det(P)$, and P is determined up to isomorphism by the pair $(n, \det P)$. For P of non-constant rank the same holds componentwise, $\operatorname{Spec} R$ having finitely many components since R is Noetherian.

***Proof*:**
Let P have constant rank $n \geq 1$ and set $L = \det(P) = \Lambda^n(P)$.

Step 1: the splitting. If $n = 1$ there is nothing to prove: $L = \Lambda^1(P) = P$ and $f = 0$. So assume $n \geq 2$. Then $n > 1 = \operatorname{kdim}(R)$, which is the hypothesis of theorem 1.1.14, and part (1) of that theorem gives $P \cong P_0 \oplus R^{n-1}$ with P_0 of constant rank 1, that is, a line bundle.

Step 2: identifying P_0 as the determinant. Apply Λ^n and expand with proposition 1.2.8(3):

$$\Lambda^n(P_0 \oplus R^{n-1}) \cong \bigoplus_{i=0}^{n} \Lambda^i(P_0) \otimes \Lambda^{n-i}(R^{n-1})$$

Since P_0 has rank 1, $\Lambda^i(P_0) = 0$ for $i \geq 2$; and $\Lambda^n(R^{n-1}) = 0$ because $n > n-1$. Only the $i = 1$ term survives, giving $L \cong P_0 \otimes \Lambda^{n-1}(R^{n-1}) \cong P_0$. Hence $P \cong L \oplus R^{n-1}$.

Step 3: the pair determines P. If $L \oplus R^{n-1} \cong L' \oplus R^{n-1}$ for line bundles L and L', applying Λ^n and the computation of Step 2 to each side gives $L \cong L'$. So rank and determinant together classify P, completing the proof.

Over an integral domain the line bundles admit a concrete description: they are exactly the *invertible ideals.*

Proposition 1.2.11: Classification of Projective Modules over Integral Domains

Let R be a commutative integral domain. L is a line bundle if and only if it is (isomorphic to) an invertible ideal. If I and J are fractional ideals and I is invertible, then $I \otimes_R J \cong IJ$. Furthermore, there is an exact sequence of abelian groups:

$$1 \to R^\times \to F^\times \xrightarrow{\text{Div}} J_K(R) \to \operatorname{Pic}(R) \to 0$$

where $J_K(R)$ is the ideal group.

***Proof*:**
If I and J are invertible ideals such that $IJ \subseteq R$, then we can interpret elements of J as homomorphisms $I \to R$. If $IJ = R$, then we can find $x_i \in I$ and $y_i \in J$ so that $x_1y_1 + \cdots + x_ny_n = 1$. The $\{x_i\}$ assemble to give a map $R^n \to I$ and the $\{y_i\}$ assemble to give a map $I \to R^n$. The composite $I \to R^n \to I$ is the identity, because it sends $r \in I$ to $\sum x_i y_i r = r$. Thus I is a summand of R^n, i.e., I is a finitely generated projective module.

As R is an integral domain and $I \subseteq F$, $\mathrm{rank}(I)$ is the constant $\dim_F(I \otimes_R F) = \dim_F(F) = 1$. Hence I is a line bundle.

This construction gives a set map $J_K(R) \to \mathrm{Pic}\,(R)$. To show that it is a group homomorphism, it suffices to show that $I \otimes_R J \cong IJ$ for invertible ideals. Suppose that I is a submodule of F which is also a line bundle over R. As I is projective, $I \otimes_R -$ is an exact functor. Thus if J is an R-submodule of F, then $I \otimes_R J$ is a submodule of $I \otimes_R F$. The map $I \otimes_R F \to F$ given by multiplication in F is an isomorphism because I is locally free and F is a field. Therefore the composite

$$I \otimes_R J \subseteq I \otimes_R F \xrightarrow{\text{multiply}} F$$

sends $\sum x_i \otimes y_i$ to $\sum x_i y_i$. Hence $I \otimes_R J$ is isomorphic to its image $IJ \subseteq F$. This proves the third assertion.

The kernel of $J_K(R) \to \mathrm{Pic}\,(R)$ is the set of invertible ideals I having an isomorphism $I \cong R$. If $f \in I$ corresponds to $1 \in R$ under such an isomorphism, then $I = fR = \mathrm{div}(f)$. This proves exactness of the sequence at $J_K(R)$.

The units $R^\times$ of R inject into $F^\times$. If $f \in F^\times$, then $fR = R$ if and only if $f \in R$ and f is in no proper ideal, i.e., if and only if $f \in R^\times$. This proves exactness at $R^\times$ and $F^\times$.

Finally, we have to show that every line bundle L is isomorphic to an invertible ideal of R. Since $\mathrm{rank}(L) = 1$, there is an isomorphism $L \otimes_R F \cong F$. This gives an injection $L \cong L \otimes_R R \subseteq L \otimes_R F \cong F$, i.e., an isomorphism of L with an R-submodule I of F. Since L is finitely generated, I is a fractional ideal. Choosing an isomorphism $L^\vee \cong J$, lemma 1.2.3 yields

$$R \cong L \otimes_R L^\vee \cong I \otimes_R J \cong IJ$$

Hence $IJ = fR$ for some $f \in F^\times$, and $I(f^{-1}J) = R$, so I is invertible.

Divisor groups, class groups and their relation to the Picard group are developed in [4, Divisor Class Group].

2

K_0 Group

We saw the Picard group allowed us to treat the collection of (isomorphism classes of) line bundles of constant rank one as a group under $\otimes$. We shall now be looking at a similar situation with the direct sum.

2.1 Algebraic Build-up

Every abelian monoid can be turned universally into an abelian group. The construction takes the free abelian group on the elements of M and imposes exactly the relations M already satisfies.

The construction and its properties belong to *Core Algebra*: $M^{-1}M$ is [15, Grothendieck Completion], realized there as $(M \times M)/\sim$ with $(m, n) \sim (m', n')$ when $m + n' + p = m' + n + p$ for some p, and written $[m] - [n]$. A monoid is a *cancellation monoid* if $m + a = n + a$ forces $m = n$.

Proposition 2.1.1: Properties of Monoid Completion

Let M be a monoid, $M^{-1}M$ be its completion. Then:

1. Every element of $M^{-1}M$ can be written as $[m] - [n]$ for $m, n \in M$
2. For each $[m], [n] \in M^{-1}M$, $[m] = [n]$ if and only if there exists a p such that $m+p = n+p$.
3. The map $M \times M \to M^{-1}M$ where $(m, n) \mapsto [m] - [n]$ is surjective
4. M injects into $M^{-1}M$ if and only if M is a cancellation monoid.
5. The Grothendieck Completion is a functor that is left adjoint to the forgetful functor from the category of abelian group to the category of abelian monoids.

***Proof*:**
Parts (1) through (3) are [15, Properties of the Grothendieck Completion](1)-(3), and part (4) is its universal property, [15, Properties of the Grothendieck Completion](4): composition with $M \to M^{-1}M$ is a natural bijection $\mathrm{Hom}_{\mathbf{Grp}}(M^{-1}M, A) \cong \mathrm{Hom}_{\mathbf{Mon}}(M, A)$ for every abelian group A, which is exactly the stated adjunction.

Augmenting a monoid with a second operation satisfying the ring axioms gives a *semiring*. The semiring $\mathbb{N}$ is the initial semiring in this category. The Grothendieck completion can then also be seen as a functor from semirings to rings.

Free R-modules play a distinguished role in $K_0(R)$, enough to warrant naming the kind of submonoid they form

Definition 2.1.2: Cofinal Monoid

Let M be a monoid. Then $L \subseteq M$ is called *cofinal* if for every $m \in M$, there exists a $m' \in M$ such that $m + m' \in L$.

The terminology *cofinal* comes from order theory, were a subset $A \subseteq B$ of an preorder $(B, \leq)$ is cofinal if for every $b \in B$, there is an $a \in A$ such that $b \leq a$.

Lemma 2.1.3: Cofinal Monoids

Let $L \subseteq m$ be a cofinal monoid. Then:

1. $L^{-1}L \subseteq M^{-1}M$ is a subgroup
2. Every element $[m] \in M^{-1}M$ can be written as $[m'] - [\ell]$ for $m' \in M$ and $\ell \in L$.
3. If $[m] = [m'] \in M^{-1}M$, then $m + \ell = m' + \ell$ for some $\ell \in L$.

***Proof*:**

1. The inclusion $L \subseteq M$ is a monoid map, so it induces $L^{-1}L \to M^{-1}M$, and the image is a subgroup. Injectivity is (3) below applied with $m, m' \in L$.
2. Given $[m_1] - [m_2] \in M^{-1}M$, cofinality gives m_2' with $\ell := m_2 + m_2' \in L$. Adding and subtracting $[m_2']$ gives

$$[m_1] - [m_2] = [m_1 + m_2'] - [m_2 + m_2'] = [m_1 + m_2'] - [\ell]$$

   so $m' := m_1 + m_2'$ and ℓ work.
3. If $[m] = [m']$ then by proposition 2.1.1(2) there is $p \in M$ with $m + p = m' + p$. Cofinality gives p' with $\ell := p + p' \in L$, and adding p' to both sides gives $m + \ell = m' + \ell$, completing the proof.

Example 2.1.4: Grothendieck Completion

1. The monoid $\mathbb{N}$ under addition is cancellative, so it injects into its completion, and $\mathbb{N}^{-1}\mathbb{N} = \mathbb{Z}$.
2. Let $M = \{0, 1\}$ with $1 + 1 = 1$. This is not cancellative, since $1 + 1 = 0 + 1$ while $1 \neq 0$, and indeed $M^{-1}M = 0$: the relation $1 = 0$ holds in the completion because $1 + 1 = 0 + 1$.
3. For a ring R, the monoid $P(R)$ of definition 2.2.1 is cancellative exactly when stably isomorphic finitely generated projectives are isomorphic, which by theorem 1.1.14 holds in ranks above the Krull dimension but can fail below it.

2.2 K_0 of a ring

Definition 2.2.1: First K-ring: $K_0(R)$

Let R be a [not necessarily commutative] ring, $P(R)$ the collection of (isomorphism classes of) finitely generated projective R-modules. Combining it with $\oplus$ and appending an identity element 0 gives an abelian monoid. Its Grothendieck completion is the group $K_0(R)$.

When R is commutative, then along with $\otimes_R$, this forms a ring.

The restriction to finitely generated modules is what makes the group nonzero: infinitely generated ones trivialize the structure. For any finitely generated projective P, admitting the infinitely generated R^∞ would give:

$$P \oplus R^\infty \cong R^\infty$$

Therefore $[P] = 0$ and $K_0(R) = 0$, which is why the finitely generated case is the right one.

If R is commutative, then $K_0(R)$ is commutative with identity $1 = [R]$. This is indeed the identity, for $\oplus$ and $\otimes$ distribute, and $P \otimes_R R \cong P$. The canonical trivial example is when $R = k$ (or any division ring). Then $K_0(k) \cong \mathbb{Z}$. Similarity, if R is a local ring or a PID, then $K_0(R) \cong \mathbb{Z}$ (as these are sufficient conditions for a projective module to be free). Nontrivial examples come once enough theory is available to identify which properties give structural results for $K_0(R)$.

It should be noted that $K_0(-)$ is a functor from **Ring** to **Ab** or from **CRing** to **CRing**, namely if $\varphi : R \to S$ is a ring homomorphism, then $\otimes_R S : P(R) \to P(S)$ is a monoid map which induces the map $K_0(R) \to K_0(S)$. If R, S are commutative, then $\otimes_R S : P(R) \to P(S)$ is a semiring map as:

$$(P \otimes_R Q) \otimes_R S \cong (P \otimes_R S) \otimes_S (Q \otimes_R S)$$

and hence a ring homomorphism. This is what transports properties from known $K_0(R)$ groups to unknown ones.

In $K_0(R)$, the free-modules play a particularity important role. First of all, they are *cofinal* in $P(R)$ (see definition 2.1.2). Then every element of $K_0(R)$ can be written as:

$$[P] - [R^n]$$

Furthermore, by lemma 2.1.3, $[P] = [Q]$ if and only if:

$$P \oplus R^n \cong Q \oplus R^m$$

that is, P and Q are stably isomorphic, and $[P] = [R^n]$ if P is stably free. If R satisfies the IBP, then $L \subseteq P(R)$ is isomorphic to $\mathbb{N}$. For such R there is a useful characterization of trivial K_0 groups:

Lemma 2.2.2: IBP and Trivial K_0

Let $\mathbb{N} \to P(R)$ be the monoid map sending $n \to R^n$ inducing the group homomorphism $\mathbb{Z} \to K_0(R)$. Then:

1. this map is injective if and only if R satisfies IBP
2. If R satisfies IBP, then

$$K_0(R) \cong \mathbb{Z} \text{ if and only if every finitely generated projective } R\text{-module is stably free}$$

***Proof*:**

1. The map $\mathbb{N} \to P(R)$, $n \mapsto R^n$, is injective exactly when $R^n \cong R^m$ forces $n = m$, which is IBP. Since the free modules are cofinal in $P(R)$, lemma 2.1.3(3) says $[R^n] = [R^m]$ in $K_0(R)$ holds only if $R^n \oplus R^k \cong R^m \oplus R^k$ for some k, that is $R^{n+k} \cong R^{m+k}$; under IBP this gives $n = m$. Conversely if $\mathbb{Z} \to K_0(R)$ is injective then $R^n \cong R^m$ implies $[R^n] = [R^m]$ implies $n = m$.
2. Suppose $K_0(R) \cong \mathbb{Z}$, generated by $[R]$ under (1). For P finitely generated projective, $[P] = [R^m]$ for some $m \geq 0$, so by lemma 2.1.3(3) there is n with $P \oplus R^n \cong R^{m+n}$, that is P is stably free. Conversely, if every finitely generated projective is stably free then every class in $K_0(R)$ is $[R^m] - [R^n]$, so $\mathbb{Z} \to K_0(R)$ is onto, and it is injective by (1).

Example 2.2.3: K_0 of some rings

1. As mentioned before, fields/division rings, PIDs, and local rings all have $K_0(R) = \mathbb{Z}$
2. If $R = R_1 \times R_2$, then by [15, Modules over a Product Ring] there is an equivalence of categories
$$(R_1 \times R_2)\textbf{-Mod} \simeq R_1\textbf{-Mod} \times R_2\textbf{-Mod}$$
carrying finitely generated projectives to pairs of finitely generated projectives. Hence $P(R) \cong P(R_1) \times P(R_2)$ as monoids, and group completion preserves finite products, so $K_0(R) \cong K_0(R_1) \times K_0(R_2)$. Thus K_0 may be computed component-wise.
3. Let $R = M_n(F)$ Then R is a simple ring, and so every $M_n(F)$-module is projective. Recall that an $M_n(F)$-module is some direct sums of F, and the *length* of an $M_n(F)$-module is the number of copies. Then the length gives a natural map:
$$K_0(M_n(F)) \cong \mathbb{Z}$$
Note that $M_n(F)$ has length n, and so the subgroup generated by the free modules have index n, so the natural inclusion map $\mathbb{Z} \to K_0(M_n(F))$ does not split!
4. Let R be a semi-simple ring so that by Artin-Wedderburn:
$$R \cong M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

Hence:

$$K_0(R) = \prod_i K_0(M_{n_i}(D_i)) \cong \mathbb{Z}^k$$

5. If R is a commutative ring, there is a natural map to a field, $R \to R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m}$; call this field F. Then this induces a natural map $K_0(R) \to K_0(F)$, that is $K_0(R) \to \mathbb{Z}$, sending $[R]$ to 1. Every nonzero commutative ring has a maximal ideal, so such a map always exists in the commutative case; for non-commutative rings it need not.

6. (If you know some algebraic geometry) Recall that a sheaf is *flasque* if all restriction maps are surjective. The key interesting property of Flasque sheaves is that they have trivial cohomology, $H^i(X, \mathcal{F}) = 0$ for $i > 0$ and $\mathcal{F}$ a flasque sheaf. The $K_0(R)$ group connects to (co)homological considerations, and the following is the appropriate purely algebraic counterpart. A ring R is *flasque* if there is an R-bimodule M that is finitely generated and projective as a right R-module, such that there is a R-bimodule isomorphism:

$$\varphi : R \oplus M \xrightarrow{\sim} M$$

For such a ring $K_0(R) = 0$. Indeed:

$$P \oplus (P \otimes_R M) \cong P \otimes_R (R \oplus M) \cong (P \otimes_R M)$$

Hence $[P] + [P \otimes_R M] = [P \otimes_R M]$, implying $[P] = [0]$. If R is flasque and the right R-module structure of M is R, then R is said to be an *infinite sum ring.* The isomorphism φ becomes $R^2 \cong R$. An example of such a ring is $\mathrm{End}_R(R^\infty)$.

7. A ring is *von Neumann regular* if for every $x \in R$ there is a $y \in R$ with $xyx = x$, a weak substitute for inverses. Every field is von Neumann regular, taking $y = x^{-1}$, as is $M_n(F)$; no integral domain other than a field is. For such a ring $K_0(R)$ is computed by the lattice of finitely generated projectives, which the idempotents make tractable; see [3] for the details.

8. let $R = \varinjlim_i R_i$ where $\{R_i\}$ is a filtered system. Then every finitely generated projective R-module is of the form:

$$R \otimes_{R_i} P_i$$

where P_i is some finitely generated projective R_i-module. Any isomorphism:

$$R \otimes_{R_i} P_i \cong R \otimes_{R_i} P_i'$$

must set a finite set of generators to a finite set of generators, and so for j large enough this reduces to:

$$R_j \otimes_{R_i} P_i \cong R_j \otimes_{R_i} P_i'$$

So $P(R)$ is a filtered colimit of the $P(R_i)$, and colimits pass through K-groups:

$$K_0(R) = \varinjlim K_0(R_i)$$

Every commutative ring is the colimit of its finitely generated subrings. As every finitely generated commutative ring is Neotherian with finite normalization (the integral closure is finitely generated over the ring), this reduces $K_0(R)$ to the $K_0(R_i)$ of rings with better properties.

The next theorem is what makes K_0 computable modulo nilpotents, and it has a geometric reading: passing to the reduction does not change K_0, so K_0 cannot see nilpotent thickenings. It needs one hypothesis beyond nilpotence.

Definition 2.2.4: I-adically Complete Ring

Let R be a ring and $I \subseteq R$ an ideal. Then R is *I-adically complete* if the canonical map

$$R \longrightarrow \varprojlim_n R/I^n$$

is an isomorphism, the limit being taken over the tower $\cdots \to R/I^2 \to R/I$.

A nilpotent ideal is the degenerate case: if $I^n = 0$ then the tower is eventually constant at R and the map is trivially an isomorphism.

Theorem 2.2.5: K-group and Nilpotence

Let R be a ring and $I \subseteq R$ be nilpotent, or more generally complete (see definition above). Then:

$$K_0(R) \cong K_0(R/I)$$

And if R is commutative, then:

$$K_0(R) \cong K_0(R_{\text{red}})$$

***Proof*:**
We'll show $P(R) \cong P(R/I)$, which shall complete the result. We first need to build-up some important result:

lemma 1: Let P_1, P_2 be finitely generated projective R-modules and $I \subseteq R$ a radical idea. Then $P_1 \cong_R P_2$ if and only if $P_1/IP_1 \cong_R P_2/IP_2$

***Proof*:**
Certainly $P_1 \cong_R P_2$ implies $P_1/IP_1 \cong_R P_2/IP_2$, so we do the converse. There is a natural map $P_1 \to P_1/IP_1$, and then there is an isomorphism $\varphi : P_1/IP_1 \to P_2/IP_2$. Finally, there is a natural projection $\pi : P_2 \to P_2/IP_2$:

$$\begin{tikzcd} P_1 \arrow[d, "\pi"'] \arrow[dr, "\varphi\circ\pi"] & P_2 \arrow[d, "\pi"] \\ P_1/IP_1 \arrow[r, "\varphi"'] & P_2/IP_2 \end{tikzcd}$$

As P_1 is projective, there is a natural lift making the following diagram commute:

$$\begin{tikzcd} P_1 \arrow[r, "\tilde{\varphi}"] \arrow[d, "\pi"'] & P_2 \arrow[d, "\pi"] \\ P_1/IP_1 \arrow[r, "\varphi"'] & P_2/IP_2 \end{tikzcd}$$

If $\widetilde{\varphi}$ is an isomorphism, the proof is complete. Note that as φ is an isomorphism, in particular it is surjective, then as I is radical it is a consequence of Nakayama's lemma that $\varphi : P_1 \to P_2$ is surjective ([8, Nakayama's Lemma I], or [24]). Let's thus show injectivity. To that end, let us localize at each $\mathfrak{p} \subseteq R$ and consider:

$$\begin{tikzcd} P_{1,\mathfrak{p}} \arrow[r, "\tilde{\varphi}_{\mathfrak{p}}"] \arrow[d, "\pi"'] & P_{2,\mathfrak{p}} \arrow[d, "\pi"] \\ P_{1,\mathfrak{p}}/I_{\mathfrak{p}}P_{1,\mathfrak{p}} \arrow[r, "\varphi_{\mathfrak{p}}"'] & P_{2,\mathfrak{p}}/I_{\mathfrak{p}}P_{2,\mathfrak{p}} \end{tikzcd}$$

As P is a finitely generated projective module, its localization at a prime ideal is free, hence each $P_{i,\mathfrak{p}}$ has a basis. The key is that the projection onto $P_{1,\mathfrak{p}}/I_{\mathfrak{p}}P_{1,\mathfrak{p}}$ is still a basis. If we show then as $\varphi_{\mathfrak{p}}$ is an isomorphism, by commutativity the image of the basis in $\widetilde{\varphi}_{\mathfrak{p}}$ must be linearly independent which shows the map is injective. To that end, if $\{e_1, ..., e_n\}$ is a basis for $P_{1,\mathfrak{p}}$, its projection spans so let's show it is linearly independent. Say

$$\sum_i r_i \overline{e}_i = 0$$

Then $\sum_i r_i e_i \in I_{\mathfrak{p}}P_{1,\mathfrak{p}}$. By properties of ideals, $r_i \in I_{\mathfrak{p}}$, but then we must have that in the quotient $r_i = 0$ for all i, which by our previous reasoning allows us to conclude injectivity.

Hence, $\widetilde{\varphi}_{\mathfrak{p}}$ is injective for each prime $\mathfrak{p} \subseteq R$, and so $\ker \widetilde{\varphi}$ vanishes locally at every prime, hence vanishes ([8, Zero Module Is A Local Property]), so $\widetilde{\varphi}$ is injective.

Now, let us show that if I is a nilpotent ideal, then there is a bijection between the finitely generated projective R-modules and the finitely generated projective R/I-modules, which shall then complete the proof. Without loss of generality we can assume that $I^2 = 0$ (we can then use induction to for the result for $I^n = 0$ for $n \geq 2$)[a]. Let us first show that an idempotent $\overline{e} \in R/I$ lifts to an idempotent $e \in R$. If r is a lift of e, then we can verify that:

$$e = r + rxr + (1 - r)y(1 - r)$$

where $x, y \in I$. Verify that $e^2 = e$ and it indeed maps to $\overline{e}$. This lift is unique up to conjugation by an element $u \in 1 + I$. With this lift, we shall show that every finitely generated R/I-module $\overline{P}$ is isomorphic to P/IP for an appropriate finitely generated projective module P. The key is that idempotent matrices correspond to summands of free modules. In particular if $\overline{P}$ is a finitely generated projective R/I-module, then there exists an idempotent $\overline{e} \in M_n(R/I)$ such that $\mathrm{im}(\overline{e}) = \overline{P}$. Then this e can be lifted to $e \in M_n(R)$ where $e^2 = e$ and $e \equiv \overline{e} \bmod I$, so by taking $\mathrm{im}(e) = P$ we get:

$$\overline{P} \cong P/IP$$

Finally, we are at the stage where we reduced our problem of finitely generated projective R-modules to modules of the form P/IP. Then by our lemma, $P_1/IP_1 \cong P_2/IP_2$ if and only if $P_1 \cong P_2$, showing that the isomorphism classes are in bijective corresponds, completing the proof.

[a] This is where the completeness generalization also comes in, as we can consider $\varprojlim R/I^n$

Example 2.2.6: 0-dimensional Commutative Ring

Let R be a commutative ring and R_{red} be its reduction. Then it is an easy exercise to show that R_{red} is Artinian if and only if $\mathrm{Spec}\,(R)$ is finite and discrete. More generally (in the "non-finite case"), if a ring is von Neumann regular (for each $x \in R$ there exists a $y \in R$ such that $xyx = x$; this is a "weak inverse"), then R has Krull dimension 0 and is reduced (in fact, this is an equivalent characterization of von-Neumann regular rings, giving a good geometric characterization of such rings[a])

[a] Note that there are many non-commutative examples, such matrix rings, left principal ideals generated by idempotents, and $\mathrm{End}_k(V)$ for a vector space, and as they are closed under direct product that makes every semi-simple von-Neumann regular ring semi-simple

2.3 Connection to H_0

We start connecting the K_0 ring to (co)homological results, starting with the most simple one of the $H_0(R)$ group, which we shall take as $\mathrm{Hom}(\mathrm{Spec}\,(R), \mathbb{Z})$. Recall from definition 1.1.13 that a finitely generated projective R-module M determines a map $\mathrm{Spec}\,(R) \to \mathbb{Z}$ sending $\mathfrak{p} \mapsto \dim_{\kappa(\mathfrak{p})} M \otimes_R \kappa(\mathfrak{p})$, its rank. To see this connection, consider the following warm-up result:

Theorem 2.3.1: Pierce's Theorem

Let R be a 0-dimensional commutative ring. Then:

$$K_0(R) = \mathrm{Hom}(\mathrm{Spec}\,(R), \mathbb{Z})$$

Note that no assumption was made on R be finitely generated, for example it may be an infinite product of fields.

***Proof*:**

Since R is 0-dimensional every prime is maximal, so $\mathrm{Spec}\,(R)$ is Hausdorff, and being quasicompact it is a Boolean space: its topology has a basis of clopen sets $D(e)$ for idempotents e.

Step 1: rank is a continuous map. For P finitely generated projective, $\operatorname{rank}(P)$ is locally constant by proposition 1.1.12, hence a continuous map $\operatorname{Spec}(R) \to \mathbb{Z}$ with $\mathbb{Z}$ discrete. This gives the homomorphism $\operatorname{rank}\colon K_0(R) \to \operatorname{Hom}(\operatorname{Spec}(R), \mathbb{Z})$.

Step 2: surjectivity. A continuous $f\colon \operatorname{Spec}(R) \to \mathbb{Z}$ has finite image, since $\operatorname{Spec}(R)$ is quasicompact, and its fibres are clopen, hence of the form $D(e_i)$ for orthogonal idempotents e_i summing to 1. The values $f(i)$ may be negative, so f need not be the rank of a module; shift first. Choose $N \geq \max_i |f(i)|$, so that $f + N$ is non-negative, and set

$$P = \bigoplus_i (e_i R)^{f(i)+N}$$

a finitely generated projective R-module with $\operatorname{rank}(P) = f + N$. Then $[P] - [R^N] \in K_0(R)$ has rank f, since rank is a homomorphism and $\operatorname{rank}(R^N)$ is the constant N.

Step 3: injectivity. Suppose $\operatorname{rank}(P) = \operatorname{rank}(Q)$. Localizing at a prime $\mathfrak{p}$, both become free of the same rank over the local ring $R_{\mathfrak{p}}$, so $P_{\mathfrak{p}} \cong Q_{\mathfrak{p}}$. Both modules are finitely presented, so such an isomorphism spreads to a basic open: lifting it to a map $P_s \to Q_s$ for some $s \notin \mathfrak{p}$, its cokernel is finitely generated and dies at $\mathfrak{p}$, hence dies after inverting a further element, and the kernel is handled the same way since the map splits. So P and Q become isomorphic on some $D(s) \ni \mathfrak{p}$.

No gluing is then required, which is what makes the Boolean hypothesis do real work. Each $D(s)$ contains a clopen $D(e)$ around $\mathfrak{p}$, since $\operatorname{Spec}(R)$ has a basis of clopen sets; quasicompactness extracts a finite subcover, and the Boolean algebra of clopen sets refines it to a finite *disjoint* partition by $D(e_1), \ldots, D(e_k)$ with $1 = \sum_i e_i$ orthogonal. Then $R \cong \prod_i Re_i$ and $P \cong \bigoplus_i Pe_i \cong \bigoplus_i Qe_i \cong Q$, with no compatibility condition on overlaps because there are none. Hence $[P] = [Q]$, and rank is injective on $K_0(R)$, completing the proof.

The relation extends to general commutative rings:

Definition 2.3.2: H_0 Group

Let R be a commutative ring. Then:

$$H_0(R) = [\operatorname{Spec}(R), \mathbb{Z}] = \operatorname{Hom}(\operatorname{Spec}(R), \mathbb{Z})$$

The group $H_0(R)$ is always free abelian, so it is never divisible: it cannot be $\mathbb{Q}$. Quasicompactness of $\operatorname{Spec}(R)$ does not by itself bound the number of connected components, as the infinite product of fields above shows. If R is Noetherian, however, $\operatorname{Spec}(R)$ has finitely many components, say n, and then:

$$H_0(R) \cong \mathbb{Z}^n$$

If on top of that R is an (integral) domain (more generally $\operatorname{Spec}(R)$ is connected), then:

$$H_0(R) \cong \mathbb{Z}$$

Lemma 2.3.3: H_0 direct summand of K-group

The group $H_0(R)$ is a direct summand of $K_0(R)$. If $\dim R = 0$, then $K_0(R) \cong H_0(R)$.

***Proof*:**
Take the monoid L of the componentwise free modules R^n. Then as we saw, L is cofinal, and $L \to P(R)$ is a semiring map which naturally gives a map $L^{-1}L \to K_0(R)$. Then as L is isomorphic to $\mathrm{Hom}(\mathrm{Spec}\,(R), \mathbb{N})$, we must have $L^{-1}L \cong H_0(R)$, and it a subgroup.

Now, there is a natural map $K_0(R) \to H_0(R)$ given by the rank, namely there is a natural map from $\mathrm{rank} : P(R) \to [\mathrm{Spec}\,(R), \mathbb{N}]$, which is a semiring homomorphism. Thus, we get $H_0(R) \to K_0(R) \to H_0(R)$, is the identity, and so $H_0(R)$ is a direct summand of $K_0(R)$, as we sought to show.

For the special case of $\dim(R) = 0$, By Pierce's Theorem we have $K_0(R) \cong H_0(R)$.

Since $K_0(R)$ splits off $H_0(R)$, the other component is the kernel of the rank map, denoted $\widetilde{K}_0(R)$, so that:

$$K_0(R) \cong H_0(R) \oplus \widetilde{K}_0(R)$$

It is shown later that $\widetilde{K}_0(R)$ is a nilradical. Since $H_0(R)$ is already reduced, it is then the nilradical of $K_0(R)$.

Lemma 2.3.4: Limit and Kernel

Let R be a commutative ring, and let $P_n(R)$ denote the subset of all modules in $P(R)$ consisting of projective modules of constant rank n. There is a map $P_n(R) \to K_0(R)$ sending

$$P \mapsto [P] - [R^n]$$

This map is compatible with the stabilization map $P_n(R) \to P_{n+1}(R)$ sending $P \mapsto P \oplus R$, and the induced map is an isomorphism:

$$\varinjlim_n P_n(R) \cong \widetilde{K}_0(R)$$

***Proof*:**
The map is well defined: P has constant rank n, so $[P] - [R^n]$ has rank 0 and lands in $\widetilde{K}_0(R) = \ker(\mathrm{rank})$. It is compatible with stabilization because $P \oplus R$ has constant rank $n+1$ and $[P \oplus R] - [R^{n+1}] = [P] - [R^n]$.

For surjectivity, an element of $\widetilde{K}_0(R)$ is $[P] - [Q]$ with $\mathrm{rank}P = \mathrm{rank}Q$. Choosing Q' with $Q \oplus Q' \cong R^k$ free, it equals $[P \oplus Q'] - [R^k]$, and $P \oplus Q'$ has constant rank k. For injectivity, if $[P] - [R^n] = [P'] - [R^n]$ with both of rank n, then $[P] = [P']$, so by lemma 2.1.3(3) P and P' become isomorphic after adding a free module, which is exactly equality in the colimit, completing the proof.

The colimit is available, but base cancellation shows the geometric content of a finitely generated projective module over a ring of dimension d already appears in rank $\leq d$:

Corollary 2.3.5: Rank Stabilizes Above the Dimension

Let R be a commutative Noetherian ring with $\mathrm{kdim} R = d$. Then for every $n > d$ the map of lemma 2.3.4 restricts to a bijection of sets

$$P_n(R) \xrightarrow{\sim} \widetilde{K}_0(R), \qquad P \mapsto [P] - [R^n]$$

so the colimit defining $\widetilde{K}_0(R)$ is already reached at every level above d.

***Proof*:**
Injectivity: if $P, Q \in P_n(R)$ have $[P] - [R^n] = [Q] - [R^n]$ then $[P] = [Q]$, so P and Q are stably isomorphic, and theorem 1.1.14(2) upgrades this to $P \cong Q$ because $n > d$.

Surjectivity: by lemma 2.3.4 every element of $\widetilde{K}_0(R)$ is $[P'] - [R^m]$ with P' of constant rank m. If $m \geq n$, apply theorem 1.1.14(1) to split off a free summand repeatedly until the rank is n, which does not change the class. If $m < n$, replace P' by $P' \oplus R^{n-m}$, again without changing the class. Either way the class is hit by an element of $P_n(R)$, completing the proof.

The H_0 group is based off of the spectrum of R, and hence is dependent on its commutativity. Let us now try to bring this interpretation to non-commutative rings. To do this, another interpretation of the kernel that is rather nice is that it is the intersection of all the kernels of map $K_0(R) \to K_0(F)$ for all maps $R \to F$ into a field: as rank is a natural map, $\widetilde{K}_0(F) = 0$, and $H_0(R)$ captures the information at each prime ideal, that is at each stalk, which is recovered by ranging over all fields admitting a map $R \to F$.

The non-commutative analogue of a field is a simple Artinian ring. A maximal commutative subring of a simple Artinian ring S is a finite product of 0-dimensional local rings, which is what makes this the right generalization. Define $\widetilde{K}_0(R)$ as the intersection of the kernels of $K_0(R) \to K_0(S)$ where S is a simple Artinian ring. If no map $R \to S$ exists, set $\widetilde{K}_0(R) = K_0(R)$. Then define:

$$H_0(R) = \frac{K_0(R)}{\widetilde{K}_0(R)}$$

In the commutative case $H_0(R)$ is free abelian. In the non-commutative case it is at least torsion-free. To see this, let X be the set of homomorphisms $R \to S_x$ through which any other $R \to S'$ factors (such a set exists by using Zorn's lemma). Then as $K_0(S_x) \to K_0(S')$ is an isomorphism. Hence $\widetilde{K}_0(R)$ is the intersection of all kernels of maps $K_0(R) \to K_0(S_x)$, and $H_0(R)$ is the image of $K_0(R)$ in the torsion-free group:

$$\prod_{x \in X} K_0(S_x) \cong \prod_{xinX} \mathbb{Z}$$

Trace and Determinant

We are in the realm of representation theory, hence we have our usual trace and determinant map that shows up. In this section, we shall see how these maps give us some interesting information on our K_0 and H_0 group.

Hattori-Stallings Trace Map

Let R be an associative ring, and let $[R, R]$ be the additive subgroup of R generated by the elements $rs - sr$ for $r, s \in R$. Then the trace provides a natural map:

$$\mathrm{tr} : M_n(R) \to \frac{R}{[R, R]}$$

where we quotient by $[R, R]$ as tr should be independent of conjugation, which shall require $a_{i\ell}b_{\ell i}$ to commute. Then it should be a simple computation to see that all traceless matrices are in $[M_n(R), M_n(R)]$ which gives us the isomorphism:

$$\frac{M_n(R)}{[M_n(R), M_n(R)]} \cong \frac{R}{[R, R]}$$

Naturally, not all elements of $\mathrm{End}_R(P)$ of a finitely generated projective R-module P is in matrix form, however it is close enough to still define the trace: as P is finitely generated and projective choose Q such that $P \oplus Q \cong R^n$. Then we have an idempotent element $e \in M_n(R)$ such that $e(R^n) = P$, so:

$$\mathrm{End}_R(P) = eM_n(R)e$$

It is not too hard to verify that any other choices of idempotent will yield an e_1 which is idempotent and conjugate to e in some $M_m(R)$. Thus, we have that the trace map is independent of representative and we have a well-defined map:

$$\mathrm{tr} : \mathrm{End}_R(P) \to \frac{R}{[R, R]}$$

The trace of the identity map on P will be the trace of the idempotent element e. We call this particular value the *trace* of P.

Now, if P' is a finitely generated projective R-module represented by an idempotent f, then $P \oplus P'$ is represented by $\begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$, and so we have its trace is $\mathrm{tr}(P) + \mathrm{tr}(P')$! Thus, the trace is an additive map on $P(R)$, and so we get a natural map:

Definition 2.3.6: Hattori-Stallings Trace Map

Let R be a (not necessarily commutative ring). Then by the above discussion, the natural map:

$$K_0(R) \to \frac{R}{[R, R]}$$

is called the *Hattori-Stallings Trace Map.*

For commutative R the commutator subgroup vanishes, so $R/[R, R] \cong R$. This gives a more explicit description of $K_0(R)$ and $H_0(R)$, $\mathrm{Spec}\,(R)$ now being available. Start by taking the trace map on $K_0(R) \to R$ and restrict to $H_0(R)$. A continuous $\mathrm{Spec}\,(R) \to \mathbb{Z}$ has clopen fibres, so it decomposes $\mathrm{Spec}\,(R)$ into finitely many components and correspondingly splits $R = R_1 \times \cdots \times R_c$ by orthogonal idempotents $e_1, \ldots, e_c$ summing to 1. Since $\mathrm{tr}(e_iR) = e_i$, the trace of a class $f \in H_0(R)$, written $f(i)$ for its constant value on the i-th component, is

$$\mathrm{tr}(f) = \sum_i f(i)\, e_i$$

Because the e_i are orthogonal idempotents, $e_i e_j = \delta_{ij} e_i$, this assignment is multiplicative: $\mathrm{tr}(fg) = \mathrm{tr}(f)\mathrm{tr}(g)$. Hence the trace factors as follows.

Proposition 2.3.7: Hattori-Stallings Map Factorization

Let R be *commutative*. Then the Hattori-Stallings trace map factors through:

$$K_0(R) \xrightarrow{\text{rank}} H_0(R) \to R$$

***Proof*:**
Consider the following diagram:

$$\begin{tikzcd}
K_0(R) \arrow[r] \arrow[d, "\mathrm{tr}"'] & \prod_{\mathfrak{p}} K_0(R_{\mathfrak{p}}) \arrow[d, "\mathrm{tr}"] \\
R \arrow[r, hook, "\mathrm{diag}"] & \prod_{\mathfrak{p}} R_{\mathfrak{p}}
\end{tikzcd}$$

The square commutes because the trace is natural in the ring. The kernel of the top arrow is $\widetilde{K}_0(R)$, by the description of $\widetilde{K}_0$ as the intersection of the kernels of the maps to the local factors. The bottom arrow is injective, so the left-hand trace kills $\widetilde{K}_0(R)$. Since $K_0(R)/\widetilde{K}_0(R) = H_0(R)$, the trace factors through $H_0(R)$, which is the claimed factorization.

Example 2.3.8: Trace and Characters on Representations

1. Let k be commutative and $R = k[G]$ of a group G. Then if $g, h \in G$ are conjugate, then $g - h \in [R, R]$ (as $g = xhx^{-1}$, so $xhx^{-1} - h$ is the commutator of xh and x^{-1}). Hence $R/[R, R]$ is free on the conjugacy classes of G. In that basis:

$$\mathrm{tr}(P) = \sum_{g \in G} r_P(g)[g]$$

where $r_P(g)$ are coefficients seen as functions of the set $G/\sim$, for each P.

2. In the special case where G is finite, then any finitely generated projective $k[G]$-module P is also a projective k-module, so the trace map $\mathrm{End}_k(P) \to k$ is available and gives the character $\chi_P \colon G \to k$:

$$\chi_P(g) = \mathrm{tr}(g)$$

Hattori proved that:

$$\chi_P(g) = |Z_G(g)| r_P(g^{-1})$$

connecting characters to the trace theory of this section. The connection is not pursued further here.

One consequence is that $\mathbb{Z}[G]$ has no non-trivial idempotents:

***Proof*:**
Let $e \in \mathbb{Z}[G]$ be idempotent and put $P = e\mathbb{Z}[G]$, a finitely generated projective $\mathbb{Z}[G]$-module. Then $\chi_P(1)$ is the rank of P as a $\mathbb{Z}$-module, so $0 \leq \chi_P(1) \leq |G| = \operatorname{rank}_{\mathbb{Z}}\mathbb{Z}[G]$. Hattori's formula at $g = 1$ reads $\chi_P(1) = |Z_G(1)|\, r_P(1) = |G|\, r_P(1)$, with $r_P(1) \in \mathbb{Z}$. The only integers $r_P(1)$ compatible with that bound are 0 and 1, forcing $\operatorname{rank}_{\mathbb{Z}} P \in \{0, |G|\}$, that is $P = 0$ or $P = \mathbb{Z}[G]$, hence $e = 0$ or $e = 1$, completing the proof.

3. Let k be a field of characteristic 0 that is a splitting field for G, a finite group; for instance k algebraically closed. Maschke's theorem makes $k[G]$ semisimple, and Wedderburn's theorem then decomposes it as a product of simple k-algebras

$$k[G] \cong S_1 \times \cdots \times S_c$$

where c is the number of conjugacy classes of G. The splitting hypothesis is what makes c the conjugacy count: over a general characteristic-zero k the factors are indexed by Galois orbits instead. Quotienting by commutators kills nothing beyond each factor's own, so

$$\frac{k[G]}{[k[G], k[G]]} \cong k^c$$

and example 2.2.3(4) gives $K_0(k[G]) \cong \mathbb{Z}^c$. The trace map is then the inclusion $\mathbb{Z}^c \subseteq k^c$ of the ranks into the factors.

Determinant

Recall definition 1.2.9 where we can define the determinant in a natural way if R is commutative on a finitely generated projective R-module. We saw in proposition 1.2.10 that we may classify all finitely generated projective R-modules over a 1-dimensional ring, and in proposition 1.2.11 we saw that the determinant of a finitely generated projective R-module P can be seen as elements of $\operatorname{Pic}(R)$. These results work when reducing to K-groups:

Proposition 2.3.9: K_0 Group and Determinant

Let R be a commutative ring. Then the determinant determines a natural map:

$$\det : K_0(R) \to \operatorname{Pic}(R)$$

***Proof*:**
The key is that the non-top wedge products disappear. By the universal properties of monoids, it suffices to show that:

$$\det(P \oplus Q) \cong \det(P) \otimes \det(Q)$$

Without loss of generality, we may assume that P, Q have constant rank n, m respectively. Then $\bigwedge^{n+m}(P \oplus Q)$ is the sum over all i, j with $i + j = n + m$ of $(\bigwedge^i P) \otimes (\bigwedge^j Q)$. If $i > n$ then $\bigwedge^i P = 0$, and if $j > m$ then $\bigwedge^j Q = 0$; since $i + j = n + m$, the only surviving term is $i = n$, $j = m$. Hence:

$$\bigwedge^{n+m} (P \oplus Q) \cong (\bigwedge^{n} P) \otimes (\bigwedge^{m} Q)$$

as we sought to show.

Corollary 2.3.10: Rank and Determinant Represents K-group

Let R be a commutative ring. Then $H_0(R) \oplus \mathrm{Pic}\,(R)$ is a ring with a square-zero ideal $\mathrm{Pic}\,(R)$ and there is a surjective homomorphism:

$$K_0(R) \twoheadrightarrow H_0(R) \oplus \mathrm{Pic}\,(R)$$

with kernel consists of equivalence classes representable by $[P] - [R^n]$ where P has constant rank n and $\bigwedge^n P \cong R$.

***Proof*:**
Write $SK_0(R)$ for the kernel of $(\mathrm{rank}, \det)$. The pair is a homomorphism by lemma 2.3.3 and proposition 2.3.9, and it is surjective: given (f, L) with $f \in H_0(R)$ and L a line bundle, the class $[L] + (f-1)[R]$ has rank f and determinant L, using $\det(R^k) \cong R$.

The ring structure on $H_0(R) \oplus \mathrm{Pic}\,(R)$ is $(f, L)(g, M) = (fg,\ L^g \otimes M^f)$, under which $\mathrm{Pic}\,(R)$ is a square-zero ideal, since a product of two elements of $\{0\} \oplus \mathrm{Pic}\,(R)$ has first coordinate 0 and hence second coordinate trivial.

Finally, an element of $SK_0(R)$ has rank 0, so by lemma 2.3.4 it is $[P] - [R^n]$ with P of constant rank n, and it has trivial determinant, so $\bigwedge^n P \cong R$. That is the stated description of the kernel, completing the proof.

Corollary 2.3.11: Rank and Determinant: One Dimensional Case

Let R be a commutative Noetherian 1-dimensional ring. Then:

$$K_0(R) \cong H_0(R) \oplus \mathrm{Pic}\,(R)$$

***Proof*:**
This follows from the above corollary and proposition 1.2.10.

2.4 Morita Equivalence

Two rings can differ in every element-level respect and still have the same modules. When that happens K_0 cannot tell them apart, since K_0 is built from the isomorphism classes of finitely generated projectives under direct sum and an equivalence of module categories carries that data across intact. This section records the invariance and the one computation it makes routine.

The theory behind it belongs to *Core Algebra* [15, Morita's Theorem], which develops generators, progenerators and Morita's theorem in full; only the statements this chapter uses are recalled.

Definition 2.4.1: Morita Equivalence

Rings R and S are *Morita equivalent* if the categories R-**Mod** and S-**Mod** are equivalent.

Morita's theorem says every such equivalence has the same shape. A left R-module P is a *progenerator* if it is finitely generated, projective, and a generator, meaning every module is a quotient of a direct sum of copies of P. The theorem ([15, Morita's theorem]) states that R and S are Morita equivalent exactly when $S \cong \mathrm{End}_R(P)^{\mathrm{op}}$ for some progenerator P over R, in which case the equivalence is $\mathrm{Hom}_R(P, R) \otimes_R -$ with quasi-inverse $P \otimes_S -$. The invariance below needs only the consequence that such an equivalence matches up finitely generated projectives.

Corollary 2.4.2: Morita Invariance of K_0

Let R and S be Morita equivalent rings. Then

$$K_0(R) \cong K_0(S)$$

***Proof*:**
Let $F\colon R\text{-}\mathbf{Mod} \to S\text{-}\mathbf{Mod}$ be an equivalence. By [15, finitely generated projectives correspond], F restricts to a bijection between the isomorphism classes of finitely generated projective R-modules and those of finitely generated projective S-modules, and it carries direct sums to direct sums.

The isomorphism classes of finitely generated projectives over a ring form a commutative monoid under $\oplus$, and K_0 of that ring is its group completion by definition 2.2.1. The bijection above is therefore an isomorphism of monoids, and group completion is functorial, so it induces an isomorphism on the completions, completing the proof.

The converse fails, and cheaply: $K_0(\mathbb{Z}) \cong \mathbb{Z} \cong K_0(\mathbb{Q})$, while $\mathbb{Z}$ and $\mathbb{Q}$ are not Morita equivalent, being non-isomorphic commutative rings ([15, commutative rings are Morita rigid]). So K_0 is a coarser invariant than the module category.

Example 2.4.3: Matrix Rings
For any ring R and any $n \geq 1$, the module R^n of columns is a progenerator with $\mathrm{End}_R(R^n)^{\mathrm{op}} \cong M_n(R)$, so R and $M_n(R)$ are Morita equivalent. By corollary 2.4.2,

$$K_0\big(M_n(R)\big) \cong K_0(R)$$

for every n. Taking R a field F gives $K_0(M_n(F)) \cong K_0(F) \cong \mathbb{Z}$, so the rank of a matrix-algebra module is the only invariant K_0 sees, and taking $R = \mathbb{Z}$ gives $K_0(M_n(\mathbb{Z})) \cong \mathbb{Z}$.

More generally, if P is any progenerator over R then $K_0(\mathrm{End}_R(P)^{\mathrm{op}}) \cong K_0(R)$, which is the form used in section 5.2 when the exact category of projectives is replaced by an equivalent one.

2.5 *Mayer-Vietoris Sequence

In algebraic topology the Mayer-Vietoris sequence computes the invariants of a space glued from two pieces from the invariants of the pieces and of their overlap. The same shape is available for K-groups, with a Milnor square (theorem 1.1.16) playing the role of the gluing datum: S and R/I are the two pieces and S/I is the overlap. As we just saw, $M_n(R)$ produces equivalent categories. More generally, in representation theory we are interested in $\mathrm{GL}_n(R)$, the general linear group. Which n should we pick? In chapter 4, we shall take the limit over all the n to define the group $\mathrm{GL}(R)$. For now, we relate these through the natural map:

$$g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

These groups shall be central in understanding K_1 in the next section. One important construction we shall use is patching together information using Milnor patching: consider the Milnor square:

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow & & \downarrow \\ R/I & \xrightarrow{\bar{f}} & S/I \end{array}$$

Recall theorem 1.1.16(3) that any finitely generated projective R-module can derive from patching, hence define

$$\partial_n : \mathrm{GL}_n(S/I) \to K_0(R)$$

where $\partial_n(g) = [P] - [R^n]$. By theorem 1.1.16(5), the first two properties of $P(-)$ give:

$$\partial_n(g) = \partial_{n+1} \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

Hence, taking the sequence of $\{\partial_n\}_{n=1}^{\infty}$ give a homomorphism:

$$\partial : \mathrm{GL}(S/I) \to K_0(R)$$

Milnor patching now gives the sequence.

Theorem 2.5.1: Mayer-Vietoris

Given a Milnor square as above, the sequence

$$GL(S/I) \xrightarrow{\partial} K_0(R) \xrightarrow{\Delta} K_0(S) \oplus K_0(R/I) \xrightarrow{\pm} K_0(S/I)$$

is exact. The image of ∂ is the double coset space

$$GL(S)\backslash GL(S/I)/GL(R/I) = GL(S/I)/\sim$$

where $x \sim gxh$ for $x \in GL(S/I), g \in GL(S)$, and $h \in GL(R/I)$.

***Proof Key ideas*:**
Exactness at $K_0(R)$ and at $K_0(S) \oplus K_0(R/I)$ is the patching correspondence of theorem 1.1.16(1) and (3): a class dies in both $K_0(S)$ and $K_0(R/I)$ exactly when the two pieces it patches are free, which is when it comes from ∂. That ∂ is a homomorphism, and the identification of its image with the double coset space, both come from theorem 1.1.16(5): $\partial(g)$ depends only on the double coset of g, since changing g on either side by an element of $\mathrm{GL}(S)$ or of $\mathrm{GL}(R/I)$ changes the patched module by an isomorphism.

The remaining verification, that the sequence is exact at $\mathrm{GL}(S/I)$ and that the connecting map is well defined on the colimit rather than on each GL_n separately, is a diagram chase in the patching category that this book does not set up. It is carried out in [3, Ch. II], and we cite it rather than reproduce it.

Before computing with it, one companion sequence is needed. It is the units-and-Picard analogue of the K-group sequence above, and it is what converts a Milnor square into a Pic computation.

Theorem 2.5.2: The Units-Pic Sequence

Let Δ denote the diagonal map and $\pm$ the difference map, sending $(s, \bar{r})$ to $\bar{s}f(\bar{r})^{-1}$ and (L', L) to $L' \otimes_S S/I \otimes_{R/I} L^{-1}$. Then for a Milnor square the following sequence is exact:

$$1 \to R^\times \xrightarrow{\Delta} S^\times \times (R/I)^\times \xrightarrow{\pm} (S/I)^\times \xrightarrow{\partial} \mathrm{Pic}\,(R) \xrightarrow{\Delta} \mathrm{Pic}\,(S) \times \mathrm{Pic}\,(R/I) \xrightarrow{\pm} \mathrm{Pic}\,(S/I)$$

***Proof*:**
As R is the pullback of S and R/I, exactness at the first two places is immediate. Theorem 1.1.16 immediately gives exactness at the last two places, leaving only exactness at $(S/I)^\times$. Given $s \in S^\times$ and $\bar{r} \in (R/I)^\times$, set $\beta = \pm(s, \bar{r}) = \bar{s}f(\bar{r})^{-1}$, where $\bar{s}$ denotes the reduction of s modulo I. Then we see that $\lambda = (s, \bar{r}) \in L_\beta \subset S \times R/I$, and every element of L_β is a multiple of λ. It follows that $L_\beta \cong R$. Conversely, suppose given $\beta \in (S/I)^\times$ with $L_\beta \cong R$. If $\lambda = (s, \bar{r})$ is a generator of L_β, then s and $\bar{r}$ are both units, which implies that $\beta = \bar{s}f(\bar{r})^{-1}$ which completes the proof. To that end, if $s' \in S$ maps to $\beta \in S/I$, then $(s', 1) \in L_\beta$, namely since $(s', 1) = (xs, x\bar{r})$ for some $x \in R$, this implies that $\bar{r} \in (R/I)^\times$. If $t \in S$ maps to $f(\bar{r})^{-1}\beta^{-1} \in S/I$, then $st \equiv 1$ modulo I. But then $I \subset sR$ because $I \times 0 \subset L_\beta$, so $st = 1 + sx$ for some $x \in R$. So $s(t - x) = 1$, and hence $s \in S^\times$ giving the desired result.

With both sequences available, the standard computation runs as follows.

Example 2.5.3: K_0 of the Nodal Cubic
Let $R = k[x, y]/(y^2 - x^3 - x^2)$ be the coordinate ring of an irreducible nodal cubic, and let $S = k[t]$ be its normalization, with $t = y/x$. The conductor ideal is $I = (x, y)$, and $R/I \cong k$ while $S/I \cong k \times k$, the two branches at the node. This is a Milnor square.

Since R is a 1-dimensional Noetherian ring, corollary 2.3.11 gives

$$K_0(R) \cong H_0(R) \oplus \mathrm{Pic}\,(R)$$

and $\mathrm{Spec}\, R$ is connected, so $H_0(R) \cong \mathbb{Z}$. For $\mathrm{Pic}\,(R)$, feed the square into theorem 2.5.2. Both $S = k[t]$ and $R/I = k$ have trivial Picard group, so the sequence forces $\mathrm{Pic}\,(R)$ to be the cokernel of

$$S^\times \times (R/I)^\times \xrightarrow{\pm} (S/I)^\times$$

that is, of $k^\times \times k^\times \to k^\times \times k^\times$ sending (a, b) to the pair of branch values, whose image is the diagonal. The cokernel is $k^\times$, so $\mathrm{Pic}\,(R) \cong k^\times$ and

$$K_0(R) \cong \mathbb{Z} \oplus k^\times$$

The unit $k^\times$ measures the gluing datum at the node: a line bundle on R is a line bundle on the normalization together with an identification of the two branch fibres, and that identification is a scalar.

3

K_0 of a Category

Every construction of K_0 so far has taken a ring as input and produced a group from its finitely generated projective modules. The relations imposed were those of *split* exact sequences, because a direct sum decomposition is all that $\mathcal{P}(R)$ offers. That is a limitation of the input, not of the idea. A category of modules over a Noetherian ring has genuine short exact sequences, most of which do not split, and imposing $[B] = [A] + [C]$ on those produces a different and often more computable group. Vector bundles on a scheme sit between the two cases: kernels and cokernels of maps of vector bundles need not be vector bundles, so the category is not abelian, but it carries a distinguished class of sequences that behave like exact ones.

This chapter takes the categorical input seriously and defines K_0 for each of these settings in turn. The reward is that the theory stops being about rings. Once K_0 is defined for an abelian category it applies to coherent sheaves, which is what chapter 6 needs. Once it is defined for an exact category it applies to vector bundles, and the same input drives the Q-construction of chapter 5. The definitions here are elementary and can be checked by hand, which is the reason for meeting them now rather than when the higher theory demands them.

3.1 K_0 of an Abelian Category

One case is already behind us, and naming it fixes what the rest of the chapter varies. A *symmetric monoidal category* ([7, Symmetric Monoidal Category]) whose morphisms are all isomorphisms has a monoid of isomorphism classes under its product, and the Grothendieck completion of that monoid (proposition 2.1.1) is its K_0. Taking $\mathcal{P}(R)$ under $\oplus$ recovers definition 2.2.1 exactly: the ring-level K_0 of chapter 2 is the symmetric monoidal case, and the relations it imposes are those of split sequences because $\oplus$ is all a monoidal product sees. The sections below vary the input rather than the recipe, and section 5.3 returns to the monoidal case with $S^{-1}S$, which performs the same completion at the level of spaces rather than of monoids.

The passage from a ring to an abelian category changes one thing: which sequences impose relations. In $\mathcal{P}(R)$ every short exact sequence splits, so the relation $[B] = [A] + [C]$ adds nothing beyond $[A \oplus C] = [A] + [C]$. In a category such as the finitely generated modules over $\mathbb{Z}$ the sequence

$$0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n \to 0$$

does not split, and forcing $[\mathbb{Z}/n] = 0$ is a genuine new relation. The group that results is sensitive to torsion in a way the projective-module version is not.

Definition 3.1.1: K_0 of an Abelian Category

Let $\mathcal{A}$ be a small abelian category. Then $K_0(\mathcal{A})$ is the abelian group with one generator $[A]$ for each isomorphism class of objects of $\mathcal{A}$, subject to the relation

$$[B] = [A] + [C]$$

for every short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$.

Smallness is what makes the isomorphism classes a set, so that the free abelian group on them exists. In practice one replaces a large category by an equivalent small one, which does not change K_0. Taking the sequence $0 \to 0 \to A \to A \to 0$ gives $[0] = 0$, and $0 \to A \to A \oplus C \to C \to 0$ gives $[A \oplus C] = [A] + [C]$, so the direct sum relation is a consequence rather than an assumption.

The definition is engineered for one purpose: $K_0(\mathcal{A})$ is the universal recipient of an invariant that is additive on short exact sequences.

Proposition 3.1.2: Universal Property of $K_0(\mathcal{A})$

Let $\mathcal{A}$ be a small abelian category and G an abelian group. Call a function χ from the objects of $\mathcal{A}$ to G *additive* if $\chi(A) = \chi(A')$ whenever $A \cong A'$, and $\chi(B) = \chi(A) + \chi(C)$ for every short exact sequence $0 \to A \to B \to C \to 0$. Then composition with $A \mapsto [A]$ is a bijection

$$\mathrm{Hom}_{\mathbf{Ab}}\big(K_0(\mathcal{A}), G\big) \xrightarrow{\sim} \{\,\text{additive functions } \mathcal{A} \to G\,\}$$

***Proof*:**
A homomorphism $\varphi\colon K_0(\mathcal{A}) \to G$ gives the function $A \mapsto \varphi([A])$, which is additive because $[-]$ is constant on isomorphism classes and satisfies the defining relations. Conversely, an additive χ determines a homomorphism from the free abelian group on isomorphism classes, and it kills each defining relator $[B] - [A] - [C]$ precisely by additivity, so it descends to $K_0(\mathcal{A})$. The two assignments are mutually inverse, since a homomorphism out of $K_0(\mathcal{A})$ is determined by its values on the generators $[A]$, completing the proof.

The invariant of a ring obtained this way has its own name, and it is not $K_0(R)$.

Definition 3.1.3: G-Theory of a Ring

Let R be a Noetherian ring and $\mathrm{Mod}_{fg}(R)$ the category of finitely generated R-modules. Then

$$G_0(R) := K_0\big(\mathrm{Mod}_{fg}(R)\big)$$

The Noetherian hypothesis is what makes $\mathrm{Mod}_{fg}(R)$ abelian: a submodule of a finitely generated module over a non-Noetherian ring need not be finitely generated, so kernels can leave the category. Note the contrast with $K_0(R)$, which is built from projectives and split sequences. There is always a map $K_0(R) \to G_0(R)$, sending a projective module to itself. It is an isomorphism whenever R is regular Noetherian, so that every finitely generated module has a finite projective resolution. That is example 5.4.2, which the resolution theorem makes available later. The converse fails, so this is a sufficient condition and not a characterisation: over the cusp $R = k[[t^2, t^3]]$, which is not regular, the relations $2[k] = 0$ and $3[k] = 0$ coming from $R/(t^2)$ and $R/(t^3)$ force $[k] = 0$, leaving $G_0(R) = \mathbb{Z} \cdot [R] \cong K_0(R)$. The case needed here, R Dedekind, is proved directly in proposition 3.1.6 and depends on nothing below.

Example 3.1.4: Finite-Length Modules

Let $\mathcal{A}$ be the category of R-modules of finite length. Every object has a composition series, and the Jordan-Hölder theorem says the multiset of simple subquotients is independent of the series. Sending M to the tuple of those multiplicities is additive, since a short exact sequence splices composition series, so by proposition 3.1.2 it factors through $K_0(\mathcal{A})$. The factorization is an isomorphism

$$K_0(\mathcal{A}) \cong \bigoplus_S \mathbb{Z}$$

the sum over isomorphism classes of simple objects, with inverse sending the basis vector at S to $[S]$. That the two composites are the identity needs one observation: applying the defining relation along a composition series of M gives $[M] = \sum_S m_S(M)[S]$ in $K_0(\mathcal{A})$, where $m_S(M)$ is the multiplicity of S. So the classes of the simple objects generate, and the multiplicity map is inverse to the assignment $e_S \mapsto [S]$ on both sides.

1. For $R = \mathbb{F}_q$ the finite-length modules are the finite-dimensional vector spaces, there is one simple object, and $K_0(\mathcal{A}) \cong \mathbb{Z}$ by dimension.

2. For $R = \mathbb{Z}$ the finite-length modules are the finite abelian groups, the simple objects are the $\mathbb{Z}/p$, and $K_0(\mathcal{A}) \cong \bigoplus_p \mathbb{Z}$, one factor for each prime.

The second computation is worth pausing on, because it shows how much larger a finite-length K_0 can be than the K_0 of the ring it sits over. Passing from finite abelian groups to *all* finitely generated abelian groups collapses it completely.

Example 3.1.5: $G_0(\mathbb{Z})$ and $G_0(\mathbb{F}_q)$

For $R = \mathbb{F}_q$ the finitely generated modules are the finite-dimensional vector spaces, so $G_0(\mathbb{F}_q) \cong \mathbb{Z}$ by dimension, agreeing with $K_0(\mathbb{F}_q)$.

For $R = \mathbb{Z}$ the sequence $0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n \to 0$ gives $[\mathbb{Z}/n] = [\mathbb{Z}] - [\mathbb{Z}] = 0$ in $G_0(\mathbb{Z})$, so every finite abelian group has class zero. A finitely generated abelian group is a finite group plus a free part, so $[M] = \mathrm{rank}(M)\,[\mathbb{Z}]$ and $G_0(\mathbb{Z})$ is generated by $[\mathbb{Z}]$. That alone leaves open whether it is $\mathbb{Z}$ or a finite cyclic quotient. What closes it is that rank is itself additive, since $- \otimes_{\mathbb{Z}} \mathbb{Q}$ is exact, so by proposition 3.1.2 it defines a homomorphism $G_0(\mathbb{Z}) \to \mathbb{Z}$ sending $[\mathbb{Z}] \mapsto 1$. A generator mapping to a generator forces

$$G_0(\mathbb{Z}) \cong \mathbb{Z}$$

which again agrees with $K_0(\mathbb{Z})$. Compare example 3.1.4(2): the whole of $\bigoplus_p \mathbb{Z}$ dies once the free modules are admitted, because each $\mathbb{Z}/p$ becomes a difference of two copies of $\mathbb{Z}$.

For a Dedekind domain the two invariants again agree, but the common value is no longer $\mathbb{Z}$, and the difference is exactly the class group.

Proposition 3.1.6: G-Theory of a Dedekind Domain

Let R be a Dedekind domain. Then

$$G_0(R) \cong \mathbb{Z} \oplus \mathrm{Pic}\,(R)$$

In particular $G_0(\mathcal{O}_K) \cong \mathbb{Z} \oplus \mathrm{Cl}\,(K)$ for the ring of integers of a number field, and $G_0(\mathbb{Z}) \cong \mathbb{Z}$ since $\mathrm{Pic}\,(\mathbb{Z}) = 0$.

***Proof*:**
The route is the comparison map $K_0(R) \to G_0(R)$ of definition 3.1.3, and it has two halves.

The comparison is an isomorphism. Over a Dedekind domain this needs no general machinery, because resolutions have length at most one. Let M be finitely generated and choose a surjection $P_0 \twoheadrightarrow M$ with P_0 finitely generated projective, for instance free. Its kernel P_1 is a submodule of a free module, hence torsion-free, and finitely generated since R is Noetherian. A finitely generated torsion-free module over a Dedekind domain is projective ([5, Classifying Torsion-Free Modules Over Dedekind Domain]). So

$$0 \to P_1 \to P_0 \to M \to 0$$

is a projective resolution of length at most one. Set $\chi(M) := [P_0] - [P_1] \in K_0(R)$.

This is well defined: given a second such resolution $0 \to P_1' \to P_0' \to M \to 0$, Schanuel's lemma ([15, Schanuel's Lemma]) gives $P_1 \oplus P_0' \cong P_1' \oplus P_0$, whence $[P_0] - [P_1] = [P_0'] - [P_1']$. Note that Schanuel genuinely applies here, the P_i being projective objects of the ambient category. That is exactly what fails in the general resolution theorem, where the preferred subcategory need not consist of projectives.

It is additive. Let $0 \to M' \to M \to M'' \to 0$ be exact and take resolutions $0 \to P_1' \to P_0' \to M' \to 0$ and $0 \to P_1'' \to P_0'' \to M'' \to 0$ of the above form. The horseshoe lemma ([13, Horseshoe Lemma]) asks for an abelian category with enough projectives, which $\mathrm{Mod}_{fg}(R)$ is: every finitely generated module is a quotient of a finitely generated free one, and finitely generated projectives are projective objects there. It returns a resolution of M with $P_i = P_i' \oplus P_i''$, which is again of length at most one because $P_i' = P_i'' = 0$ for $i \geq 2$, and which stays finitely generated. Hence $\chi(M) = \chi(M') + \chi(M'')$. By proposition 3.1.2, χ therefore descends to a homomorphism $G_0(R) \to K_0(R)$.

The two maps are mutually inverse. On a projective P the resolution $0 \to 0 \to P \to P \to 0$ gives $\chi(P) = [P]$, so the composite on $K_0(R)$ is the identity; and applying the defining relation of $G_0(R)$ to $0 \to P_1 \to P_0 \to M \to 0$ gives $[M] = [P_0] - [P_1]$ there, so the composite on $G_0(R)$ is the identity too. Hence $K_0(R) \xrightarrow{\sim} G_0(R)$.

Computing $K_0(R)$. Assume R is not a field, so that $\dim R = 1$. A field has $\mathrm{Pic} = 0$ and $G_0 \cong \mathbb{Z}$, and the statement holds trivially there. Corollary 2.3.11 then applies, R being commutative Noetherian of dimension 1, and gives $K_0(R) \cong H_0(R) \oplus \mathrm{Pic}\,(R)$. A domain has irreducible, hence connected, spectrum, so $H_0(R) = \mathrm{Hom}(\mathrm{Spec}\,R, \mathbb{Z}) \cong \mathbb{Z}$, and the two halves combine to $G_0(R) \cong \mathbb{Z} \oplus \mathrm{Pic}\,(R)$, completing the proof.

The isomorphism is *not* induced by rank and determinant on G_0 directly. The determinant of definition 1.2.9 is defined on projectives, and the naive extension $M \mapsto \det(M/M_{\mathrm{tors}})$ is not additive: over $R = \mathbb{Z}[\sqrt{-5}]$ with $\mathfrak{p} = (2, 1+\sqrt{-5})$ the sequence $0 \to \mathfrak{p} \to R \to R/\mathfrak{p} \to 0$ would force $[\mathfrak{p}] = 0$ in $\mathrm{Pic}\,(R)$, which is false. What the isomorphism transports is the determinant *computed from a projective resolution*, $\det(M) = \det(P_0) \otimes \det(P_1)^{-1}$, under which $\det[R/\mathfrak{p}] = [\mathfrak{p}]^{-1}$.

Exercise 3.1.1

1. Let $\mathcal{A}$ be a small abelian category and $A \in \mathcal{A}$ an object with a finite filtration $0 = A_0 \subseteq A_1 \subseteq \cdots \subseteq A_m = A$. Show that $[A] = \sum_j [A_j/A_{j-1}]$ in $K_0(\mathcal{A})$.
2. Let G be a finite group and k a field of characteristic zero that is a splitting field for G. Show that K_0 of the category of finite-dimensional k-representations of G is free abelian of rank the number of conjugacy classes of G. Which result of this section does the computation reduce to, and where is the splitting hypothesis used?
3. Let R be Noetherian and $I \subseteq R$ a nilpotent ideal. Show that the functor sending an R/I-module to itself induces a surjection $G_0(R/I) \to G_0(R)$. (Filter by powers of I.)
4. Show that $\mathrm{Mod}_{fg}(R)$ is abelian if and only if R is Noetherian. For the harder direction, take a non-finitely-generated ideal $I \subseteq R$ and consider $R \to R/I$; exhibit the failure, and say why non-finite-generation of the module kernel really does obstruct a categorical kernel in the full subcategory.
5. Show that $\chi(M) = \dim_k M$ is additive on finite-dimensional $k[x]$-modules, and deduce that it factors through K_0 of that category. Compute that K_0 using example 3.1.4, and say what the simple objects are.

3.2 K_0 of an Exact Category

Between the two settings so far sits a third, and it is the one algebraic geometry presents. Vector bundles on a scheme are not an abelian category: the kernel of a map of bundles need not be a bundle, since the rank can jump on a closed subset. But they are not merely a monoid either, because a short exact sequence of bundles that happens to have bundle kernel and cokernel is a genuine constraint one wants K_0 to respect. An exact category axiomatizes precisely this: an additive category together with a nominated class of sequences that behave like short exact ones.

The starting datum is an additive category in which one knows which short exact sequences are allowed. The axioms are Quillen's, and they isolate exactly what the Q-construction of chapter 5 needs. They are not developed anywhere else in this series, so the list below is self-contained; [2] is the standard survey, and the reader who wants the redundancies among the axioms sorted out will find them there.

Definition 3.2.1: Exact Category

An *exact category* is an additive category $\mathscr{E}$ together with a class $\mathscr{M}$ of *admissible* short exact sequences

$$0 \to M' \xrightarrow{i} M \xrightarrow{j} M'' \to 0$$

closed under isomorphism, in which i is a kernel of j and j is a cokernel of i, satisfying:

1. the split sequence $0 \to M' \to M' \oplus M'' \to M'' \to 0$ is admissible for all objects M', M'' of $\mathscr{E}$;
2. the class of *admissible monomorphisms* (those i occurring as the left map of an admissible sequence, written $i\colon M' \rightarrowtail M$) is closed under composition, and the pushout of an admissible monomorphism along an arbitrary map exists and is again an admissible monomorphism;
3. the class of *admissible epimorphisms* (those j occurring as the right map, written $j\colon M \twoheadrightarrow M''$) is closed under composition, and the pullback of an admissible epimorphism along an arbitrary map exists and is again an admissible epimorphism;
4. the pullback of an admissible *monomorphism* along an admissible *epimorphism* is again an admissible monomorphism.

Clause (4) is redundant: it is derivable from (1) through (3), and many treatments omit it for that reason. The derivation is [2, Prop. 2.15], whose survey also shows that Quillen's original list carried a further axiom which is likewise redundant. Clause (4) is stated here because the composition law of the Q-construction (definition 5.2.1) uses it directly, and deriving it at that point would interrupt the construction with an argument about the axioms rather than about K-theory. Nothing below depends on the redundancy, only on the clause.

The axioms encode the operations a short exact sequence supports without committing to an ambient abelian category. Clauses (2) through (4) are what the whole theory uses: together they guarantee that a square built from one admissible mono and one admissible epi can be completed with both new legs admissible, and that completion is what defines composition in the Q-construction. An admissible monomorphism plays the role of a subobject inclusion and an admissible epimorphism the role of a quotient projection, and only those inclusions and quotients in the distinguished class $\mathscr{M}$ impose relations on K-theory.

Definition 3.2.2: The Exact Category $\mathscr{P}(R)$

Let R be a ring. The exact category $\mathscr{P}(R)$ has as objects the finitely generated projective R-modules and as morphisms all R-module homomorphisms. A short exact sequence of R-modules

$$0 \to P' \xrightarrow{i} P \xrightarrow{j} P'' \to 0$$

with P', P, P'' finitely generated projective is declared *admissible*.

Because every object of $\mathscr{P}(R)$ is projective, every short exact sequence of objects splits: the surjection j admits a section, so $P \cong P' \oplus P''$. The admissible monomorphisms are therefore exactly the split injections onto direct summands, and the admissible epimorphisms exactly the split projections off direct summands. The pullback and pushout axioms hold because the pullback of a split surjection is a split surjection, and its domain is again finitely generated projective: pulling $j\colon P \twoheadrightarrow P''$ back along $f\colon Q \to P''$ gives $P \times_{P''} Q \cong \ker j \oplus Q$, a direct sum of finitely generated projectives. Pushouts are dual, and the category is additive with $\oplus$ as biproduct. Thus $\mathscr{P}(R)$ is an exact category in the sense of definition 3.2.1, and it is the category whose K-theory will be the K-theory of the ring R.

Example 3.2.3: The Split Exact Category over a Field

Let F be a field. An object of $\mathscr{P}(F)$ is a finite-dimensional F-vector space, and up to isomorphism it is determined by its dimension, so the isomorphism classes are $\{F^n : n \geq 0\}$. Every short exact sequence

$$0 \to F^a \to F^n \to F^b \to 0$$

of finite-dimensional vector spaces splits (a basis of the image lifts), and the rank-nullity relation forces $n = a + b$. Hence the admissible sequences of $\mathscr{P}(F)$ are precisely the sequences realizing $F^n \cong F^a \oplus F^b$ with $n = a + b$, and an admissible monomorphism $F^a \rightarrowtail F^n$ is the inclusion of an a-dimensional subspace as a direct summand, which (since F is a field) means any injective linear map. The single numerical invariant, dimension, is additive on admissible sequences. This is the simplest exact category with nontrivial K_0, and the one against which the abstract statements of this chapter are most cheaply tested.

With the input fixed, K_0 is defined exactly as before, with the admissible sequences giving the relations.

Definition 3.2.4: K_0 of an Exact Category

Let $\mathscr{E}$ be a small exact category, or one equivalent to a small category, as in definition 3.1.1. Then $K_0(\mathscr{E})$ is the abelian group with one generator $[M]$ for each isomorphism class of objects, subject to

$$[M] = [M'] + [M'']$$

for every *admissible* short exact sequence $0 \to M' \to M \to M'' \to 0$.

Everything turns on which sequences were nominated. Choosing a smaller class $\mathscr{M}$ imposes fewer relations and remembers more. Choosing only the split sequences remembers nothing beyond direct-sum decompositions. The universal property is the same as in the abelian case, with the same one-line proof.

Proposition 3.2.5: Universal Property of $K_0(\mathscr{E})$

Let $\mathscr{E}$ be a small exact category and G an abelian group. Call $\chi\colon \operatorname{Ob}(\mathscr{E}) \to G$ *additive* if it is constant on isomorphism classes and $\chi(M) = \chi(M') + \chi(M'')$ for every admissible sequence. Then composition with $M \mapsto [M]$ is a bijection from $\operatorname{Hom}_{\mathbf{Ab}}(K_0(\mathscr{E}), G)$ to the set of additive functions, natural in G.

***Proof*:**

Identical to proposition 3.1.2, with "short exact sequence" replaced throughout by "admissible short exact sequence": an additive χ kills each defining relator and so descends, a homomorphism out of $K_0(\mathscr{E})$ restricts to an additive function, and the two assignments are mutually inverse because the classes $[M]$ generate. Naturality in G is immediate, the bijection being composition with a fixed map, completing the proof.

The first thing to check is that the new definition does not disagree with the old one where both apply.

Proposition 3.2.6: K_0 of $\mathscr{P}(R)$ Recovers $K_0(R)$

Let R be a ring. Then $K_0(\mathscr{P}(R)) \cong K_0(R)$, the Grothendieck group of definition 2.2.1, by an isomorphism carrying $[P]$ to $[P]$.

***Proof*:**
Both groups corepresent the same functor, so it is enough to identify the two notions of additive function.

Every admissible sequence of $\mathscr{P}(R)$ splits, so a function χ on isomorphism classes is additive in the sense of proposition 3.2.5 exactly when $\chi(P' \oplus P'') = \chi(P') + \chi(P'')$. Such a χ also sends 0 to 0, by additivity applied to the admissible sequence $0 \to 0 \to P \to P \to 0$, so it is exactly a homomorphism of monoids from $(P(R), \oplus)$ to G. So $\mathrm{Hom}_{\mathbf{Ab}}(K_0(\mathscr{P}(R)), G) \cong \mathrm{Hom}_{\mathbf{Mon}}(P(R), G)$ naturally in G, by proposition 3.2.5.

On the other side, $K_0(R)$ is by definition 2.2.1 the Grothendieck completion of $(P(R), \oplus)$, and [15, Properties of the Grothendieck Completion](4) gives $\mathrm{Hom}_{\mathbf{Ab}}(K_0(R), G) \cong \mathrm{Hom}_{\mathbf{Mon}}(P(R), G)$, again naturally in G. Two objects corepresenting the same functor are canonically isomorphic by Yoneda, and chasing the identity map shows the isomorphism sends $[P]$ to $[P]$, completing the proof.

Functoriality needs the maps that respect the nominated sequences.

Proposition 3.2.7: Exact Functors Induce Homomorphisms

Let $F\colon \mathscr{E} \to \mathscr{E}'$ be an *exact functor* of small exact categories, that is, an additive functor carrying admissible sequences to admissible sequences. Then $[M] \mapsto [FM]$ defines a homomorphism $K_0(F)\colon K_0(\mathscr{E}) \to K_0(\mathscr{E}')$, and $K_0(G \circ F) = K_0(G) \circ K_0(F)$ with $K_0(\mathrm{id}) = \mathrm{id}$.

***Proof*:**
The assignment $M \mapsto [FM]$ is constant on isomorphism classes, F being a functor, and carries an admissible sequence of $\mathscr{E}$ to one of $\mathscr{E}'$, so $[FM] = [FM'] + [FM'']$ holds in $K_0(\mathscr{E}')$. It is therefore additive, and proposition 3.2.5 gives the homomorphism. Functoriality is immediate from the formula on generators, completing the proof.

Exercise 3.2.1

1. Show that an abelian category is exact when every short exact sequence is declared admissible, by checking the three axioms of definition 3.2.1, and that definition 3.2.4 then agrees with definition 3.1.1.

2. Let $\mathscr{E}$ be an additive category with only the split sequences declared admissible. Show that $K_0(\mathscr{E})$ is the Grothendieck completion of the monoid of isomorphism classes under $\oplus$.

3. Exhibit one additive category carrying two different exact structures whose K_0 groups differ, and compute both. (Finitely generated abelian groups will do.)
4. Show that $- \otimes_R S$ is an exact functor $\mathscr{P}(R) \to \mathscr{P}(S)$ for a ring map $R \to S$, and identify the induced map of proposition 3.2.7 with the one already described after definition 2.2.1.
5. Let $\mathscr{E}$ be a small exact category and $\mathscr{E}_0 \subseteq \mathscr{E}$ a full additive subcategory closed under admissible extensions, with the inherited structure. Show that the inclusion is exact, and give an example where the induced map on K_0 is not injective.

4

K_1 and K_2 Group

The K_0 group let us classify the vector bundles over a ring or (paracompact) topological space. As shown in the topological K-theory of [22], for (say complex) vector bundles over X:

$$K_0(X) = [X, \mathbb{Z} \times BU]$$

Showing there is a relation between homotopy groups and $K_0(X)$. From the algebraic perspective, we found the isomorphism classes of the projective modules. It is now natural to ask for the automorphism classes. We shall focus on the algebraic perspective, but shall see that it will naturally tie with the topological by taking suspension (recall from [20, chapter 5] that there is a natural map $\pi_k(X) \to \pi_{k+1}(\Sigma X)$).

4.1 Defining K_1

Let R be a ring (with unit). Take the maps:

$$G_n(R) \to G_{n+1}(R) \qquad g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

The limit of these maps shall be denoted $\mathrm{GL}(R)$. Then:

Definition 4.1.1: K_1 Group

Let R be a ring (not necessarily commutative). Then:

$$K_1(R) = \frac{GL(R)}{[GL(R), GL(R)]}$$

Given a ring homomorphism $f : R \to S$, applying f to each matrix entry gives $\mathrm{GL}_n(R) \to \mathrm{GL}_n(S)$ compatibly with stabilization, hence an induced map $\mathrm{GL}(R) \to \mathrm{GL}(S)$. A group homomorphism carries commutators to commutators, so this descends to the abelianizations and gives a natural map:

$$K_1(R) \to K_1(S)$$

And hence, $K_1(-)$ is a functor as well.

Before looking at examples, let us have a better grasp of the commutator. As it turns out, these shall always be generated by elementary matrices.

Definition 4.1.2: Elementary Matrices Group of Gl(R)

Let $E_n(R)$ denote the elementary matrices of $\mathrm{GL}_n(R)$. Then $E(R)$ denotes the limit of the stabilization morphisms.

Lemma 4.1.3: Elementary Matrix Group is Perfect

For $n \geq 3$, $E_n(R)$ is a perfect group, that is $E_n(R) = [E_n(R), E_n(R)]$

The restriction to $n \geq 3$ is needed: $E_2(\mathbb{F}_2) = \mathrm{SL}_2(\mathbb{F}_2) \cong S_3$, which is not perfect. Its commutator subgroup is $A_3 \cong \mathbb{Z}/3$, of index 2, so its abelianization is $\mathbb{Z}/2$.

***Proof*:**
For distinct i, j, k, we get:

$$[e_{ij}(r), e_{k\ell}(s)] = \begin{cases} 1 & \text{if } j \neq k \text{ and } i \neq \ell, \\ e_{i\ell}(rs) & \text{if } j = k \text{ and } i \neq \ell, \\ e_{kj}(-sr) & \text{if } j \neq k \text{ and } i = \ell \end{cases}$$

and so:

$$e_{ij}(r) = [e_{ik}(r), e_{kj}(1)]$$

Theorem 4.1.4: Whitehead's Lemma

$$K_1(R) = \frac{\mathrm{GL}(R)}{E(R)}$$

***Proof*:**
The commutator subgroup contains $E(R)$ by lemma 4.1.3. Conversely, every commutator in $GL_n(R)$ can be expressed as a product in $GL_{2n}(R)$:

$$[g, h] = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} h & 0 \\ 0 & h^{-1} \end{pmatrix} \begin{pmatrix} (hg)^{-1} & 0 \\ 0 & hg \end{pmatrix} \tag{4.1}$$

And certainly each of these terms is in $E_{2n}(R)$, giving the desired equivalence.

Example 4.1.5: K_1 Groups

1. Let R be commutative. Then there is a group homomorphism:

$$\det : \mathrm{GL}(R) \to R^\times$$

This induces a surjective map $\det\colon K_1(R) \to R^\times$ whose kernel is written $SK_1(R)$. There is also a natural inclusion $R^\times \to \mathrm{GL}(R)$ sending u to $\mathrm{diag}(u, 1, 1, \dots)$, and the determinant splits it. With $\mathrm{SL}(R) = \varinjlim_n \mathrm{SL}_n(R)$ as the kernel, this is exactly the situation of [15, Recognition Theorem For Semi-Direct Products]:

$$\mathrm{GL}(R) \cong \mathrm{SL}(R) \rtimes R^\times$$

Abelianizing therefore gives:

$$K_1(R) \cong R^\times \oplus SK_1(R)$$

2. For $R = k$ a field with $k \notin \{\mathbb{F}_2, \mathbb{F}_3\}$, one has $\mathrm{SL}(k) = [\mathrm{GL}(k), \mathrm{GL}(k)]$, which reduces the previous item to:

$$K_1(F) \cong F^\times$$

One inclusion is immediate: $[\mathrm{GL}(k), \mathrm{GL}(k)] \leqslant \mathrm{SL}(k)$, since each commutator $[A, B] = ABA^{-1}B^{-1}$ has determinant 1. For the reverse, every matrix in $\mathrm{SL}_n(k)$ must be written as a product of commutators, and over a field it suffices to show that all elementary matrices can be written as a product of commutators. But this is now a matter of computations, for example type I elementary matrices $A = I + tE_{ij}$ and $B = I + E_{jk}$ where $k \neq i, j$ (which is possible since $n \geq 3$), when taking their their commutator is:

$$[A, B] = ABA^{-1}B^{-1} = (I + tE_{ij})(I + E_{jk})(I - tE_{ij})(I - E_{jk}) = I + tE_{ik}$$

notice how it was importance for $n \geq 3$ to insure the computations follow.

3. If $R = R' \times R''$ is a product of rings, then $\mathrm{GL}(R) = \mathrm{GL}(R') \times \mathrm{GL}(R'')$, and the commutator subgroups decompose likewise, so K_1 respects finite products.

4. The K_1 group is invariant under Morita equivalence. First, as R and $M_n(R)$ are Morita equivalent:

$$M_{mn}(R) = \mathrm{End}_R(R^m \otimes R^n) \qquad M_m(M_n(R)) = \mathrm{End}_{M_n(R)}((M_n(R))^m)$$

There is therefore a natural isomorphism:

$$\mathrm{GL}_{mn}(R) \cong \mathrm{GL}_m(M_n(R))$$

which give the same "stabilization" morphism (the map between $\mathrm{GL}_n(R)$ and $\mathrm{GL}_{n+1}(R)$), and so:

$$K_1(R) \cong K_1(M_n(R))$$

5. Let $R = D$ be a division ring. Then for the same reason as field, every invertible matrix may be reduced to:

$$\mathrm{diag}(r, 1, ..., 1)$$

and hence:

$$\frac{\mathrm{GL}_n(D)}{E_n(D)} \cong D^\times/W(D)$$

where $W(D)$ is the subgroup generated by the $(1+rs)(1+sr)^{-1}$, introduced below. A division ring is semilocal, so by lemma 4.1.8 together with Vaserstein's identification $W(D) = [D^\times, D^\times]$ this becomes:

$$\frac{\mathrm{GL}_n(R)}{E_n(D)} \cong D^\times/[D^\times, D^\times]$$

for all $n > 1$.

6. Let $F/\mathbb{Q}$ be a finite field extension and $\mathcal{O}_F$ its ring of integers, the integral closure of $\mathbb{Z}$ in F. Then it is a result of Bass and Milnor that

$$SK_1(\mathcal{O}_F) = 1$$

and so $K_1(\mathcal{O}_F) = \mathcal{O}_F^\times$. By [5, Dirichlet Unit Theorem], the group $\mathcal{O}_F^\times$ is isomorphic to:

$$\mu(F) \times \mathbb{Z}^{r_1+r_2-1}$$

where $\mu(F)$ is the finite cyclic group of roots of unity in F, and r_1 and r_2 count the real embeddings and the pairs of conjugate complex embeddings of F, so that $r_1 + r_2$ is the number of archimedean places. For $F = \mathbb{Q}$ this gives $K_1(\mathbb{Z}) = \{\pm 1\}$, and for an imaginary quadratic F it gives $K_1(\mathcal{O}_F) = \mu(F)$, both finite; the rank is positive exactly when F has more than one archimedean place.

Let us take a moment to look at the K_1 group of semi-local rings due to their importance in representation theory (they are the "local rings" in the non-commutative case).

Definition 4.1.6: Semi-local Ring

Let R be a ring. Then R is *semilocal* if $R/J(R)$ is semisimple, where $J(R)$ is the Jacobson radical of R.

Note that all division rings are semi-local, hence the discussion about semi-local rings generalizes the above discussion.

The commutator subgroup $[R^\times, R^\times]$ admits a concrete description for semilocal rings. If $r, s \in R$ are such that $1 + rs$ is a unit, then so is $1 + sr$ because $(1 + sr)(1 - s(1 + rs)^{-1}r) = 1$. Then define subgroup $W(R) \leqslant R^\times$ generated by the $(1 + rs)(1 + sr)^{-1}$. It is not too hard to show that this subgroup belongs to $E_2(R)$. For $R = M_2(\mathbb{F}_2)$, $W(R) = R^\times \cong \Sigma_3$ and $K_1(R) = 1$ but $R^\times \neq [R^\times, R^\times]$. If T is the subring of upper triangular matrices in $M_2(\mathbb{F}_2)$, its group of units is abelian ($T^\times \cong \mathbb{Z}/2$), but $K_1(T) = 1$ since

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = (1 + rs)(1 + sr)^{-1} \text{ for } r = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } s = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Now, a mathematician called *Vaserstein* showed that $W(R) = [R^\times, R^\times]$ if $\Lambda = R/\mathrm{rad}(R)$ is a product of matrix rings, none of which is $M_2(\mathbb{F}_2)$, and at most one of the factors in Λ is $\mathbb{F}_2$. In particular, $W(R) = [R^\times, R^\times]$ for every local ring R. The same applies for semilocal rings:

Proposition 4.1.7: K1 of Semilocal Rings

If R is a semilocal ring, then the natural inclusion of $R^\times = GL_1(R)$ into $GL(R)$ induces an isomorphism $K_1(R) \cong R^\times/W(R)$.
If R is a commutative semilocal ring, then

$$SK_1(R) = 1 \quad \text{and} \quad K_1(R) = R^\times.$$

***Proof*:**
By lemma 4.1.8 it suffices to prove that $R^\times = \mathrm{GL}_1(R)$ maps onto $K_1(R)$. This will follow by induction after showing that $GL_n(R) = E_n(R)GL_{n-1}(R)$. Let J denote the Jacobson radical of R, so that R/J is a finite product of matrix algebras over division rings. We will show that $\mathrm{GL}_n(R) \twoheadrightarrow \mathrm{GL}_n(R/I)$. Indeed, if $I \subseteq R$ is an ideal, each homomorphism $E_n(R) \to E_n(R/I)$ is onto as the generators $e_{ij}(r)$ of $E_n(R)$ map onto the generators of $E_n(R/I)$ as I is radical. Hence, $(R/J)^\times$ maps onto $K_1(R/J)$. Every $\bar{g} \in GL_n(R/J)$ is a product $\bar{e}\bar{g}_1$, where $\bar{e} \in E_n(R/J)$ and $\bar{g}_1 \in GL_1(R/J)$.

Now, given $g \in GL_n(R)$, its reduction $\bar{g}$ in $GL_n(R/J)$ may be decomposed into $\bar{g} = \bar{e}\bar{g}_1$. Then we can lift $\bar{e}$ to an element $e \in E_n(R)$. The matrix $e^{-1}g$ is congruent to the diagonal matrix $\bar{g}_1$ modulo J, so its diagonal entries are all units and its off-diagonal entries lie in J. Since the off-diagonal entries lie in J and the diagonal entries are units, elementary row operations $e_{ij}(r)$ with $r \in J$ clear the off-diagonal entries one at a time, reducing $e^{-1}g$ to a diagonal matrix $D = \mathrm{diag}(r_1, \dots, r_n)$. The matrix $\mathrm{diag}(1, \dots, 1, r_n, r_n^{-1})$ is in $E_{n+1}(R)$ by the Whitehead identity of (4.1). Multiplying D by this matrix yields a matrix in $\mathrm{GL}_{n-1}(R)$. The induction starts at $n = 1$, where the claim is vacuous, completing the proof.

Lemma 4.1.8: Determinant on semilocal Rings

Let R be a noncommutative semilocal ring ($R/J(R)$ is a semisimple ring). Then there exists a unique determinant map from

$$\det : GL_n(R) \twoheadrightarrow R^\times/W(R)$$

where:

1. $\det(e) = 1$ for every elementary matrix e,
2. if $\rho = \mathrm{diag}(r, 1, \dots, 1)$ and $g \in \mathrm{GL}_n(R)$, then $\det(\rho \cdot g) = r \cdot \det(g)$.

It is moreover a group homomorphism, $\det(gh) = \det(g)\det(h)$. In particular, when reducing to $K_1(R)$:

$$K_1(R) \cong R^\times/W(R)$$

***Proof*:**
Since R is semilocal, the previous proof gives $\mathrm{GL}_n(R) = E_n(R) \cdot \mathrm{GL}_1(R)$: every matrix is a product of elementary matrices and a single diagonal matrix $\mathrm{diag}(r, 1, \dots, 1)$. Define the map $\det : GL_n(R) \to R^\times/W(R)$ as follows. For any elementary matrix e, we set $\det(e) = 1$. For a

diagonal matrix $\rho = \mathrm{diag}(r_1, r_2, \ldots, r_n)$, define

$$\det(\rho) = r_1 r_2 \cdots r_n \cdot W(R)$$

For a general matrix $g \in GL_n(R)$, express g as a product of elementary matrices and diagonal matrices, and define $\det(g)$ as the product of the determinants of these factors. Since the decomposition $\mathrm{GL}_n(R) = E_n(R) \cdot \mathrm{GL}_1(R)$ exhausts $\mathrm{GL}_n(R)$, and properties (1) and (2) prescribe the value on each factor, any map satisfying them agrees with this one, so it is unique. And indeed, let $\rho = \mathrm{diag}(r, 1, \ldots, 1)$ and $g \in GL_n(R)$. Express g as a product $g = e_1 e_2 \cdots e_k \cdot d$, where each e_i is an elementary matrix and d the diagonal matrix. Then $\rho \cdot g = \rho \cdot e_1 e_2 \cdots e_k \cdot d$. Then this is equivalent to multiplying the first row of g by r, which multiplies the determinant by r. Hence,

$$\det(\rho \cdot g) = r \cdot \det(g)$$

Next, we must show det is a group homomorphism

$$\det(gh) = \det(g)\det(h) \text{ for all } g, h \in GL_n(R)$$

It suffices to establish this for elementary and diagonal matrices. For elementary matrices e_1 and e_2, we have $\det(e_1 e_2) = 1 = \det(e_1)\det(e_2)$. For diagonal matrices $d_1 = \mathrm{diag}(r_1, \ldots, r_n)$ and $d_2 = \mathrm{diag}(s_1, \ldots, s_n)$, we have $\det(d_1 d_2) = \det(\mathrm{diag}(r_1 s_1, \ldots, r_n s_n)) = r_1 s_1 \cdots r_n s_n \cdot W(R) = (r_1 \cdots r_n \cdot W(R))(s_1 \cdots s_n \cdot W(R)) = \det(d_1)\det(d_2)$.

Finally, by definition $K_1(R)$ is the abelianization of $GL(R) = \varinjlim_n GL_n(R)$. The determinant map we constructed gives a homomorphism from $GL_n(R)$ to $R^\times/W(R)$ for each n, which are compatible under stabilization, so they assemble into a homomorphism $\mathrm{GL}(R) \to R^\times/W(R)$. Its target is abelian, so it factors through the abelianization $K_1(R)$. It is surjective because $\mathrm{GL}_1(R) = R^\times$ already surjects, and its kernel is exactly $W(R)$ by property (1), every elementary matrix having determinant 1. Hence

$$K_1(R) \cong R^\times/W(R)$$

as we sought to show.

With the basic properties of K_1 established on rings. Let us start by applying it to finitely generated projective R-modules. As $K_1(R)$ is defined as an amalgamation of all the "automorphisms" representable as matrices, the natural question is how $\mathrm{Aut}(P)$ relates to $K_1(R)$ for a finitely generated projective P. Since P is projective there exists a finitely generated projective R-module Q with

$$P \oplus Q = R^n$$

This defines a natural map $\mathrm{Aut}(P) \to \mathrm{GL}_n(R)$, $\alpha \mapsto \alpha \oplus 1_Q$, which is the basis of the map to $K_1(R)$:

Lemma 4.1.9: Automorphism Map

There exits a well-defined map $\mathrm{Aut}(P) \to \mathrm{GL}(R)$ up to inner automorphism of $\mathrm{GL}(R)$, and hence a well-defined homomorphism:

$$\mathrm{Aut}(P) \to K_1(R)$$

***Proof*:**
First, let's show it is defined up to inner automorphism. Fix Q and n. Then Two different isomorphisms between $P \oplus Q$ and R^n must differ by an automorphism of R^n, i.e., by an element $g \in GL_n(R)$. Thus if $\alpha \in \mathrm{Aut}(P)$ maps to the matrices A and B, respectively, we have:

$$A = gBg^{-1}$$

Next we observe that we can stabilize; replacing Q by $Q \oplus R^m$ and $P \oplus Q \cong R^n$ by $P \oplus (Q \oplus R^m) \cong R^{n+m}$, then $GL_n(R) \to GL(R)$ factors through $GL_{n+m}(R)$. Finally, suppose given a second isomorphism $P \oplus Q' \cong R^m$. Since $Q \oplus R^m \cong R^n \oplus Q'$, we may stabilize both Q and Q' to make them isomorphic and invoke the above argument.

But then, we get a map $\mathrm{Aut}(P) \to \mathrm{GL}(R)$, and as its defined up to inner automorphism we have a well-defined map:

$$\mathrm{Aut}(P) \to K_1(R)$$

as we sought to show.

Corollary 4.1.10: Tensor and $K_1(S)$ as $K_0(R)$ module

Let R, S be rings. Then there is a natural tensor product:

$$K_0(R) \otimes K_1(S) \to K_1(R \otimes S)$$

If R is commutative and S an R-algebra, there is a further natural map:

$$K_0(R) \otimes K_1(S) \to K_1(S)$$

Hence, $K_1(S)$ is a module over the ring $K_0(R)$.

***Proof*:**
For each finitely generated projective R-module P and each m, lemma 4.1.9 gives us:

$$\mathrm{Aut}(P \otimes S^m) \to K_1(R \otimes S)$$

Next, for each $\beta \in GL_m(S)$, let $[P] \cdot \beta$ denote the image of the automorphism $1_P \otimes \beta$ of $P \otimes S^m$ under this map. Fixing β and m, the isomorphism

$$(P \oplus P') \otimes S^m \cong (P \otimes S^m) \oplus (P' \otimes S^m)$$

yields the identity

$$[P \oplus P'] \cdot \beta = [P] \cdot \beta + [P'] \cdot \beta$$

in $K_1(R \otimes S)$. Hence $P \mapsto [P] \cdot \beta$ is an additive function of $P \in \mathbf{P}(R)$, and so it factors through $K_0(R)$.

Now fix P. then the map $GL_m(S) \to K_1(R \otimes S)$ given by $\beta \mapsto [P] \cdot \beta$ is compatible with stabilization in m. Thus it factors through a map $GL(S) \to K_1(R \otimes S)$ and through a map $K_1(S) \to K_1(R \otimes S)$. Hence, we see the product is well-defined and bilinear.

When R is commutative, $K_0(R)$ is a ring (as we saw in chapter 2). If S is an R-algebra, there is a ring map $R \otimes S \to S$. Composing the external product with $K_1(R \otimes S) \to K_1(S)$ yields a natural product operation $K_0(R) \otimes K_1(S) \to K_1(S)$. Associativity of the tensor product then gives

$$[P \otimes_R Q] \cdot \beta = [P] \cdot ([Q] \cdot \beta)$$

completing the proof.

Section 2.3 connected K_0 with H_0. The same can be done for K_1. Recall first from [20, Hurewicz Theorem in Dimension One] that $H_1(G; \mathbb{Z})$ is naturally isomorphic to $G/[G, G]$. Taking $G = \mathrm{GL}(R)$ gives:

$$K_1(R) = H_1(\mathrm{GL}(R); \mathbb{Z}) = \varinjlim_n H_1(\mathrm{GL}_n(R); \mathbb{Z})$$

Lemma 4.1.9 gives the composition:

$$H_1(\mathrm{Aut}(P); \mathbb{Z}) \to H_1(\mathrm{GL}_n(R); \mathbb{Z}) \to K_1(R)$$

Morita invariance of $K_1(-)$ comes next, and it needs the homological interpretation in a different form. Let $\mathbf{P}(R)$ be the category of finitely generated projective R-modules and let $t\mathbf{P}$ (the objects are the same, the morphisms between P and P' are the isomorphism classes of Q such that $P \oplus Q = P'$). Then $P \mapsto H_1(\mathrm{Aut}(P); \mathbb{Z})$ is a well-defined functor from $t\mathbf{P}$ to $\mathbf{Ab}$, so filtered colimits of it are available (recall [7]). But then, by taking advantage of the fact that free modules are cofinal in $t\mathbf{P}$:

Lemma 4.1.11: Morita Buildup (Bass)

$$K_1(R) \cong \varinjlim_{P \in t\mathbf{P}} H_1(\mathrm{Aut}(P); \mathbb{Z})$$

***Proof*:**
The functor $P \mapsto H_1(\mathrm{Aut}(P); \mathbb{Z})$ on $t\mathbf{P}(R)$ was constructed above, and the free modules R^n are cofinal in $t\mathbf{P}(R)$: every finitely generated projective P admits Q with $P \oplus Q \cong R^n$, which is a morphism $P \to R^n$ in $t\mathbf{P}(R)$. A colimit over a filtered category is unchanged by restricting to a cofinal subcategory, so

$$\varinjlim_{P \in t\mathbf{P}} H_1(\mathrm{Aut}(P); \mathbb{Z}) \cong \varinjlim_n H_1(\mathrm{GL}_n(R); \mathbb{Z})$$

and the right side is $H_1(\mathrm{GL}(R); \mathbb{Z}) = K_1(R)$, since group homology commutes with the filtered colimit defining $\mathrm{GL}(R)$ and H_1 is abelianization, completing the proof.

Since $\mathbf{P}(R)$ and $\mathbf{P}(S)$ are Morita equivalent, the above corollary gives:

Proposition 4.1.12: Morita Invariance of K_1

Let R, S be Morita-equivalent rings. Then:

$$K_1(R) \cong K_1(S)$$

***Proof*:**
Looking at the usual case of R and $M_n(R)$ using lemma 4.1.11, and that we have the Morita-equivalent categories to get:

$$K_1(R) \cong K_1(M_n(R))$$

A ring homomorphism $f\colon R \to S$ induces the same map as for K_0, but there is a second map, going the other way, with better applications in cohomology. Whenever S has a finite resolution by finitely generated projective R-modules there is a transfer:

$$f_* : K_1(S) \to K_1(R)$$

which has no K_0 analogue built from $- \otimes_R S$ alone.

Lemma 4.1.13: Simple Transfer Functor

Let $T : \mathbf{S} \to \mathbf{R}$ be an additive functor. Then it induces a homomorphism

$$K_1(T)\colon K_1(S) \to K_1(R)$$

And $T_1 \oplus T_2$ induces the sum $K_1(T_1) + K_1(T_2)$.

***Proof*:**
First, the functor T induces an evident functor $t\mathbf{P}(S) \to t\mathbf{P}(R)$. Next, if P is a finitely generated projective S-module, by cohomology theory T also induces a homomorphism $\mathrm{Aut}_S(P) \to \mathrm{Aut}_R(TP)$ and hence $H_1(\mathrm{Aut}_S(P);\mathbb{Z}) \to H_1(\mathrm{Aut}_R(TP);\mathbb{Z})$. As P varies, these assemble to give a natural transformation of functors from the translation category $t\mathbf{P}(S)$ to abelian groups. Since $K_1(S) = \varinjlim_{P\in\mathbf{P}(S)} H_1(\mathrm{Aut}_S(P);\mathbb{Z})$ by lemma 4.1.11 taking the direct limit over $t\mathbf{P}(S)$ yields:

$$K_1(S) \to \varinjlim_{P\in\mathbf{P}(S)} H_1(\mathrm{Aut}_R(P);\mathbb{Z}) \to \varinjlim_{Q\in\mathbf{P}(R)} H_1(\mathrm{Aut}_R(Q);\mathbb{Z}) = K_1(R)$$

as we sought to show.

Corollary 4.1.14: $K_1(-)$ Functor

Suppose that S is finitely generated projective as an R-module. Then the forgetful functor $\mathbf{P}(S) \to \mathbf{P}(R)$ induces a natural transfer homomorphism $f_* : K_1(S) \to K_1(R)$. If R is commutative, the composite

$$K_1(R) \xrightarrow{f^*} K_1(S) \xrightarrow{f_*} K_1(R)$$

is multiplication by $[S] \in K_0(R)$.

***Proof*:**

Let T be the forgetful map. Then the map $K_1(S) \xrightarrow{f_*} K_1(R)$ is the transfer map. We compute the composite $f_* \circ f^*$ by computing its effects on $\alpha \in GL_n(R)$. We see that the matrix $f^*(\alpha) = 1_S \otimes_R \alpha$ lies in $GL_n(S)$. To apply f_* we consider $1_S \otimes_R \alpha$ as an element of the group $\mathrm{Aut}_R(S^n) = \mathrm{Aut}_R(S \otimes_R R^n)$, which we then map into $GL(R)$. But this is just the product

$[S] \cdot \alpha$, completing the proof.

4.2 Relative K-Groups and Homology

The next process should be familiar to any reader with some knowledge of relative homology, and will give us our first deep connection between the K_n groups and homology. Consider an ideal $I \subseteq R$. Then given a map $R \to R/I$, we get a natural map $\mathrm{GL}(R) \to \mathrm{GL}(R/I)$. Let $\mathrm{GL}(I)$ be its kernel. Now, let $E(R, I)$ be the smallest normal subgroup containing the elementary matrices $e_{ij}(x)$ with $x \in I$. Note that $E(R, I) = \bigcup_n E_n(R, I)$. Then we have

Theorem 4.2.1: Whitehead's Lemma for Relative Groups

Let R be a ring and $I \subseteq R$ an ideal. Then $E(R, I) \trianglelefteq \mathrm{GL}(I)$ and contains the commutator subgroup of $\mathrm{GL}(I)$.

***Proof*:**
For any matrix $g = 1 + \alpha \in GL_n(I)$, the identity

$$\begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & g^{-1}\alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -g\alpha & 1 \end{pmatrix}$$

shows that the matrix $\begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix}$ is in $E_{2n}(R, I)$. (The product of the first three matrices is in $E_{2n}(R, I)$.) Hence if $h \in E_n(R, I)$, then the conjugate

$$\begin{pmatrix} ghg^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} h & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g^{-1} & 0 \\ 0 & g \end{pmatrix}$$

is in $E(R, I)$. Finally, if $g, h \in GL_n(I)$, then $[g, h]$ is in $E_{2n}(R, I)$ by (4.1).

Thus, the following is well-defined:

Definition 4.2.2: Relative K_1-group

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_1(R, I) = \frac{\mathrm{GL}(I)}{E(R, I)}$$

Proposition 4.2.3: Snake Lemma For K-theory

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_1(R, I) \hookrightarrow K_1(R) \to K_1(R/I) \xrightarrow{\partial} K_0(I) \to K_0(R) \to K_0(R/I)$$

***Proof*:**
First, we shall prove a simpler exact sequence:

$$1 \to GL(I) \to GL(R) \to GL(R/I) \xrightarrow{\partial} K_0(I) \to K_0(R) \to K_0(R/I)$$

First, consider the short exact sequence:

$$0 \to I \to R \to R/I \to 0$$

where I is an ideal in R. This induces a sequence of group homomorphisms

$$1 \to GL(I) \to GL(R) \to GL(R/I)$$

where $GL(I)$ consists of matrices of the form $1 + \alpha$ with α having entries in I. This sequence is exact since any matrix that maps to the identity in $GL(R/I)$ must have the form $1 + \alpha$ with α having entries in I.

To extend this to the K-theory groups, we define the boundary map $\partial : GL(R/I) \to K_0(I)$ as follows. For any $\bar{g} \in GL(R/I)$, we can lift it to a matrix $g \in M_n(R)$ (not necessarily invertible). The matrix g induces an R-module homomorphism $R^n \to R^n$. Let C be the cokernel of this homomorphism. Since $\bar{g}$ is invertible, the reduction of C modulo I is zero, which means C is an I-module. We define $\partial(\bar{g}) = [C] \in K_0(I)$. Verifying exactness is now a standard procedure.

Thus, we have the complete exact sequence

$$1 \to GL(I) \to GL(R) \to GL(R/I) \xrightarrow{\partial} K_0(I) \to K_0(R) \to K_0(R/I)$$

Since the K_1 groups are quotients of the GL groups and $E(R)$ maps onto $E(R/I)$, this gives exactness except at $K_1(R)$. Suppose $g \in GL(R)$ maps to zero under $GL(R) \to K_1(R) \to K_1(R/I)$. Then the reduction $\bar{g}$ of g mod I is in $E(R/I)$. Since $E(R)$ maps onto $E(R/I)$, there is a matrix e in $E(R)$ mapping to $\bar{g}$, i.e., ge^{-1} is in the kernel $GL(I)$ of $GL(R) \to GL(R/I)$. Hence the class of ge^{-1} in $K_1(R, I)$ is defined and maps to the class of g in $K_1(R)$, giving exactness as we sought to show.

4.3 The Steinberg Group and K_2

The K_0 and K_1 group we have covered so far had natural definitions motivated outside of homology. However, they both naturally where related to homology, namely they were tied to H_0 and H_1 of the group homology for matrix rings (where R can be seen as a 1-dimensional matrix ring). We may ask if we can further extend this and define higher K-groups via this method. We shall explore this in more detail in the next chapter, but in this chapter we shall cover more "naturally" how we may discover finding K_2 without simply saying $K_2(R) = H_2(E(R); \mathbb{Z})$. Given the group $E(R)$, a natural question to ask is what extensions we may have. Given we want to quotient by the centralizer, we are particularity interested in *central extensions*. Even better if we can find a *universal central extension*. Topologically, this would be associated to the *universal cover* of $BE(R)$ (similarly as to what we saw in topological K-theory). The group associated to this is called the *Steinberg Group*, and is often denoted $\mathrm{st}(R)$. To stay concrete, we shall define everything "low-brow" directly with elements instead of as the consequence of universal properties. The advantage of this approach is to make it easier to "play" with the objects and get a feeling to the information they store:

Definition 4.3.1: Steinberg Group

Let R be a ring. For $n \geq 3$ the *Steinberg group* $\mathrm{st}_n(R)$ is the group presented by generators $x_{ij}(r)$, one for each $r \in R$ and each pair $i \neq j$ with $1 \leq i, j \leq n$, subject to the *Steinberg relations*:

1. $$x_{ij}(r)x_{ij}(s) = x_{ij}(r+s)$$

2. $$[x_{ij}(r), x_{k\ell}(s)] = \begin{cases} 1 & j \neq k,\ i \neq \ell \\ x_{i\ell}(rs) & j = k,\ i \neq \ell \\ x_{kj}(-sr) & j \neq k,\ i = \ell \end{cases}$$

Recall from lemma 4.1.3 that the elementary matrices of $E(R)$ satisfy the same relationships, hence there is a canonical surjection

$$\mathrm{st}_n(R) \to E_n(R) \qquad x_{ij}(r) \mapsto e_{ij}(r)$$

The group $\mathrm{st}_n(R)$ embeds in $\mathrm{st}_{n+1}(R)$ in the usual way, so $\mathrm{st}(R)$ is defined as the colimit. Its kernel over $E(R)$ is the next definition:

Definition 4.3.2: K2 Group

Let R be a ring. Then the $K_2(R)$ is defined as the kernel of the map $\mathrm{st}(R) \to E(R)$.

Thus, it fits in the exact sequence:

$$1 \to K_2(R) \to \mathrm{st}(R) \to \mathrm{GL}(R) \to K_1(R) \to 1$$

That $K_2(R)$ is the centre of the Steinberg group is the next theorem.

Theorem 4.3.3: K2 and Center of Steinberg

Let R be a ring. Then $K_2(R)$ is an abelian group and is the center of $\mathrm{st}(R)$.

***Proof*:**
If $x \in \mathrm{st}(R)$ commutes with every element of $\mathrm{st}(R)$, then $\varphi(x)$ commutes with all of $E(R)$, since φ is surjective. The center of $E(R)$ is trivial: a matrix commuting with every $e_{ij}(r)$ commutes in particular with $e_{ij}(1)$ for all $i \neq j$, and conjugating by these forces every off-diagonal entry to vanish and all diagonal entries to agree, so the matrix is a scalar λ; but $E(R)$ is a union of the $E_n(R)$ under the stabilization $g \mapsto \mathrm{diag}(g, 1)$, and a scalar $\lambda \neq 1$ is not fixed by that map. Hence $\varphi(x) = 1$, that is $x \in K_2(R)$, and the center of $\mathrm{st}(R)$ is contained in $K_2(R)$.

Conversely, suppose that $y \in \mathrm{st}(R)$ satisfies $\varphi(y) = 1$. Then in $E(R)$ we have

$$\varphi([y,p]) = \varphi(y)\varphi(p)\varphi(y)^{-1}\varphi(p)^{-1} = \varphi(p)\varphi(p)^{-1} = 1$$

for every $p \in \mathrm{st}(R)$. Choosing an integer n large enough so that y can be expressed as a word in the symbols $x_{ij}(r)$ with $i, j < n$, for each element $p = x_{kn}(s)$ with $k < n$ and $s \in R$, the Steinberg relations implies that the commutator $[y, p]$ is an element of the subgroup P_n of $\mathrm{st}(R)$ generated by the symbols $x_{in}(r)$ with $i < n$. On the other hand, we can see that φ maps P_n injectively into $E(R)$. Since $\varphi([y, p]) = 1$, this implies that $[y, p] = 1$. Hence y commutes with every generator $x_{kn}(s)$ with $k < n$.

By symmetry, we get that y also commutes with every generator $x_{nk}(s)$ with $k < n$. Hence y commutes with all of $St_n(R)$, since it commutes with every $x_{kl}(s) = [x_{kn}(s), x_{nl}(1)]$ with $k, l < n$. Since n can be arbitrarily large, this proves that y is in the center of $\mathrm{st}(R)$ showing it is indeed the center, completing the proof.

Example 4.3.4: K_2 groups

The computation of K_2 group becomes a little non-trivial, however here are some interesting results:

1. Let $R = \mathbb{Q}$. Then it is somewhat remarkable that:

$$K_2(\mathbb{Q}) \cong \{\pm 1\} \times \bigoplus_{p \text{ odd prime}} \mathbb{F}_p^\times$$

where this was proven following a proof similar to Gauss's proof of the Quadratic Reciprocity Law!

2. It was proven by R.K. Dennis and J. Sylvester that $K_2(\mathbb{F}[t]) = K_2(\mathbb{F})$, showing that $K_2(-)$ is invariant under extension by an algebraically independent element, so it extracts think of it as extracting "relational" or "number-theoretic" information.

3. More generally, Matsumoto's theorem (see [19]) states that:

$$K_2(k) \cong k^\times \otimes_{\mathbb{Z}} k^\times / \langle a \otimes (1 - a) \big| a \in k^\times \setminus \{1\} \rangle$$

The full proof is long, but the concrete definition above makes its outline visible. The argument draws on section 4.4 to show that this group shall satisfy the same universal property of central extensions. For $a \in k^\times$, define the element $w_{ij}(a) \in \mathrm{st}(k)$ by:

$$w_{ij}(a) = x_{ij}(a)x_{ji}(-a^{-1})x_{ij}(a)$$

For $a, b \in k^\times$, define the Steinberg symbol $\{a, b\} \in K_2(k)$ as:

$$\{a, b\} = h_{ij}(a)h_{ij}(b)h_{ij}(ab)^{-1}$$

where $h_{ij}(a) = w_{ij}(a)w_{ij}(1)^{-1}$. Then by the linearity properties given in definition 4.3.1, $\{a, b\}$ satisfy the following bilinearity relations:

$$\{a_1a_2, b\} = \{a_1, b\}\{a_2, b\}$$
$$\{a, b_1b_2\} = \{a, b_1\}\{a, b_2\}$$

The crucial step is proving that: $\{a, 1 - a\} = 1$ for all $a \in k^\times \setminus \{1\}$. Establishing it uses explicit matrices in $\mathrm{SL}_3(k)$ and their lifts to the Steinberg group. Consider the elements:

$$u(a) = x_{12}(a)x_{23}(1)x_{12}(-a)x_{23}(-1)$$
$$v(a) = x_{13}(a(1 - a))x_{23}(-a)x_{12}(-(1 - a))$$

A direct calculation in $\mathrm{st}(k)$, expanding both words with the Steinberg relations, gives $u(a)v(a)^{-1} \in K_2(k)$ with value $\{a, 1-a\}^{-1}$, and separately $u(a)v(a)^{-1} = 1$; together these give $\{a, 1-a\} = 1$. Both are long word manipulations in $\mathrm{st}_3(k)$ with no further idea in them, and are carried out in [3, Ch. III, Thm. 6.1]. Next, let $\mathcal{M}(k)$ denote the abstract group:

$$\mathcal{M}(k) = k^\times \otimes_{\mathbb{Z}} k^\times / \langle a \otimes (1-a) \big| a \in k^\times \setminus \{1\} \rangle$$

Define a homomorphism $\varphi : \mathcal{M}(k) \to K_2(k)$ by mapping $a \otimes b \mapsto \{a, b\}$. The bilinearity relations and the Steinberg relation ensure that φ is well-defined, and it remains to show φ is an isomorphism. The inverse is built from a homomorphism $\psi : K_2(k) \to \mathcal{M}(k)$ and show that $\psi \circ \varphi = \mathrm{id}_{\mathcal{M}(k)}$ and $\varphi \circ \psi = \mathrm{id}_{K_2(k)}$.

The inverse ψ comes from the universal property of $\mathrm{st}(k)$ as the universal central extension of $\mathrm{SL}(k)$ (section 4.4): one builds a central extension

$$1 \to \mathcal{M}(k) \to E \to \mathrm{SL}(k) \to 1$$

and checks that every central extension of $\mathrm{SL}(k)$ factors uniquely through it, so that E and $\mathrm{st}(k)$ are both universal and therefore isomorphic over $\mathrm{SL}(k)$, giving $\mathcal{M}(k) \cong K_2(k)$. Constructing E is the substantial half of Matsumoto's theorem and needs the full Steinberg-symbol calculus; it is carried out in [3, Ch. III, Thm. 6.1]. What is established here is the direction that is cheap from the presentation, namely that φ is well defined, which is what the bilinearity relations and $\{a, 1-a\} = 1$ give.

4. Let $R = \mathbb{Z}$. Then $K_2(R) \cong \mathbb{Z}/2\mathbb{Z}$. By Matsumoto's theorem:

$$K_2(\mathbb{Q}) \cong \mathbb{Q}^\times \otimes_{\mathbb{Z}} \mathbb{Q}^\times / \langle a \otimes (1-a) \big| a \in \mathbb{Q}^\times \setminus \{1\} \rangle$$

Every element in $\mathbb{Q}^\times$ can be written as $\pm p_1^{n_1} \cdots p_k^{n_k}$ where p_i are distinct primes and $n_i \in \mathbb{Z}$. Thus, $K_2(\mathbb{Q})$ is generated by symbols of the form:

$$\{-1, -1\}, \{-1, p\}, \{p, q\}$$

where p, q are primes. The next step combines theorem 4.4.6 which is a generalization of the snake lemma, and [21, chapter 5] from class field theory to get:

$$K_2(\mathcal{O}_F) \to K_2(F) \xrightarrow{\partial} \bigoplus_{\mathfrak{p}} \kappa(\mathfrak{p})^\times \to K_1(\mathcal{O}_F) \to K_1(F)$$

For $\mathbb{Z} \subset \mathbb{Q}$, this becomes:

$$K_2(\mathbb{Z}) \to K_2(\mathbb{Q}) \xrightarrow{\partial} \bigoplus_{p \text{ prime}} \mathbb{F}_p^\times \to K_1(\mathbb{Z}) \to K_1(\mathbb{Q})$$

where the component of ∂ at a prime ℓ is the *tame symbol*

$$\partial_\ell(\{a, b\}) = (-1)^{v_\ell(a) v_\ell(b)} \cdot \left(\frac{a^{v_\ell(b)}}{b^{v_\ell(a)}} \right) \bmod \ell$$

with v_ℓ the ℓ-adic valuation. Write p and q for distinct primes. Since $v_\ell(-1) = 0$ for every ℓ, both exponents vanish on $\{-1,-1\}$ and the generators evaluate as:

$$\partial_\ell(\{-1,-1\}) = 1 \quad \text{for every prime } \ell$$

$$\partial_\ell(\{-1,q\}) = \begin{cases} -1 \bmod q & \text{if } \ell = q \\ 1 & \text{otherwise} \end{cases}$$

$$\partial_\ell(\{p,q\}) = \begin{cases} q^{-1} \bmod p & \text{if } \ell = p \\ p \bmod q & \text{if } \ell = q \\ 1 & \text{if } \ell \neq p, q \end{cases}$$

For the last of these: at $\ell = p$ one has $v_p(p) = 1$ and $v_p(q) = 0$, so the sign is $+1$ and the fraction is $p^0/q^1 = q^{-1}$; at $\ell = q$ the roles reverse and the fraction is $p^1/q^0 = p$.

Exactness of the sequence at $K_2(\mathbb{Q})$, together with injectivity of $K_2(\mathbb{Z}) \to K_2(\mathbb{Q})$, identifies $K_2(\mathbb{Z})$ with $\ker\partial$. Computing that kernel generator by generator does not work, for two reasons worth recording. First, a generator can escape detection: $\partial_p(\{p,q\}) = q^{-1} \bmod p$ is trivial whenever $q \equiv 1 \pmod p$, as with $\partial_3(\{3,7\}) = 7^{-1} \equiv 1$, so $\{p,q\}$ is *not* in general detected at $\ell = p$; one must order the primes $p < q$ and use $\ell = q$, where $p \not\equiv 1 \pmod q$. Second, and more basically, $\ker\partial$ need not be generated by those generators that happen to lie in it, so no enumeration of generators settles the question.

What does settle it is Tate's computation, quoted in item (1) above: $K_2(\mathbb{Q}) \cong \{\pm 1\} \times \bigoplus_{\ell \text{ odd}} \mathbb{F}_\ell^\times$, in which the $\mathbb{F}_\ell^\times$ factor is exactly the tame symbol at ℓ. So ∂ is the projection onto the second factor and

$$K_2(\mathbb{Z}) \;\cong\; \ker\partial \;\cong\; \{\pm 1\} \;\cong\; \mathbb{Z}/2\mathbb{Z}$$

generated by $\{-1,-1\}$. The Steinberg symbol relations give $\{-1,-1\}^2 = \{(-1)^2,-1\} = \{1,-1\} = 1$, so its order divides 2; that it is not trivial is exactly the statement that the $\{\pm 1\}$ factor in Tate's decomposition is nonzero, and is not something the symbol relations alone can give.

These examples show how directly $K_2(-)$ is tied to number theory.

4.4 Universal Central Extensions

As described earlier, $\mathrm{st}(R)$ can naturally be seen as being the *universal central extension* of $E(R)$. Let us dive into what this means.

Let G be a group and A an abelian group. Then a *central extension* of G by A[1] if there exists a short exact sequence

$$1 \to A \to X \to G \to 1$$

such that A maps into the center of X. The extension splits if there is a section map $G \to X$, or

[1] remember that in group theory we usually reverse the order when we say "X" is extending "Y", hence why we say G is extended by A

equivalently that it is isomorphic to the extension:

$$1 \to A \to A \times G \xrightarrow{\pi_2} G \to 1$$

By theorem 4.3.3, we have a central extension:

$$1 \to K_2(R) \to \mathrm{st}(R) \to E(R) \to 1$$

If X, Y are extensions of G, they are said to be equivalent if there is an isomorphism $f : X \to Y$ which is the identity on A and induces the identity map on G. Then it is a known result that $H^2(G; A)$ classifies all the isomorphism classes (see [16] or [18]). A *universal central extension* of G would then be a central extension $X \to G$ such that for any other central extension $Y \to G$ it must factor through $X \to G$ uniquely. We have yet to show that a universal central extension always exists. We build-up to this result now:

Lemma 4.4.1: Universal Central Extension and Perfect Groups I

Let G have a universal central extension $X \to G$. Then G and X are perfect, that is $G = [G, G]$ and $X = [X, X]$.

***Proof*:**
Assume $B = \frac{X}{[X,X]}$ is nontrivial. Then there would be at least two homomorphisms over G from X to the central extension:

$$1 \to B \to B \times G \to G \to 1$$

namely $(0, \pi)$ and (pr, π) where $pr : X \to B$ is the natural projection, contradicting the uniqueness of universality.

Lemma 4.4.2: Universal Central Extension and Perfect Groups II

Let G have two central extension X, Y, and assume X is a perfect group. Then there is at most one homomorphism over G from X to Y.

***Proof*:**
If f and f' are two such homomorphisms, then for any x and x' in X we can write $f'(x) = f(x)c$, $f'(x') = f(x')c'$ for elements c and c' in the center of Y. Therefore $f'(xx'x^{-1}(x')^{-1}) = f(xx'x^{-1}(x')^{-1})$. Since the commutators $[x, x'] = xx'x^{-1}(x')^{-1}$ generate X, we must have $f' = f$, completing the proof.

Theorem 4.4.3: Universal Central Extensions of Perfect Groups

Let G be a perfect group. Then G has a universal central extension:

$$1 \to H_2(G;\mathbb{Z}) \to \frac{[F,F]}{[R,F]} \to G \to 1$$

where F is a free group mapping onto G and $R \subseteq F$ is the kernel of this map (i.e. the relations).

If X is any central extension of G, then the following are equivalent:

1. X is a universal central extension
2. X is perfect, and every central extension of X splits
3. $H_1(X;\mathbb{Z}) = H_2(X;\mathbb{Z}) = 0$

***Proof*:**
First, let $F \twoheadrightarrow G$ be a presentation of G with kernel R. Then $[R,F] \trianglelefteq F$, and so we have a short exact sequence:

$$1 \to \frac{R}{[R,F]} \to \frac{F}{[R,F]} \to G \to 1$$

By the universal properties of commutators with some diagram chasing, we obtain:

$$1 \to \frac{R \cap [F,F]}{[R,F]} \to \frac{[F,F]}{[R,F]} \to [G,G] \to 1$$

If G is perfect, $G = [G,G]$, and Hopf's formula gives $\frac{R\cap[F,F]}{[R,F]} \cong H_2(G;\mathbb{Z})$ ([18]), giving the exact sequence. The formula has no owner upstream of this book in the series: it is proved in `NateGaloisCohom`, which is a sibling rather than a prerequisite, so it is cited externally here pending a placement decision.

The sequence just built is universal. Given any central extension $X \to G$, the surjection $\varphi\colon F \to G$ lifts to $\Phi\colon F \to X$ because F is free. Then $\Phi(R)$ lies in the centre of X, since R maps to 1 in G and the extension is central, so $\Phi([R,F]) = 1$ and Φ descends to a map $[F,F]/[R,F] \to X$ over G. It is unique by lemma 4.4.2, the source being perfect.

(1) ⇒ (2). If X is universal it is perfect by lemma 4.4.1. Were some central extension $Y \to X$ non-split, then $Y \to G$ would be a central extension admitting two distinct maps from X over G, contradicting uniqueness.

(2) ⇒ (3). X perfect gives $H_1(X;\mathbb{Z}) = X/[X,X] = 0$. If $H_2(X;\mathbb{Z})$ were nonzero, the Hopf sequence applied to X would produce a non-split central extension of X, against (2).

(3) ⇒ (1). With $H_1(X;\mathbb{Z}) = H_2(X;\mathbb{Z}) = 0$, the Hopf construction applied to X returns X itself, so X is its own universal central extension; composing with $X \to G$ makes X universal over G, giving the desired result.

Corollary 4.4.4: K2 Group and Homology

The Steinberg group $\mathrm{st}(R)$ is the universal central extension of $E(R)$, and hence:

$$K_2(R) \cong H_2(E(R); \mathbb{Z})$$

***Proof*:**

We shall show that for $n \geq 5$, every central extension $Y \xrightarrow{\varphi} \mathrm{st}_n(R)$ splits, which by theorem 4.4.3 gives the desired result.

First, let's show that if $j \neq k$ and $l \neq i$, then every two elements $y, z \in Y$ with $\varphi(y) = x_{ij}(r)$ and $\varphi(z) = x_{kl}(s)$ must commute in Y. Pick t distinct from i, j, k, l and choose $y', y'' \in Y$ with $\varphi(y') = x_{it}(1)$ and $\varphi(y'') = x_{tj}(r)$. Then the Steinberg relations imply that both $[y', z]$ and $[y'', z]$ are in the center of Y, and since $\varphi(y) = \varphi([y', y''])$, this implies that z commutes with $[y', y'']$ and y.

Next, choose distinct indices i, j, k, l and elements $u, v, w \in Y$ with

$$\varphi(u) = x_{ij}(1), \quad \varphi(v) = x_{jk}(s), \quad \text{and} \quad \varphi(w) = x_{kl}(r)$$

If $G \leqslant Y$ is generated by u, v, w, then its commutator subgroup $[G, G]$ is generated by elements mapping under φ to $x_{ik}(s)$, $x_{jl}(sr)$, or $x_{il}(sr)$. From the first paragraph of this proof it follows that $[u, w] = 1$ and that $[G, G]$ is abelian. By group theory, we have $[[u, v], w] = [u, [v, w]]$, and so if $\varphi(y) = x_{ik}(s)$ and $\varphi(z) = x_{jl}(sr)$, we have $[y, w] = [u, z]$. Taking $s = 1$, this identity proves that the element

$$y_{il}(r) = [u, z], \quad \text{where } \varphi(u) = x_{ij}(1), \varphi(z) = x_{jl}(r),$$

does not depend upon the choice of j, nor upon the lifts u and z of $x_{ij}(1)$ and $x_{jl}(r)$.

Finally, the elements $y_{ij}(r)$ satisfy the Steinberg relations, so that there is a group homomorphism $St_n(R) \to Y$ sending $x_{ij}(r)$ to $y_{ij}(r)$. Such a homomorphism will provide the desired splitting of the extension φ. By the first paragraph of this proof, if $j \neq k$ and $l \neq i$, then $y_{ij}(r)$ and $y_{kl}(s)$ commute. The identity $[y, w] = [u, z]$ above may be rewritten as

$$[y_{ik}(r), y_{kl}(s)] = y_{il}(rs) \quad \text{for } i, k, l \text{ distinct}$$

The remaining relation $y_{ij}(r)y_{ij}(s) = y_{ij}(r + s)$ follows the same way, from $y_{ij}(r) = [u, z]$ and additivity of $z \mapsto \varphi(z)$ in the second Steinberg generator. Since every central extension of $\mathrm{st}_n(R)$ splits for $n \geq 5$, and $\mathrm{st}(R) = \varinjlim_n \mathrm{st}_n(R)$ with central extensions commuting with this filtered colimit, every central extension of $\mathrm{st}(R)$ splits. With $\mathrm{st}(R)$ perfect, theorem 4.4.3(2) makes it the universal central extension of $E(R)$, so $K_2(R) = H_2(E(R); \mathbb{Z})$, completing the proof.

$K_2(-)$ is also Morita invariant, proved in [3] and not reproduced here. The section instead closes by extending the snake lemma, for use in example 4.3.4. For any ideal $I \subseteq R$, define the augmented ring $R \oplus I$ via:

$$(r, x)(s, y) = (rs, ry + xs + xy)$$

There are two natural maps $\pi, + : R \oplus I \to R$ where $\pi(r, x) = r$ and $+(r, x) = r + x$. Let $\mathrm{st}'(R, I)$ be the normal subgroup of $\mathrm{st}(R \oplus I)$ generated by all $x_{ij}(0, v)$ where $v \in I$. Then there is certainly a map

$$\mathrm{st}'(R,I) \to E(R \oplus I, 0 \oplus I)$$

where $E(R \oplus I, 0 \oplus I) \leqslant \mathrm{GL}(R \oplus I)$. From this define:

Definition 4.4.5: Relative Steinberg Group

Let R be a ring and $I \subseteq R$ an ideal. Then the *relative Steinberg group* $\mathrm{st}(R,I)$ is the quotient of $\mathrm{st}'(R,I)$ by the normal subgroup generated by all cross-commutators:

$$[x_{ij}(0,u), x_{kl}(v,-v)] \qquad u,v \in I$$

This gives:

Theorem 4.4.6: Snake Lemma For K-theory Extended

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_2(R,I) \to K_2(R) \to K_2(R/I) \to K_1(R,I) \to K_1(R) \to \cdots \to K_0(R) \to K_0(R/I) \to K_0(I)$$

where there is no $\to 0$ at the end due to possible negative K-groups[a].

[a] Negative K-theory is not covered here; see [3] for the details.

***Proof*:**
This follows by doing some diagram chasing on:

$$\begin{tikzcd}
K_2(R,I) \arrow[r] \arrow[d] & St(R,I) \arrow[r] \arrow[d, "+"] & GL(I) \arrow[r] \arrow[d, "\text{inj.}"] & K_1(R,I) \arrow[d] \\
K_2(R) \arrow[r] \arrow[d] & St(R) \arrow[r] \arrow[d, "\text{surj.}"] & GL(R) \arrow[r] \arrow[d] & K_1(R) \arrow[d] \\
K_2(R/I) \arrow[r] & St(R/I) \arrow[r] & GL(R/I) \arrow[r] & K_1(R/I)
\end{tikzcd}$$

and using snake's lemma (with proposition 4.2.3).

5

Higher K-Theory

As we have been discovering in the last 2 chapters, K-groups seem to be representable as the homology of certain groups. In the 1960s and 70s, a large amount of effort has been put in trying to find the right way to unify these results and give a single cohomology theory to unify these groups. As it turns out, the "right" approach was in fact to see each $K_n(R)$ as $\pi_n K(R)$ for a topological space $K(R)$, that is the *homotopy* groups rather than the homology groups. The two are linked by the Hurewicz map $\pi_n \to H_n$, an isomorphism in the first nonvanishing degree but neither injective nor surjective in general. The connection is less surprising alongside topological K-theory ([22]), where the K-groups of a space are likewise homotopy classes of maps into a universal space. In particular, there has been a push to represent the information many algebraic objects as "spaces" (in many ways, the current abstract definition of topology was made to accommodate the more odd geometric behaviour these "classifying" spaces had) and we extracted information from these spaces using algebraic topology. In this chapter, we shall go over the basics of how we may generalize the K-theory we have been developing into higher order K-theory, and give the beginning of how these results link to the K-groups we have studied earlier.

5.1 Defining the Space: BGL+

Let G be any group. [20, Universal Bundle and Classifying Space] constructs a connected topological space BG whose fundamental group is $\pi_1(BG) \cong G$ and which has $\pi_n(BG) = 0$ for $n > 1$, that is, a $K(G, 1)$ ([20, $\mathbb{R}\mathrm{P}^\infty$ as $K(\mathbb{Z}/2, 1)$ and the Classifying Space BG]). The homology of BG (with coefficients in a G-module M) coincides with the algebraic homology of G (with coefficients in M). For $G = \mathrm{GL}(R)$, this gives $\mathrm{BGL}(R)$.

Definition 5.1.1: $K_n(R)$

Let $\mathrm{BGL}(R)$ be the classifying space of $\mathrm{GL}(R)$, and let $\mathrm{BGL}(R)^+$ denote any CW complex X which has a distinguished map $\mathrm{BGL}(R) \to \mathrm{BGL}(R)^+$ such that the following holds:

1. $\pi_1(\mathrm{BGL}(R)^+) \cong K_1(R)$ and the natural map from $\mathrm{GL}(R) = \pi_1(\mathrm{BGL}(R))$ to $\pi_1(\mathrm{BGL}(R)^+)$ is surjective with kernel $E(R)$
2. the distinguished map induces an isomorphism $H_*(\mathrm{BGL}(R); M) \xrightarrow{\cong} H_*(\mathrm{BGL}(R)^+; M)$ for every $K_1(R)$-module M. The map matters: lemma 5.1.8 shows an abstract isomorphism of the two sides would not pin the homotopy type down

The space X is sometimes called the *model* for $\mathrm{BGL}(R)^+$.

For $n \geq 1$,

$$K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$$

Any two models are homotopy equivalent, so $\mathrm{BGL}(R)^+$ is well defined up to homotopy and the groups $K_n(R)$ are well defined up to canonical isomorphism. Clause (2) says exactly that $\mathrm{BGL}(R) \to \mathrm{BGL}(R)^+$ is an *acyclic* map (lemma 5.1.8), and uniqueness is then theorem 5.1.7(3). No model X has been given yet. One is described below.

The definition omits $K_0(R)$. The following amends it:

Definition 5.1.2: $K(R)$

Let R be a ring, $K_0(R)$ the K-group given in definition 2.2.1. Then:

$$K(R) := K_0(R) \times \mathrm{BGL}(R)^+$$

Namely, $K(R)$ is the disjoint union of copies of the connected space $\mathrm{BGL}(R)^+$ with each element of $K_0(R)$. Then by construction:

$$K_0(R) = \pi_0(K(R)) \qquad \pi_n(K(R)) = \pi_n(\mathrm{BGL}(R)^+) = K_n(R),\ n \geq 1$$

recovering all the K-groups.

The maps $K_n(-)$ and $K(-)$ are functorial. The former is a functor from rings to abelian groups, the latter a functor from rings to the homotopy category hTop, whose objects are spaces and whose morphisms are *homotopy classes* of continuous maps ([20, The Homotopy Category hTop]). The weak equivalences are what one inverts to form such a category, not what its morphisms are. Let $R \to R'$ be a ring map. It induces $\mathrm{GL}(R) \to \mathrm{GL}(R')$, hence $\mathrm{BGL}(R) \to \mathrm{BGL}(R')$, and by theorem 5.1.7(2) a map $\mathrm{BGL}(R)^+ \to \mathrm{BGL}(R')^+$ well defined up to homotopy. Homotopic maps agree on homotopy groups, so the induced $K_n(R) \to K_n(R')$ is well defined *on the nose*. It is only the map of spaces that is defined up to homotopy. A similar argument for composition makes $K_n(-)$ a functor.

Existence of a model is theorem 5.1.7(1), so nothing below is needed to make the definition legitimate. What follows instead is the machinery of *acyclic* maps, which is what makes the model unique and what computes π_2. A concrete model, the Volodin space, is described afterwards as an illustration. Call a space acyclic[1] when its reduced homology vanishes, and note first how such spaces relate to

[1] A space is *acyclic* when it has trivial reduced homology, that is $\tilde{H}_*(X; \mathbb{Z}) = 0$.

perfect groups.

Lemma 5.1.3: Acyclic and Perfect Groups

Let E be an acyclic space. Then E is connected, its fundamental group $G = \pi_1(E)$ is a perfect group, and $H_2(G; \mathbb{Z}) = 0$.

The proof uses the *Serre spectral sequence* in its homology, local-coefficient form ([20, Homology Serre Spectral Sequence], with [20, Remark: The Local-Coefficients Serre Spectral Sequence]).

***Proof*:**
First, the acyclic space E must be connected, as $H_0(E) = \mathbb{Z}$. Indeed, $G/[G,G] = H_1(E; \mathbb{Z}) = 0$, so we have $G = [G, G]$, i.e., G is a perfect group. To calculate $H_2(G)$, observe that the universal covering space $\tilde{E}$ has $H_1(\tilde{E}; \mathbb{Z}) = 0$. We claim the homotopy fibre F of the canonical map $E \to BG$ is homotopy equivalent to $\tilde{E}$. Writing the long exact sequence of the fibration,

$$\cdots \to \pi_{n+1}(BG) \xrightarrow{\partial} \pi_n(F) \to \pi_n(E) \to \pi_n(BG) \xrightarrow{\partial} \pi_{n-1}(F) \to \cdots$$

Since BG is a $K(G, 1)$ and $E \to BG$ induces an isomorphism on π_1, the segment $\pi_2(BG) = 0 \to \pi_1(F) \to \pi_1(E) \xrightarrow{\cong} \pi_1(BG)$ forces $\pi_1(F) = 1$, while $\pi_n(BG) = 0$ for $n \geq 2$ gives $\pi_n(F) \cong \pi_n(E)$ in those degrees. So F is simply connected and maps to E by a weak equivalence on all higher homotopy, which characterises the universal cover: $F \simeq \tilde{E}$.

Then the *Serre spectral sequence* for this homotopy fibration is $E^2_{pq} = H_p(G; H_q(\tilde{E}; \mathbb{Z})) \Rightarrow H_{p+q}(E; \mathbb{Z})$ and the conclusion that $H_2(G; \mathbb{Z}) = 0$ follows from the associated exact sequence of low degree terms:

$$H_2(E; \mathbb{Z}) \to H_2(G; \mathbb{Z}) \xrightarrow{d^2} H_1(\tilde{E}; \mathbb{Z})_G \to H_1(E; \mathbb{Z}) \to H_1(G; \mathbb{Z})$$

The third term carries *co*invariants, not invariants: it is $E^2_{0,1} = H_0(G; H_1(\tilde{E}; \mathbb{Z}))$, and $H_0(G; M) = M_G$ by definition of group homology. Both outer terms vanish because E is acyclic, and $H_1(\tilde{E}; \mathbb{Z}) = 0$ kills the middle, giving $H_2(G; \mathbb{Z}) = 0$ as claimed.

Definition 5.1.4: Volodin Space

Let R be a ring. The *Volodin space* $X(R)$ is the subspace of $\mathrm{BGL}(R)$ constructed as follows:

1. For each n ,let $T_n(R) \leqslant \mathrm{GL}_n(R)$ consist of upper triangular matrices with 1's on the diagonal. Let $T(R) = \varinjlim_{n\to\infty} T_n(R) \leqslant \mathrm{GL}(R)$
2. View the permutation group S_n as the subgroups of $\mathrm{GL}_n(R)$ by their representation as permutation matrices, and their union as S_∞.
3. For each $\sigma \in S_n$, let $T_n^\sigma(R)$ denote the subgroup of $\mathrm{GL}_n(R)$ obtained by conjugating $T_n(R)$ by σ. Then $BT_n(R)$ and $BT_n(R)^\sigma$ are subspaces of $\mathrm{BGL}_n(R)$, and hence of $\mathrm{BGL}(R)$
4. The *Volodin space* is defined to be:

$$X(R) = \bigcup_{n,\sigma} BT_n(R)^\sigma$$

Two facts make this an example rather than a construction carried out here. First, $X(R)$ is *acyclic*. That is a theorem of Suslin's, cited rather than proved ([3, Ch. IV, Ex. 1.3.2], which attributes it to him). It is the one input in this section that the book does not establish. Second, granting acyclicity, $\mathrm{BGL}(R)/X(R)$ is a model for $\mathrm{BGL}(R)^+$, *not* for $K(R)$: the space $\mathrm{BGL}(R)$ is connected and so any quotient of it is, whereas $\pi_0(K(R)) = K_0(R)$ is $\mathbb{Z}$ already for a field. That identification is [3, Ch. IV, Ex. 1.6].

What can be checked directly is the K_1 level. The image of $\pi_1(X(R)) \to \mathrm{GL}(R)$ is $E(R)$: the group $\pi_1(X(R))$ is generated by the images of the $\pi_1(BT_n(R)^\sigma)$, the image of the composition

$$\pi_1(BT_n^\sigma(R)) \to \pi_1(X(R)) \to \pi_1(\mathrm{BGL}(R)) = \mathrm{GL}(R)$$

is the subgroup $T_n^\sigma(R) \leqslant E(R)$, and conversely every generator $e_{ij}(r)$ of $E(R)$ lies in some $T_n^\sigma(R)$. So the subgroup killed by passing to $\mathrm{BGL}(R)/X(R)$ is exactly the one *definition* 5.1.1(1) asks to be killed.

Identifying $\pi_1(X(R))$ itself needs the machinery of acyclic maps developed next. The answer is corollary 5.1.10 below.

Definition 5.1.5: Acyclic Maps

Let X, Y be based connected CW complexes. Then a map $f : X \to Y$ is called *acyclic* if the homotopy fiber $F(f)$ of f is acyclic (i.e. Has trivial homology, the homology of a point).

Note that this implies that $F(f)$ is connected and $\pi_1(F(f))$ is a perfect group. Then from the exact sequence $\pi_1(F(f)) \to \pi_1(X) \to \pi_1(Y) \to \pi_0(F(f))$ of homotopy groups shows that if $X \to Y$ is acyclic, then $\pi_1(X) \twoheadrightarrow \pi_1(Y)$, and its kernel P is a perfect normal subgroup of $\pi_1(X)$. Such groups have a name:

Definition 5.1.6: +-Construction

Let P be a perfect normal subgroup of $\pi_1(X)$ where X is a based connected CW complex. Then an acyclic map $f : X \to Y$ is called a *+-construction* on X (relative to P) if P is the kernel of $\pi_1(X) \to \pi_1(Y)$.

For any perfect normal subgroup of the fundamental group, there exists such a group.

Theorem 5.1.7: Existence of +-construction

Let $P \leqslant \pi_1(X)$ be a perfect normal subgroup. Then:

1. There exists a +-construction $f : X \to Y$ relative to P
2. Let $f : X \to Y$ be a +-construction relative to P, and let $g : X \to Z$ be a map st P vanishes in $\pi_1(Z)$. Then there is a map $h : Y \to Z$ such that $g = hf$ and h is unique up to homotopy
3. If g is another +-construction relative to P, then the map h in (2) is a homotopy equivalence, namely $h : Y \xrightarrow{\sim} Z$

***Proof*:**
See [17]

Let us now show why these are called acyclic maps:

Lemma 5.1.8: Acyclic Maps and Homology

Let X, Y be connected CW complexes. Then $f : X \to Y$ is acyclic if and only if the *induced* map

$$f_* \colon H_*(X; M) \longrightarrow H_*(Y; M)$$

is an isomorphism for every $\pi_1(Y)$-module M. It is not enough that the two sides be abstractly isomorphic: the degree-2 map $S^2 \to S^2$ has $\pi_1 = 1$, so the coefficient modules are just abelian groups and both sides are $(M, 0, M)$ for every M, yet its homotopy fibre has $\pi_1 = \mathbb{Z}/2$ and so is not acyclic.

***Proof*:**
Suppose first that f is acyclic, with homotopy fiber $F(f)$. Then $\pi_1 F(f) \to \pi_1(Y)$ is trivial, and $\pi_1(F(f))$ acts trivially upon M. By [20, Universal Coefficient Theorem For Homology], $H_q(F(f); M) = 0$ for $q \neq 0$ and $H_0(F(f); M) = M$.

Therefore $E^2_{pq} = 0$ for $q \neq 0$ in the Serre spectral sequence for f:

$$E^2_{pq} = H_p(Y; H_q(F(f); M)) \Rightarrow H_{p+q}(X; M)$$

The spectral sequence collapses, yielding that f_* is an isomorphism $H_p(X; M) \to H_p(Y; M)$ for all p.

Conversely, suppose first that $\pi_1(Y) = 0$ and that

$$f_*\colon H_*(X;\mathbb{Z}) \longrightarrow H_*(Y;\mathbb{Z})$$

is an isomorphism. The induced map is essential here and not a stylistic point: the degree-2 map $S^2 \to S^2$ of the counterexample above has $\pi_1(Y) = 1$, so it falls inside this very case, and it has abstractly isomorphic homology without being acyclic. By the Comparison Theorem for the Serre spectral sequences (see [1]) for $F(f) \to X \xrightarrow{f} Y$ and $* \to Y \xrightarrow{=} Y$, we have $\tilde{H}_*(F(f);\mathbb{Z}) = 0$. Hence $F(f)$ and f are acyclic.

The general case reduces to this by the following trick: Let $\tilde{Y}$ denote the universal covering space of Y, and let $\tilde{X} = X \times_Y \tilde{Y}$ denote the corresponding covering space of X. Then there are natural isomorphisms

$$H_*(\tilde{Y};\mathbb{Z}) \cong H_*(Y;M) \qquad \text{and} \qquad H_*(\tilde{X};\mathbb{Z}) \cong H_*(X;M)$$

where $M = \mathbb{Z}[\pi_1(Y)]$. The assumption that f_* is an isomorphism $H_*(X;M) \to H_*(Y;M)$ implies that the map $\tilde{f} : \tilde{X} \to \tilde{Y}$ induces isomorphisms on integral homology. In degree 0 this gives $H_0(\tilde{X};\mathbb{Z}) \cong H_0(\tilde{Y};\mathbb{Z}) = \mathbb{Z}$, so $\tilde{X}$ is connected and the special case applies to it. Since $\pi_1(\tilde{Y}) = 0$, that case makes the homotopy fiber $F(\tilde{f})$ of $\tilde{f}$ an acyclic space. But by path lifting we have $F(\tilde{f}) \cong F(f)$, so $F(f)$ is acyclic. Thus f is an acyclic map.

Now, recall from section 4.4 that every perfect group P has a universal central extension $E \to P$ whose kernel is the abelian group $H_2(P;\mathbb{Z})$, namely theorem 4.4.3. This gives:

Theorem 5.1.9: 2-homotopy group and K_2

let P be a perfect normal subgroup of G with corresponding $+$-construction $f : BG \to BG^+$. Then if $F(f)$ is the homotopy fiber of f, then $\pi_1(F(f))$ is the universal central extension of P, and

$$\pi_2(BG^+) \cong H_2(P;\mathbb{Z})$$

For $G = \mathrm{GL}(R)$:

$$K_2(R) \cong H_2(E(R);\mathbb{Z})$$

***Proof*:**
Consider the exact sequence from homotopy theory:

$$\pi_2(BG) \to \pi_2(BG^+) \to \pi_1(F(f)) \to G \to G/P \to 1$$

Here $\pi_2(BG) = 0$ because BG is a $K(G,1)$. That the image of $\partial\colon \pi_2(BG^+) \to \pi_1(F(f))$ is *central* is the substantial input, and it is what makes the extension central at all. [3, Ch. IV, Prop. 1.7] asserts it in exactly this situation, deriving it by citation from a general criterion of Whitehead's. We take it in the form Weibel states. Thus $\pi_1(F(f))$ is a central extension of P with kernel $\pi_2(BG^+)$. Since $F(f)$ is acyclic, lemma 5.1.3 makes $\pi_1(F(f))$ perfect with $H_2\big(\pi_1(F(f));\mathbb{Z}\big) = 0$. Note this is the homology of the *group*, which is what theorem 4.4.3 uses, not the homology of the space. That theorem then makes $\pi_1(F(f))$ the universal central extension of P.

The last result comes from applying this to $G = \mathrm{GL}(R)$ so that $P = E(R)$, completing the proof.

Corollary 5.1.10: Fundamental Group of Volodin Space

The fundamental group $\pi_1(X(R))$ of the Volodin space is the Steinberg group $\mathrm{st}(R)$.

***Proof*:**
The shape of the argument is theorem 5.1.9 applied to $G = \mathrm{GL}(R)$ and $P = E(R)$: it identifies π_1 of the homotopy fibre of the $+$-construction with the universal central extension of $E(R)$, which is $\mathrm{st}(R)$ by corollary 4.4.4. What that needs, and what the two facts recorded with definition 5.1.4 do *not* give, is that the homotopy fibre of $\mathrm{BGL}(R) \to \mathrm{BGL}(R)/X(R)$ is $X(R)$ itself. It is tempting to read this off from acyclicity, but "the homotopy fibre of $X \to X/A$ is A" is false in general: for A acyclic with $\pi_1(A) = P$ and $X = A \vee S^1$, the quotient is S^1 and the fibre is the infinite cyclic cover $\bigvee_{n\in\mathbb{Z}} A$, whose fundamental group is a free product of copies of P rather than P. The identification is therefore a genuine theorem, stated without proof by Weibel, and we cite it: [3, Ch. IV, Cor. 1.7.2].

5.2 The Q-Construction

The plus construction of the previous section produced the higher K-groups $K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$ of definition 5.1.1 by surgery on the classifying space of $\mathrm{GL}(R)$. That construction is tied to a single ring R and to the general linear group. It says nothing, on its face, about how K-theory behaves for an arbitrary class of modules glued by short exact sequences, and it offers no direct handle on $K_0(R)$, which had to be appended by hand in definition 5.1.2. Quillen's Q-construction repairs both deficiencies at once. It takes as input not a ring but an *exact category* (a category of modules, or sheaves, or bundles, together with a distinguished class of short exact sequences) and outputs a single space $BQ\mathscr{E}$ whose homotopy groups are the K-groups of that category. This section builds the Q-construction from scratch, forms its classifying space, and proves the first comparison theorem: the fundamental group $\pi_1(BQ\mathscr{P}(R))$ recovers the Grothendieck group $K_0(R)$ of definition 2.2.1. The agreement of these Quillen K-groups with the plus-construction groups of definition 5.1.1 is the subject of the next section.

Its input is an exact category, whose axioms and running examples were set up in section 3.2: the class $\mathscr{M}$ of admissible short exact sequences (definition 3.2.1), the split exact category $\mathscr{P}(R)$ of finitely generated projective modules (definition 3.2.2), and the vector-space case (example 3.2.3) against which every abstract statement below should first be tested. The pushout and pullback clauses of definition 3.2.1 are the ones the construction uses: they guarantee that a square built from one admissible mono and one admissible epi can be completed, and it is exactly that completion which defines composition in $Q\mathscr{E}$.

With an exact category in hand, the Q-construction produces a new category. Its objects are unchanged, but a morphism is no longer a single map: it is a subobject-then-quotient datum, a way of relating M to P through a common intermediary N that sits inside P and surjects onto M.

Definition 5.2.1: The Q-Construction

Let $\mathscr{E}$ be an exact category. The category $Q\mathscr{E}$ has the same objects as $\mathscr{E}$. A morphism from M to P in $Q\mathscr{E}$ is an isomorphism class of diagrams

$$M \twoheadleftarrow N \rightarrowtail P$$

consisting of an admissible epimorphism $N \twoheadrightarrow M$ and an admissible monomorphism $N \rightarrowtail P$, where two such diagrams $M \twoheadleftarrow N \rightarrowtail P$ and $M \twoheadleftarrow N' \rightarrowtail P$ are identified when there is an isomorphism $N \xrightarrow{\sim} N'$ commuting with both legs. The composite of $M \twoheadleftarrow N \rightarrowtail P$ with $P \twoheadleftarrow N' \rightarrowtail S$ is the diagram $M \twoheadleftarrow N'' \rightarrowtail S$ in which N'' is the pullback

$$\begin{tikzcd} N'' \arrow[r, tail] \arrow[d, two heads] & N' \arrow[d, two heads] \\ N \arrow[r, tail] & P \end{tikzcd}$$

of $N \rightarrowtail P$ along $N' \twoheadrightarrow P$, with the leg $N'' \twoheadrightarrow M$ obtained by composing $N'' \twoheadrightarrow N \twoheadrightarrow M$ and the leg $N'' \rightarrowtail S$ obtained by composing $N'' \rightarrowtail N' \rightarrowtail S$. The identity morphism of M is the class of $M \twoheadleftarrow M \rightarrowtail M$ with both legs the identity.

The pullback in the composition law exists and has the stated legs by clauses (3) and (4) of definition 3.2.1, the second of which was listed there for exactly this purpose: the top map $N'' \rightarrowtail N'$ is the pullback of the admissible monomorphism $N \rightarrowtail P$ along $N' \twoheadrightarrow P$, hence an admissible monomorphism, and the left map $N'' \twoheadrightarrow N$ is the pullback of the admissible epimorphism $N' \twoheadrightarrow P$ along $N \rightarrowtail P$, hence an admissible epimorphism. Composing $N'' \twoheadrightarrow N$ with the admissible epimorphism $N \twoheadrightarrow M$ gives an admissible epimorphism by axiom (3), and composing $N'' \rightarrowtail N'$ with the admissible monomorphism $N' \rightarrowtail S$ gives an admissible monomorphism by axiom (2). Thus the composite is again a legitimate morphism of $Q\mathscr{E}$. That this composition is associative up to the identifications by isomorphism follows from the universal property of pullbacks, which makes the iterated pullback of three composable morphisms canonically independent of the order of formation; this verification is recorded as part of the next proposition.

Proposition 5.2.2: $Q\mathscr{E}$ is a Category

Let $\mathscr{E}$ be an exact category. Then $Q\mathscr{E}$ of definition 5.2.1 is a category: composition is well-defined on isomorphism classes, associative, and unital.

***Proof*:**
Step 1: Well-definedness. Suppose $M \twoheadleftarrow N \rightarrowtail P$ is replaced by an isomorphic representative $M \twoheadleftarrow \tilde{N} \rightarrowtail P$ via $\theta\colon N \xrightarrow{\sim} \tilde{N}$ commuting with both legs, and likewise $P \twoheadleftarrow N' \rightarrowtail S$ by $\theta'\colon N' \xrightarrow{\sim} \tilde{N}'$. Forming the pullback of $\tilde{N} \rightarrowtail P$ along $\tilde{N}' \twoheadrightarrow P$ yields an object $\tilde{N}''$, and the pair (θ, θ') induces an isomorphism $N'' \xrightarrow{\sim} \tilde{N}''$ by the universal property of the pullback, since θ and θ' are compatible with the maps to P. This isomorphism commutes with the two legs to M and to S because θ, θ' do. Hence the composite class is independent of the chosen representatives.

Step 2: Unitality. Composing $M \twoheadleftarrow N \rightarrowtail P$ with the identity $P \twoheadleftarrow P \rightarrowtail P$ forms the pullback of $N \rightarrowtail P$ along $P \xrightarrow{\text{id}} P$, which is N itself with the identity legs; the resulting diagram is

$M \twoheadleftarrow N \rightarrowtail P$ again. The same holds on the other side, so the class of $M \twoheadleftarrow M \rightarrowtail M$ is a two-sided identity.

Step 3: Associativity. Given three composable morphisms $M \twoheadleftarrow N_1 \rightarrowtail P$, $P \twoheadleftarrow N_2 \rightarrowtail S$, $S \twoheadleftarrow N_3 \rightarrowtail T$, both bracketings produce, as the intermediary, the limit of the diagram $N_1 \rightarrowtail P \twoheadleftarrow N_2 \rightarrowtail S \twoheadleftarrow N_3$. Forming the limit by first taking the left pullback and then the right, or first the right and then the left, yields canonically isomorphic objects, because iterated pullbacks compute the same limit regardless of order (the pasting law for pullback squares). The induced legs to M and to T agree under this canonical isomorphism, so the two composites are equal as isomorphism classes, as we sought to show.

The proposition says $Q\mathscr{E}$ is a genuine category, so it has a nerve. The reader should keep two special kinds of morphism in view. Taking $N = M$ with the left leg the identity gives an *injective* morphism $i_!: M \rightarrowtail P$, recording an admissible monomorphism $M \rightarrowtail P$; taking $N = P$ with the right leg the identity gives a *surjective* morphism $j^!: M \to P$ from an admissible epimorphism $P \twoheadrightarrow M$ (note the direction: a quotient map $P \twoheadrightarrow M$ becomes a Q-morphism from M to P). Every morphism of $Q\mathscr{E}$ factors as $i_! \circ j^!$ through its intermediary N, and these two families of morphisms are the edges from which the fundamental group will be assembled.

The next step passes to topology. Recall from [12, Nerve of a Small Category] that the nerve $N\mathscr{C}$ of a (small) category $\mathscr{C}$ is the simplicial set whose n-simplices are the chains $C_0 \to C_1 \to \cdots \to C_n$ of n composable morphisms, with faces given by composing or dropping end maps and degeneracies by inserting identities. Its geometric realization $B\mathscr{C} = |N\mathscr{C}|$, the classifying space of [12, Classifying Space of a Small Category], is a CW complex whose homotopy groups are the homotopy invariants of $\mathscr{C}$. This applies to $Q\mathscr{E}$.

Definition 5.2.3: Quillen K-Groups

Let $\mathscr{E}$ be a (small) exact category with a chosen zero object 0. Its *classifying space* is the geometric realization $BQ\mathscr{E}$ of the nerve of the category $Q\mathscr{E}$ from definition 5.2.1, based at the vertex 0. The *Quillen K-groups* of $\mathscr{E}$ are

$$K_n(\mathscr{E}) := \pi_{n+1}(BQ\mathscr{E}, 0) \qquad (n \geq 0)$$

For a ring R, the Quillen K-groups of R are $K_n(R) := K_n(\mathscr{P}(R))$, the K-groups of the exact category $\mathscr{P}(R)$ of definition 3.2.2.

The degree shift is the price of the construction: $BQ\mathscr{E}$ is connected (every object is joined to 0 by a zigzag of injective and surjective morphisms), so π_0 carries no information, and the index $n+1$ realigns the homotopy groups so that K_0 lands where one wants it. The space $BQ\mathscr{E}$ is one connected space carrying all the K-groups simultaneously, in contrast to the plus construction, which built $\mathrm{BGL}(R)^+$ for the higher groups and then formally adjoined $K_0(R)$ in definition 5.1.2. Here $K_0(\mathscr{E})$ emerges intrinsically as $\pi_1(BQ\mathscr{E})$, and the next theorem identifies it with the Grothendieck group already defined. The smallness hypothesis is a set-theoretic convenience: $\mathscr{P}(R)$ is equivalent to a small category (restrict to the projective summands of the modules R^n, $n \geq 0$), and one works with such a skeleton so that the nerve is a genuine simplicial set. The homotopy type is unaffected by the choice of skeleton.

Example 5.2.4: The Classifying Space $BQ\mathscr{P}(F)$ for a Field

For a field F, the isomorphism classes of objects of $\mathscr{P}(F)$ are $F^0, F^1, F^2, \ldots$, one for each dimension. In $Q\mathscr{P}(F)$ there is an injective edge $i_!\colon F^a \rightarrowtail F^n$ for every admissible monomorphism, that is for every $a \leq n$ (realizing F^a as a summand of F^n), and a surjective edge $j^!\colon F^b \to F^n$ for every admissible epimorphism $F^n \twoheadrightarrow F^b$, that is for every $b \leq n$. The vertex $F^0 = 0$ is the basepoint. The one-dimensional skeleton of $BQ\mathscr{P}(F)$ already connects every F^n to the basepoint: the chain $0 \rightarrowtail F^n$ is a single injective edge from 0 to F^n. Loops at 0 are built from these edges, and the relations imposed by the 2-simplices of the nerve will, by theorem 5.2.6 below, collapse π_1 to the free abelian group on the one generator $[F^1]$, recovering $\mathbb{Z}$. The whole space is thus a concrete model whose fundamental group is computed entirely from the additive arithmetic of dimensions.

Before the comparison theorem, two structural facts about the maps that make $Q\mathscr{E}$ tractable must be isolated. Each records one of the two edge families, and each will contribute a family of loops to π_1.

Lemma 5.2.5: Edges and Their Composition in $Q\mathscr{E}$

Let $\mathscr{E}$ be an exact category. In $Q\mathscr{E}$ the following hold.

1. Every morphism $M \to P$ factors as $i_! \circ j^!$, where $j^!\colon M \to N$ comes from the admissible epimorphism $N \twoheadrightarrow M$ and $i_!\colon N \to P$ from the admissible monomorphism $N \rightarrowtail P$, and the intermediary N is determined up to isomorphism.

2. For composable admissible monomorphisms $M \rightarrowtail N \rightarrowtail P$ one has $(i'i)_! = i'_! \circ i_!$, and for composable admissible epimorphisms $j'\colon P \twoheadrightarrow N$ and $j\colon N \twoheadrightarrow M$ one has $(jj')^! = j'^! \circ j^!$. Note the order reverses: $j^!$ runs $M \to N$ and $j'^!$ runs $N \to P$, so it is $j'^!$ that is applied second.

3. For an admissible short exact sequence $0 \to M' \xrightarrow{i} M \xrightarrow{j} M'' \to 0$ the injective edge $i_!\colon M' \to M$ and the surjective edge $j^!\colon M'' \to M$ are distinct edges of $Q\mathscr{E}$ with common target M. No edge $u\colon M' \to M''$ with $j^! \circ u = i_!$ exists when $M' \neq 0$. What ties them instead is the morphism $\tau = (0 \twoheadleftarrow M' \rightarrowtail M)$ from 0 to M, which factors in two ways,
$$\tau \ = \ i_! \circ \sigma_{M'} \ = \ j^! \circ \iota_{M''}$$
where $\sigma_{M'}\colon 0 \to M'$ is the surjective edge of $M' \twoheadrightarrow 0$ and $\iota_{M''}\colon 0 \to M''$ the injective edge of $0 \rightarrowtail M''$. These are two distinct 2-simplices sharing the long edge τ, on vertex sets $\{0, M', M\}$ and $\{0, M'', M\}$.

***Proof*:**

Part (1). A morphism $M \to P$ is by definition an isomorphism class of $M \twoheadleftarrow N \rightarrowtail P$. The datum $N \twoheadrightarrow M$ is a surjective morphism $j^!\colon M \to N$ in $Q\mathscr{E}$ (the diagram $M \twoheadleftarrow N \xrightarrow{\mathrm{id}} N$), and $N \rightarrowtail P$ is an injective morphism $i_!\colon N \to P$ (the diagram $N \xleftarrow{\mathrm{id}} N \rightarrowtail P$). Composing $j^!$ then $i_!$ forms the pullback of $N \xrightarrow{\mathrm{id}} N$, which returns N with the original legs, so $i_! \circ j^! = (M \twoheadleftarrow N \rightarrowtail P)$. The intermediary N is determined up to isomorphism by the definition of equality of morphisms in $Q\mathscr{E}$.

Part (2). Composing $i_!\colon M \to N$ (diagram $M \xleftarrow{\mathrm{id}} M \rightarrowtail N$) with $i'_!\colon N \to P$ (diagram $N \xleftarrow{\mathrm{id}} N \rightarrowtail P$) forms the pullback of $M \rightarrowtail N$ along id_N, which is M, with composite leg $M \rightarrowtail N \rightarrowtail P$ the admissible monomorphism $i'i$. Thus $i'_! i_! = (i'i)_!$. The epimorphism statement is dual, composing $j^!\colon M \to N$ and $j'^!\colon N \to P$ and forming the pullback along id.

Part (3). That $i_!$ and $j^!$ are edges with target M is immediate from their definitions. Suppose an edge $u\colon M' \to M''$ satisfied $j^! \circ u = i_!$. Writing $u = (M' \twoheadleftarrow N \rightarrowtail M'')$, the composite $j^! \circ u$ is, by definition 5.2.1, the pullback of the mono leg $N \rightarrowtail M''$ along the epi leg $j\colon M \twoheadrightarrow M''$ of $j^!$, with mono leg into M, namely $j^{-1}(N) \rightarrowtail M$. For the composite to equal $i_! = (M' \xleftarrow{\mathrm{id}} M' \xrightarrow{i} M)$, the two mono legs into M must agree, so $j^{-1}(N) = i(M') = \ker j$ as subobjects of M. Applying j, which is epimorphic, gives $N = j\big(j^{-1}(N)\big) = j(\ker j) = 0$. The epi leg of u is then $0 \twoheadrightarrow M'$, which forces $M' = 0$. So no such u exists once $M' \neq 0$. Note what the argument does *not* say: $j^{-1}(N) \cong M'$ on its own would not force $N \cong M'$, and indeed $N = 0$ also has $j^{-1}(N) = \ker j \cong M'$. It is the agreement of the mono legs, not the abstract isomorphism, that pins N down.

For the two factorizations of τ, compose directly. Pulling the mono leg $0 \rightarrowtail M''$ of $\iota_{M''}$ back along j gives $j^{-1}(0) = \ker j = M'$, so $j^! \circ \iota_{M''} = (0 \twoheadleftarrow M' \rightarrowtail M) = \tau$; and pulling $\mathrm{id}_{M'}$ back along id gives $i_! \circ \sigma_{M'} = (0 \twoheadleftarrow M' \rightarrowtail M) = \tau$ as well. These two 2-simplices are what force the additivity relation computed in theorem 5.2.6, completing the proof.

The content of the lemma is bookkeeping, but it is the bookkeeping that drives the computation of π_1. Part (1) says the two edge families generate all morphisms. Part (2) says each family composes functorially. Part (3) records that an admissible sequence produces two distinct edges into M tied by a 2-simplex, the source of the additivity relation. The next theorem turns this into an isomorphism.

Theorem 5.2.6: π_1 of $Q\mathscr{E}$ Recovers K_0

Let $\mathscr{E}$ be a small exact category with zero object 0. Then there is a natural isomorphism

$$\pi_1(BQ\mathscr{E}, 0) \cong K_0(\mathscr{E})$$

where $K_0(\mathscr{E})$ is the Grothendieck group of $\mathscr{E}$: the free abelian group on the isomorphism classes $[M]$ of objects, modulo the relations $[M] = [M'] + [M'']$ for every admissible short exact sequence $0 \to M' \to M \to M'' \to 0$. In particular, for a ring R,

$$\pi_1(BQ\mathscr{P}(R), 0) \cong K_0(R)$$

the Grothendieck group of definition 2.2.1.

***Proof*:**
The fundamental group of a classifying space is computed from the 1- and 2-simplices of its nerve. The category $Q\mathscr{E}$ is connected, since the chain $0 \rightarrowtail C$ joins every object C to the zero object, so [12, The Fundamental Group of a Classifying Space] applies: relative to a choice of spanning tree T in the 1-skeleton, $\pi_1(BQ\mathscr{E}, 0)$ is generated by the directed edges of the nerve, subject to the tree relations $\gamma = 1$ for each edge of T and the composition relations $\gamma_{02} = \gamma_{12}\,\gamma_{01}$ imposed by the 2-simplices $C_0 \xrightarrow{\gamma_{01}} C_1 \xrightarrow{\gamma_{12}} C_2$, products being written so that $\gamma_{12}\,\gamma_{01}$ traverses γ_{01} first. Step 3 makes the choice of T. The presentation, and this convention, are [3, Ch. IV, Lem. 3.4], stated there for a small connected category with a maximal tree, where $[f] \cdot [g] = [f \circ g]$. It is

opposite to the one fixed in [12, The Fundamental Group of a Classifying Space], under which $[f][g] = [g \circ f]$. The two presentations are opposite groups, so a reader importing that convention will obtain $\ell_M = \ell_{M''}\ell_{M'}$ below. Nothing depends on which is chosen, only on choosing one and keeping it. The proof constructs mutually inverse homomorphisms Φ and Ψ between $K_0(\mathscr{E})$ and this group.

Step 1: The map Φ. For each object M let $\iota_M \colon 0 \rightarrowtail M$ be the injective edge from the unique admissible monomorphism $0 \rightarrowtail M$ (admissible by axiom (1) of definition 3.2.1 applied to $0 \to M \xrightarrow{\mathrm{id}} M$), and let $\sigma_M \colon 0 \to M$ be the surjective edge from the admissible epimorphism $M \twoheadrightarrow 0$. As edges of the nerve both run between the vertices 0 and M, so the product

$$\ell_M := \iota_M^{-1}\,\sigma_M$$

is a based loop at 0. Define $\Phi[M] := [\ell_M]$. To see that this respects the defining relations of $K_0(\mathscr{E})$, let $0 \to M' \xrightarrow{i} M \xrightarrow{j} M'' \to 0$ be admissible. The injective edge $i_! \colon M' \to M$ and surjective edge $j^! \colon M'' \to M$ of lemma 5.2.5(3), together with the chosen edges ι and σ from 0, span 2-simplices of the nerve forcing

$$\iota_M = i_!\,\iota_{M'} \quad \text{and} \quad \sigma_M = j^!\,\sigma_{M''}$$

by lemma 5.2.5(2) applied to $0 \rightarrowtail M' \rightarrowtail M$ and to $M \twoheadrightarrow M'' \twoheadrightarrow 0$. Write $g_e = \iota_B^{-1}\,e\,\iota_A$ for the loop attached to an edge $e \colon A \to B$ after contracting the tree of the ι_N, so that $\ell_M = g_{\sigma_M}$. Three short computations give the relation.

First, $g_{i_!} = \iota_M^{-1} i_! \iota_{M'} = (i_!\iota_{M'})^{-1} i_!\iota_{M'} = 1$: every injective edge contributes a trivial loop. Second, the two factorizations of $\tau = (0 \twoheadleftarrow M' \rightarrowtail M)$ from lemma 5.2.5(3) give

$$g_{j^!} \;=\; \iota_M^{-1}\,j^!\,\iota_{M''} \;=\; \iota_M^{-1}\,\tau \;=\; \iota_M^{-1}\,i_!\,\sigma_{M'} \;=\; \iota_{M'}^{-1}\sigma_{M'} \;=\; \ell_{M'}$$

using $\iota_M^{-1} i_! = \iota_{M'}^{-1}$ from the first relation. Third, inserting $\iota_{M''}\iota_{M''}^{-1}$ into $\ell_M = \iota_M^{-1}\sigma_M = \iota_M^{-1} j^! \sigma_{M''}$,

$$\ell_M \;=\; \big(\iota_M^{-1}\,j^!\,\iota_{M''}\big)\big(\iota_{M''}^{-1}\sigma_{M''}\big) \;=\; g_{j^!}\,\ell_{M''} \;=\; \ell_{M'}\,\ell_{M''}$$

Only 2-simplices that exist are used, and no simplicial machinery beyond the edge-path presentation. This multiplicative relation among the loops holds in $\pi_1(BQ\mathscr{E}, 0)$ for every admissible sequence. Once π_1 is shown abelian in Step 3 it reads additively as $[\ell_M] = [\ell_{M'}] + [\ell_{M''}]$, the defining relation of $K_0(\mathscr{E})$, so the assignment $\Phi[M] = [\ell_M]$ is then a well-defined homomorphism on $K_0(\mathscr{E})$.

Step 2: The map Ψ. Build a homomorphism in the reverse direction by assigning a K_0-class to each edge and checking the 2-simplex relations. To an injective edge $i_! \colon N \rightarrowtail P$ assign $0 \in K_0(\mathscr{E})$. To a surjective edge $j^! \colon M \to N$ coming from an admissible epimorphism $N \twoheadrightarrow M$ with admissible kernel $0 \to K \to N \to M \to 0$ assign $[K]$; and to a general edge $M \xrightarrow{(i_! j^!)} P$ assign the sum, namely $[\ker(N \twoheadrightarrow M)]$ for its intermediary N. Extend to formal inverses by negation. This assignment respects every 2-simplex relation $\gamma_{02} = \gamma_{12}\gamma_{01}$: composing two morphisms forms the pullback intermediary of definition 5.2.1, and the kernel of the composite admissible epimorphism is an extension of the two constituent kernels, so its class is their sum by the additivity of $K_0(\mathscr{E})$, while injective edges contribute trivial kernels. Therefore the assignment descends to a homomorphism

$$\Psi \colon \pi_1(BQ\mathscr{E}, 0) \to K_0(\mathscr{E})$$

(the target $K_0(\mathscr{E})$ is abelian, so the edge-path relations suffice to define Ψ). On a generator,

$$\Psi(\ell_M) = \Psi(\iota_M^{-1}\sigma_M) = -\Psi(\iota_M) + \Psi(\sigma_M) = -0 + [\ker(M \twoheadrightarrow 0)] = [M]$$

since the kernel of $M \twoheadrightarrow 0$ is M itself. Thus $\Psi \circ \Phi = \mathrm{id}$ on the generators $[M]$ of $K_0(\mathscr{E})$, in particular Φ is injective.

Step 3: Surjectivity, commutativity, and conclusion. Every edge of $Q\mathscr{E}$ factors as $i_! \circ j^!$ by lemma 5.2.5(1), and the injective edges ι_N from 0 form a spanning tree of the 1-skeleton (each object is joined to 0 by the single chosen ι_N). Contracting this tree, π_1 is generated by the loops g_e attached to *all* the remaining edges, not only by the σ_M, so two reductions from Step 1 are needed. An injective edge has $g_{i_!} = 1$. A general surjective edge $j^! \colon M \to N$, coming from an admissible epimorphism $j\colon N \twoheadrightarrow M$ with kernel K, has $g_{j^!} = \ell_K$ by the second computation of Step 1 applied to $0 \to K \to N \to M \to 0$. A general edge is neither, but factors as $e = i_! \circ j^!$ by lemma 5.2.5(1), and the 2-simplex relation gives $g_{i_! \circ j^!} = g_{i_!} g_{j^!}$, so its loop is a product of the two cases. Hence every generator is trivial or of the form ℓ_K, and all lie in the image of Φ, so Φ is surjective. One check remains for Φ itself: it is defined on isomorphism classes, and an isomorphism $\theta\colon M \to M'$ satisfies $\theta_! \iota_M = \iota_{M'}$ and $\theta_! \sigma_M = \sigma_{M'}$, so $\ell_M = \ell_{M'}$. Together with injectivity from Step 2, Φ is an isomorphism with inverse Ψ. Applying the additivity relation of Step 1 to the two split sequences $0 \to M' \to M' \oplus M'' \to M'' \to 0$ and $0 \to M'' \to M' \oplus M'' \to M' \to 0$ shows $[\ell_{M'}]$ and $[\ell_{M''}]$ commute, so $\pi_1(BQ\mathscr{E}, 0)$ is abelian, consistent with its being a copy of $K_0(\mathscr{E})$.

Naturality. An exact functor $F\colon \mathscr{E} \to \mathscr{E}'$ induces $QF\colon Q\mathscr{E} \to Q\mathscr{E}'$ (it preserves admissible monomorphisms, admissible epimorphisms, and the pullbacks of definition 5.2.1), hence a based map BQF carrying ℓ_M to ℓ_{FM}. Thus Φ commutes with the maps induced by F on K_0 and on π_1, so the isomorphism is natural in $\mathscr{E}$.

The ring case. For $\mathscr{E} = \mathscr{P}(R)$ the Grothendieck group $K_0(\mathscr{P}(R))$ is the free abelian group on isomorphism classes of finitely generated projective R-modules modulo the admissible sequences. Every such sequence splits (each object is projective), so its relations are exactly $[P] = [P'] + [P'']$ whenever $P \cong P' \oplus P''$, which is the defining relation of $K_0(R)$ in definition 2.2.1. Hence $\pi_1(BQ\mathscr{P}(R), 0) \cong K_0(\mathscr{P}(R)) = K_0(R)$, completing the proof.

The theorem is the first payoff of the Q-construction: the Grothendieck group, defined in definition 2.2.1 as a purely algebraic completion of a monoid, reappears as the fundamental group of a space, and the defining relations $[M] = [M'] + [M'']$ are realized geometrically as the loops contributed by admissible short exact sequences. This is the pattern the whole theory follows. Each $K_n(\mathscr{E})$ measures the $(n+1)$-dimensional loops of $BQ\mathscr{E}$, and the relations one would impose by hand at the level of K_0 are encoded once and for all in the simplicial structure of the nerve. The plus construction reached the same $K_0(R)$ only by the artificial device of definition 5.1.2. Here it is intrinsic.

Functoriality, established inside the proof, deserves a separate statement because it is what lets K-theory transport along ring maps and along exact functors between categories of modules.

Corollary 5.2.7: Functoriality of the Quillen K-Groups

Let $F\colon \mathscr{E} \to \mathscr{E}'$ be an exact functor between small exact categories. Then F induces a based map $BQF\colon BQ\mathscr{E} \to BQ\mathscr{E}'$ and hence homomorphisms

$$K_n(F)\colon K_n(\mathscr{E}) \to K_n(\mathscr{E}') \qquad (n \geq 0)$$

functorial in F. In particular, a ring homomorphism $\varphi\colon R \to S$ induces the exact functor $- \otimes_R S\colon \mathscr{P}(R) \to \mathscr{P}(S)$, and hence homomorphisms $K_n(R) \to K_n(S)$ for all $n \geq 0$, agreeing on K_0 with the map of definition 2.2.1 under the isomorphism of theorem 5.2.6.

***Proof*:**
An exact functor preserves admissible monomorphisms and admissible epimorphisms and commutes with the pullbacks defining composition in $Q\mathscr{E}$, so $M \twoheadleftarrow N \rightarrowtail P \mapsto FM \twoheadleftarrow FN \rightarrowtail FP$ is a functor $QF\colon Q\mathscr{E} \to Q\mathscr{E}'$. A functor of small categories induces a continuous map on classifying spaces ([12, Classifying Space of a Small Category]), so QF gives a based map $BQF\colon BQ\mathscr{E} \to BQ\mathscr{E}'$ fixing the zero object, inducing $K_n(F) = \pi_{n+1}(BQF)$. Functoriality in F follows from functoriality of the nerve and of π_{n+1}. For a ring map $\varphi\colon R \to S$, the functor $-\otimes_R S$ sends a finitely generated projective R-module to a finitely generated projective S-module and preserves split exact sequences (it is additive and right exact, and split sequences are preserved by any additive functor), hence is exact on $\mathscr{P}(R)$. On K_0 the induced map sends $[P]$ to $[P \otimes_R S]$, which is the map of definition 2.2.1; under theorem 5.2.6 this is K_0 of the induced map, as we sought to show.

This is the structural counterpart, for the Q-construction, of the functoriality of $\mathrm{BGL}(R)^+$ recorded after definition 5.1.2. There it required passing through homotopy because no model had been fixed. Here the functor QF is honest, defined before any homotopy theory, and the induced maps on K_n are immediate. The price paid for this directness is the degree shift and the more elaborate morphisms of $Q\mathscr{E}$, but the reward is a theory that accepts any exact category as input, opening the door to the resolution, dévissage, and localization theorems of the next section, none of which has a natural formulation in the plus-construction picture.

The section closes with the computation promised in example 5.2.4, recording the case $R = \mathbb{Z}$, the two computations the theorem most directly delivers.

Example 5.2.8: $\pi_1(BQ\mathscr{P}(F)) = \mathbb{Z}$ and $\pi_1(BQ\mathscr{P}(\mathbb{Z})) = \mathbb{Z}$

1. Let F be a field. Every finitely generated projective F-module is a finite-dimensional vector space F^n, and $[F^n] = n[F]$ in $K_0(F)$ because the standard flag $0 \subset F \subset F^2 \subset \cdots \subset F^n$ gives admissible sequences forcing $[F^n] = [F^{n-1}] + [F]$ inductively. The dimension homomorphism $K_0(F) \to \mathbb{Z}$, $[F^n] \mapsto n$, is thus an isomorphism, with inverse $1 \mapsto [F]$. By theorem 5.2.6,

$$\pi_1(BQ\mathscr{P}(F), 0) \cong K_0(F) \cong \mathbb{Z}$$

generated by the loop ℓ_F attached to the line F.

2. Let $R = \mathbb{Z}$. Since $\mathbb{Z}$ is a principal ideal domain, every finitely generated projective $\mathbb{Z}$-module is free, of the form $\mathbb{Z}^n$, and as in part (1) the rank gives $K_0(\mathbb{Z}) \cong \mathbb{Z}$ with $[\mathbb{Z}^n] \mapsto n$. By theorem 5.2.6,

$$\pi_1(BQ\mathscr{P}(\mathbb{Z}), 0) \cong K_0(\mathbb{Z}) \cong \mathbb{Z}$$

generated by $\ell_{\mathbb{Z}}$. The two computations are formally identical because both $\mathscr{P}(F)$ and $\mathscr{P}(\mathbb{Z})$ have the property that every object is free of a well-defined rank and every admissible sequence splits; the invariant computing π_1 is the rank in each case.

Exercise 5.2.1

1. Verify directly that $\mathscr{P}(R)$ of definition 3.2.2 satisfies the pullback axiom (3) of definition 3.2.1: given an admissible epimorphism $j\colon P \twoheadrightarrow P''$ of finitely generated projectives and an arbitrary map $f\colon Q \to P''$ with Q finitely generated projective, show the pullback $Q \times_{P''} P$ exists in $\mathscr{P}(R)$ and that its projection to Q is an admissible epimorphism.
2. In $Q\mathscr{P}(R)$, describe explicitly the composite of the surjective edge $j^!\colon R \to R^2$ coming from the projection $R^2 \twoheadrightarrow R$ onto the first coordinate with the injective edge $i_!\colon R^2 \rightarrowtail R^3$ coming from the inclusion of the first two coordinates. Identify the intermediary N and both legs of the resulting diagram $R \twoheadleftarrow N \rightarrowtail R^3$.
3. Let F be a field. Using theorem 5.2.6, compute $\pi_1(BQ\mathscr{P}(F \times F), 0)$. (Hint: a finitely generated projective module over $F \times F$ is a pair of vector spaces.)
4. Show that the loop ℓ_0 attached to the zero object is trivial in $\pi_1(BQ\mathscr{E}, 0)$, consistently with $[0] = 0$ in $K_0(\mathscr{E})$.
5. Let $\mathscr{E}$ be an exact category in which the only admissible sequences are the split ones. Prove that $K_0(\mathscr{E})$ is the Grothendieck group of the monoid $(\mathrm{Obj}\mathscr{E}/\cong, \oplus)$, and conclude that $K_0(\mathscr{P}(R)) = K_0(R)$ matches definition 2.2.1 without invoking the general theorem 5.2.6 machinery.
6. An exact functor $F\colon \mathscr{E} \to \mathscr{E}'$ that is an equivalence of categories preserving and reflecting admissible sequences induces an isomorphism $K_n(\mathscr{E}) \cong K_n(\mathscr{E}')$ for all n. Use this to show that Morita equivalent rings R and $M_n(R)$ have isomorphic Quillen K-groups (section 2.4).

5.3 The Plus-Equals-Q Theorem

Two definitions of higher algebraic K-theory now stand side by side. The first, from definition 5.1.1, sets $K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$ using the plus construction on the classifying space of $\mathrm{GL}(R)$. The second, from section 5.2, sets $K_n(R) = \pi_{n+1}(BQ\mathcal{P}(R))$ using the nerve of the Q-construction on the exact category $\mathcal{P}(R)$ of finitely generated projective modules. The plus construction is built from the linear group and sees only the matrices. The Q-construction is built from the category of modules and exact sequences, and sees the whole monoidal structure of direct sum together with the data of admissible monomorphisms and epimorphisms. That these two routes, starting from such different data, land on the same groups is the central structural theorem of the subject. This section states that agreement, lays out the architecture of its proof in full, and deduces from it that the higher theory genuinely extends the classical groups K_0, K_1, and K_2 computed in the previous chapters.

The agreement is what licenses the name "$K_n(R)$" for both objects. Without it, the plus-construction groups and the Q-construction groups would be two unrelated invariants that happen to coincide in degrees 0, 1, and 2. The theorem promotes that low-degree coincidence into an isomorphism in every degree, and (this is the operative point) does so naturally in the ring R. From a computational standpoint the two definitions have complementary strengths: the plus construction is concrete enough to read off K_1 and K_2 by hand, while the Q-construction is the one that carries the long exact sequences (localization, dévissage, resolution) that make higher K-groups computable. The theorem lets each definition lend its strengths to the other.

The full proof occupies a substantial part of Quillen's foundational paper and of any modern treatment. The present section gives the architecture in complete structural detail and defers the longest homotopy-theoretic verifications to [3], where they are carried out in full. This is a deferral to a complete published proof, not an omission: every step is named, the role of each tool is stated, and the only content sent elsewhere is the chain of simplicial homotopy equivalences whose verification is mechanical but lengthy.

Two tools of Quillen's own drive the argument, and stating them precisely is the bulk of the structural work. The first is a criterion for when a functor between (nerves of) categories is a homotopy equivalence, and the second is a criterion identifying the homotopy fibre of such a functor. For a small category $\mathcal{C}$ write $B\mathcal{C}$ for the geometric realization of its nerve, the classifying space whose n-simplices are chains of n composable morphisms.

Theorem 5.3.1: Quillen's Theorems A and B

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor between small categories. For an object d of $\mathcal{D}$, let F/d denote the comma category whose objects are pairs (c, u) with c an object of $\mathcal{C}$ and $u\colon F(c) \to d$ a morphism of $\mathcal{D}$, and whose morphisms are the evident triangles.

1. *(Theorem A.)* If $B(F/d)$ is contractible for every object d of $\mathcal{D}$, then $BF\colon B\mathcal{C} \to B\mathcal{D}$ is a homotopy equivalence.
2. *(Theorem B.)* If for every morphism $d \to d'$ in $\mathcal{D}$ the induced functor $F/d \to F/d'$ is a homotopy equivalence on classifying spaces, then for each object d the square

$$\begin{tikzcd} B(F/d) \arrow[r] \arrow[d] & B\mathcal{C} \arrow[d, "BF"] \\ B(\mathcal{D}/d) \arrow[r] & B\mathcal{D} \end{tikzcd}$$

is homotopy cartesian. Since $B(\mathcal{D}/d)$ is contractible, $B(F/d)$ is the homotopy fibre of BF over the point d.

***Proof*:**
Both statements are theorems of homotopy theory whose proofs proceed by analysing the bisimplicial set associated to F and applying the realization lemma for bisimplicial sets. The complete argument is given in [3, Chapter IV]. The deferral is to that complete published proof.

Theorem A says that to check a functor induces a homotopy equivalence it suffices to check that each comma category F/d is contractible, a condition that is often combinatorially accessible: contractibility of $B(F/d)$ frequently follows from the existence of an initial or terminal object, or from a natural transformation to a constant functor. Theorem B is the engine that produces fibrations: it identifies the homotopy fibre of BF with the classifying space of a comma category, which converts a long exact sequence of homotopy groups into a statement about categories one can manipulate algebraically. The hypothesis of Theorem B (invariance of F/d along morphisms of $\mathcal{D}$) is exactly what is needed to make the fibre independent of the chosen point, so that the comma category deserves to be called “the” fibre.

The second ingredient is Quillen’s group-completion construction $S^{-1}S$, which performs at the level of categories the same Grothendieck completion that turned the monoid $\mathcal{P}(R)$ into $K_0(R)$ in chapter 2, but now homotopically rather than on π_0 alone. Let S be a symmetric monoidal category in which every morphism is an isomorphism, with monoidal product written $\oplus$. The case of interest is $S = \mathrm{iso}\,\mathcal{P}(R)$, the category of finitely generated projective R-modules and their isomorphisms, under direct sum.

Definition 5.3.2: The $S^{-1}S$ Construction

Let $(S, \oplus)$ be a symmetric monoidal category ([7, Symmetric Monoidal Category]) whose morphisms are all isomorphisms. The category $S^{-1}S$ has objects the pairs (a, b) of objects of S. A morphism $(a, b) \to (a', b')$ is an equivalence class of data

$$\big(s,\ \alpha\colon s \oplus a \xrightarrow{\sim} a',\ \beta\colon s \oplus b \xrightarrow{\sim} b'\big)$$

where s is an object of S, two such data being identified when they differ by an isomorphism of the chosen s. Composition is by $\oplus$ of the connecting objects. The monoidal product $\oplus$ makes $S^{-1}S$ a symmetric monoidal category, and translation by (a, a) is invertible up to natural isomorphism, so $\pi_0(B(S^{-1}S))$ is the group completion of the monoid $\pi_0(BS)$.

The construction formalizes “formal differences $a \ominus b$” as honest objects, with a morphism recording a common stabilization s that makes two such differences equal. On π_0 it is precisely the passage from the monoid of isomorphism classes to its Grothendieck group, so $\pi_0(B(S^{-1}S)) = K_0(R)$ when $S = \mathrm{iso}\,\mathcal{P}(R)$. The content of group completion is that it does the right thing in higher degrees too: under a homology-stability hypothesis the higher homotopy of $B(S^{-1}S)$ computes the higher K-groups. The translation-invertibility clause is what distinguishes this from the bare monoid. It is the categorical form of inverting $[R]$ in the Grothendieck group, and it is what forces $B(S^{-1}S)$ to be an infinite loop space rather than an unstructured space.

Example 5.3.3: Group Completion of the Natural Numbers

Take R a ring satisfying the invariant basis property (definition 1.1.2) and let $S = \mathcal{F}(R)$ be the symmetric monoidal category of finitely generated free modules R^n $(n \geq 0)$ with their isomorphisms, under $\oplus$. The set of isomorphism classes is $\mathbb{N}$, with $\oplus$ giving addition, so $\pi_0(BS) = \mathbb{N}$ and $\pi_0(B(S^{-1}S)) = \mathbb{Z}$, the Grothendieck group of $\mathbb{N}$. The automorphism group of the object R^n is $\mathrm{GL}_n(R)$, so BS is the disjoint union $\coprod_{n\geq 0} B\mathrm{GL}_n(R)$. Forming $S^{-1}S$ stabilizes the matrix size: the object (R^a, R^b) has automorphisms recording isomorphisms after adding a common R^s, and as $s \to \infty$ the automorphism monoid stabilizes to $\mathrm{GL}(R) = \varinjlim_n \mathrm{GL}_n(R)$. Concretely, the component of $B(S^{-1}S)$ containing $(0, 0)$ has fundamental group $\mathrm{GL}(R)/E(R) = K_1(R)$, the abelianization of the stabilized linear group. The group-completion theorem (Step 2 of the proof of theorem 5.3.4) then identifies the whole space:

$$B(S^{-1}S) \simeq \mathbb{Z} \times B\mathrm{GL}(R)^+$$

already in this free-module case, with the factor $\mathbb{Z} = K_0$ of the free theory and $B\mathrm{GL}(R)^+$ the plus construction of definition 5.1.1. Replacing free modules by all finitely generated projectives upgrades $\mathbb{Z}$ to $K_0(R)$.

The example exhibits, in the simplest case, the two phenomena the general theorem packages: the $S^{-1}S$ space splits as π_0 times a connected piece, and that connected piece is the plus construction. The general statement replaces $\mathcal{F}(R)$ by $\mathcal{P}(R)$ and identifies the connected piece's homotopy with the Q-construction groups. The bridge between $S^{-1}S$ and the Q-construction is a homotopy equivalence $B(S^{-1}S) \simeq \Omega BQ\mathcal{P}(R)$, Quillen's $S^{-1}S$ theorem, proven by applying Theorem B of theorem 5.3.1 to an auxiliary functor with target $Q\mathcal{P}(R)$. With the architecture in place, the main theorem can be stated.

Theorem 5.3.4: The Plus-Equals-Q Theorem

Let R be a ring and let $\mathcal{P}(R)$ be the exact category of finitely generated projective R-modules of definition 3.2.2. There is a homotopy equivalence

$$\Omega BQ\mathcal{P}(R) \;\simeq\; K_0(R) \times \mathrm{BGL}(R)^+$$

where the left side is the loop space of the nerve of the Q-construction and the right side is the space $K(R)$ of definition 5.1.2. Consequently, for every $n \geq 1$ there is an isomorphism

$$\pi_n\big(\mathrm{BGL}(R)^+\big) \;\xrightarrow{\;\sim\;}\; \pi_{n+1}\big(BQ\mathcal{P}(R)\big)$$

so that the plus-construction group $K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$ of definition 5.1.1 coincides with the Q-construction group $K_n(R) = \pi_{n+1}(BQ\mathcal{P}(R))$ of definition 5.2.3. The two definitions of $K_n(R)$ agree for all $n \geq 0$.

The equivalence is the precise sense in which "plus equals Q". Looping the Q-construction space once recovers, up to homotopy, the space $K(R) = K_0(R) \times \mathrm{BGL}(R)^+$ that the plus construction produced. Taking π_n of both sides and using $\pi_n(\Omega X) = \pi_{n+1}(X)$ converts this into the degree-shifted isomorphism of homotopy groups: the loop shift Ω is exactly what accounts for the index gap between π_n on the plus side and π_{n+1} on the Q side. On π_0 both sides give $K_0(R)$, and on π_n for $n \geq 1$ the connected component $\mathrm{BGL}(R)^+$ gives the higher groups. The splitting off of $K_0(R)$ as a discrete factor matches the way definition 5.1.2 appended K_0 to the plus construction by hand. What makes the agreement usable is that a ring map $R \to R'$ induces the same map on K_n whichever definition is used, so theorems proved in one model transfer to the other. That naturality is a statement about the *groups*. It should not be read as naturality of the displayed equivalence of spaces: $\mathrm{BGL}(R)^+$ is only well defined up to homotopy, so the induced maps of spaces are too, and the splitting of $K(R)$ into the discrete factor $K_0(R)$ and the H-space $\mathrm{BGL}(R)^+$ is not natural either, the connecting map $K_1(R/I) \to K_0(R, I)$ being nontrivial in general ([3, Ch. IV, 1.1.2]).

***Proof Architecture, with the homotopy-theoretic core deferred*:**
The proof assembles the three tools above into a chain of equivalences. We record the structure of each link and indicate precisely which verifications are carried out in full in [3, Ch. IV]. The theorem itself is [3, Ch. IV, Cor. 7.2].

Step 1: Group completion of the projective monoid. Let $S = \mathrm{iso}\,\mathcal{P}(R)$ be the symmetric monoidal category of finitely generated projective R-modules and their isomorphisms, under $\oplus$. Its classifying space is

$$BS \;\simeq\; \coprod_{[P]} B\mathrm{Aut}_R(P)$$

the disjoint union over isomorphism classes, and $\pi_0(BS)$ is the monoid of definition 2.2.1 whose group completion is $K_0(R)$. Forming $S^{-1}S$ of definition 5.3.2 produces a symmetric monoidal category with $\pi_0(B(S^{-1}S)) = K_0(R)$. The group-completion theorem identifies the homology of $B(S^{-1}S)$ with the localization of the homology of BS at the multiplicative action of π_0. This is the homological input and is proven in full in [3, Ch. IV, Thm. 4.8]. That theorem carries two hypotheses worth naming, since neither is automatic: every morphism of S must be an isomorphism, and the translations $\mathrm{Aut}(a) \to \mathrm{Aut}(s \oplus a)$ must be injective. Both hold for $S = \mathrm{iso}\ \mathbb{C}\mathrm{P}(R)$, the first by construction and the second because a stabilized automorphism determines its restriction.

Step 2: The plus construction appears as a component. Restricting to the cofinal subcategory of free modules and stabilizing, the automorphism groups assemble to $\mathrm{GL}(R) = \varinjlim_n \mathrm{GL}_n(R)$. The group-completion theorem of Step 1 yields a homology equivalence

$$\mathbb{Z} \times B\mathrm{GL}(R)^+ \xrightarrow{\sim} B\big(S_{\mathcal{F}}^{-1} S_{\mathcal{F}}\big) \qquad \text{where } S_{\mathcal{F}} = \mathrm{iso}\ \mathcal{F}(R)$$

the plus construction entering precisely because it is, by definition 5.1.1 and theorem 5.1.7, the space with the homology of $B\mathrm{GL}(R)$ and abelianized fundamental group, which is what the localized homology computes. Passing from free to projective modules replaces the discrete factor $\mathbb{Z}$ by $K_0(R)$, since projectives are cofinal over frees and the extra components are indexed by $K_0(R)$ ([3, Ch. IV, Thm. 4.9 and Cor. 4.11.1]). This gives

$$B(S^{-1}S) \;\simeq\; K_0(R) \times \mathrm{BGL}(R)^+$$

The verification that the relevant maps are homology isomorphisms and that the target is simple enough for a homology isomorphism to be a homotopy equivalence is carried out in [3, Chapter IV].

Step 3: $S^{-1}S$ as a loop space of the Q-construction. Quillen's $S^{-1}S$ theorem provides a homotopy equivalence

$$\Omega BQ\mathcal{P}(R) \;\simeq\; B(S^{-1}S)$$

The mechanism is Theorem B of theorem 5.3.1, stated there in the F/d form dual to [3, Ch. IV, Thm. 3.8], which uses $d\backslash F$: one constructs an auxiliary functor whose target is $Q\mathcal{P}(R)$ and whose comma categories are, after passing to classifying spaces, models for $B(S^{-1}S)$, so that Theorem B identifies the homotopy fibre of the map to $BQ\mathcal{P}(R)$ with $B(S^{-1}S)$. The homotopy fibre of a map is the based loop space of its target whenever the *total space* is contractible, and that is the step here: the classifying space of the auxiliary category is contractible, so the fibre over the zero object is $\Omega BQ\mathcal{P}(R)$, giving the displayed equivalence. Connectivity of $BQ\mathcal{P}(R)$ is not what does this, and would not suffice: the identity functor of $Q\mathcal{P}(R)$ has connected target and a point for its homotopy fibre, while $\Omega BQ\mathcal{P}(R)$ has $\pi_0 = K_0(R)$. The hypothesis of Theorem B (homotopy-invariance of the relevant comma categories along the morphisms of $Q\mathcal{P}(R)$) holds because every morphism in the Q-construction is built from admissible monics and epis, along which the comma categories are homotopy equivalences by an explicit functorial deformation. The construction of the auxiliary functor and this deformation are [3, Ch. IV, Thm. 7.1 and Thm. 7.8].

Step 4: Assembly. Combining Steps 2 and 3,

$$\Omega BQ\mathcal{P}(R) \;\simeq\; B(S^{-1}S) \;\simeq\; K_0(R) \times \mathrm{BGL}(R)^+ \;=\; K(R)$$

which is the asserted equivalence. Applying π_n and using $\pi_n(\Omega X) \cong \pi_{n+1}(X)$ for the first space and $\pi_n(K_0(R) \times \mathrm{BGL}(R)^+) \cong \pi_n(\mathrm{BGL}(R)^+)$ for $n \geq 1$ (the discrete factor contributing only to

π_0) gives

$$\pi_{n+1}(BQ\mathcal{P}(R)) \;\cong\; \pi_n(\Omega BQ\mathcal{P}(R)) \;\cong\; \pi_n(\mathrm{BGL}(R)^+)$$

for $n \geq 1$, which is the degree-shifted isomorphism. On π_0 both sides return $K_0(R)$, completing the agreement in all degrees. Each link is induced by a construction functorial in the ring (group completion, the Q-construction, the comma-category identifications), with one exception: the plus construction is functorial only up to homotopy, since $\mathrm{BGL}(R)^+$ is defined only up to homotopy equivalence. That is enough here, because homotopic maps agree on homotopy groups, so the composite isomorphism commutes with the maps induced by $R \to R'$. That is naturality of the *groups*. It is not naturality of the displayed equivalences of spaces: the Step 2 link splits $K(R)$ into the discrete $K_0(R)$ and the H-space $\mathrm{BGL}(R)^+$, and that splitting is *not* natural, the connecting map $K_1(R/I) \to K_0(R, I)$ being nontrivial ([3, Ch. IV, 1.1.2]). This completes the proof.

The architecture should be read as a relay: Step 1 group-completes the monoid of projectives homologically, Step 2 recognizes the connected part of that group completion as the plus construction, Step 3 recognizes the same group completion as the loop space of the Q-construction via Theorem B, and Step 4 chains the two recognitions. The only content deferred to [3] is the homotopy-theoretic verification that the named maps are equivalences (the group-completion theorem and the explicit deformation of comma categories). The logical skeleton is complete here. The reason the deferral is appropriate rather than a gap is that these verifications are simplicial-homotopy computations with no further conceptual content, and they are reproduced in full in the cited source.

A first consequence is that the higher theory is genuinely an extension of the classical groups. The plus construction was designed in definition 5.1.1 to reproduce K_1 on π_1, and theorem 5.1.9 identified $\pi_2(\mathrm{BGL}(R)^+)$ with K_2. The plus-equals-Q theorem now says the Q-construction reproduces these same groups, so both definitions extend the classical K_0, K_1, K_2.

Corollary 5.3.5: Low-Degree Agreement of the Two Definitions

Let R be a ring. Under the isomorphism of theorem 5.3.4, the higher K-groups recover the classical groups in low degrees:

1. $K_0(R) = \pi_1(BQ\mathcal{P}(R)) = \pi_0(K(R))$ is the Grothendieck group of definition 2.2.1.
2. $K_1(R) = \pi_1(\mathrm{BGL}(R)^+) = \pi_2(BQ\mathcal{P}(R))$ is the Whitehead group $\mathrm{GL}(R)/E(R)$ of definition 4.1.1.
3. $K_2(R) = \pi_2(\mathrm{BGL}(R)^+) = \pi_3(BQ\mathcal{P}(R))$ is the group of definition 4.3.2, the kernel of $\mathrm{st}(R) \to E(R)$.

***Proof*:**
We treat the three degrees in turn, in each case computing the plus-construction side from the previous chapters and transporting across theorem 5.3.4.

1. *Degree* 0. By theorem 5.3.4 the space $\Omega BQ\mathcal{P}(R)$ is homotopy equivalent to $K(R) = K_0(R) \times \mathrm{BGL}(R)^+$, so its set of path components is $\pi_0(K(R)) = K_0(R)$, the Grothendieck group of definition 2.2.1. Equivalently $\pi_1(BQ\mathcal{P}(R)) = \pi_0(\Omega BQ\mathcal{P}(R)) = K_0(R)$, since $BQ\mathcal{P}(R)$ is connected and looping drops the degree by one. This is the statement that the Q-construction recovers K_0, recorded as theorem 5.2.6.

2. *Degree* 1. The plus construction was built in definition 5.1.1 so that $\pi_1(\mathrm{BGL}(R)^+)$ is the abelianization of $\pi_1(\mathrm{BGL}(R)) = \mathrm{GL}(R)$ relative to the perfect normal subgroup $E(R)$; by theorem 5.1.7 the map $\mathrm{GL}(R) = \pi_1(\mathrm{BGL}(R)) \to \pi_1(\mathrm{BGL}(R)^+)$ is surjective with kernel $E(R)$, hence
$$\pi_1(\mathrm{BGL}(R)^+) \;\cong\; \mathrm{GL}(R)/E(R)$$
By theorem 4.1.4 the right side is exactly $K_1(R) = \mathrm{GL}(R)/[\mathrm{GL}(R), \mathrm{GL}(R)]$ of definition 4.1.1, the commutator subgroup and $E(R)$ coinciding by Whitehead's lemma. Transporting across theorem 5.3.4 with $n = 1$ gives $\pi_2(BQ\mathcal{P}(R)) \cong \pi_1(\mathrm{BGL}(R)^+) \cong \mathrm{GL}(R)/E(R)$, as claimed.

3. *Degree* 2. Apply theorem 5.1.9 with $G = \mathrm{GL}(R)$ and the perfect normal subgroup $P = E(R)$: the homotopy fibre of the plus construction $\mathrm{BGL}(R) \to \mathrm{BGL}(R)^+$ has fundamental group the universal central extension of $E(R)$, and
$$\pi_2(\mathrm{BGL}(R)^+) \;\cong\; H_2(E(R); \mathbb{Z})$$
Now theorem 4.3.3 identifies $K_2(R)$ of definition 4.3.2 (the kernel of $\mathrm{st}(R) \to E(R)$) with the center of $\mathrm{st}(R)$, and by theorem 4.4.3 the Steinberg group $\mathrm{st}(R)$ is the universal central extension of the perfect group $E(R)$, whose kernel is $H_2(E(R); \mathbb{Z})$. Hence
$$\pi_2(\mathrm{BGL}(R)^+) \;\cong\; H_2(E(R); \mathbb{Z}) \;\cong\; K_2(R)$$
Transporting across theorem 5.3.4 with $n = 2$ gives $\pi_3(BQ\mathcal{P}(R)) \cong \pi_2(\mathrm{BGL}(R)^+) \cong K_2(R)$, as required.

In each degree the classical group of the earlier chapters is recovered on both sides of the plus-equals-Q isomorphism, so the higher theory extends K_0, K_1, and K_2, completing the proof.

The corollary closes the loop opened at the start of chapter 5: the higher K-groups were introduced as homotopy groups of a space precisely so that K_0, K_1, K_2 would reappear as π_0, π_1, π_2, and now both constructions of that space deliver on the promise. The degree-1 case is worth isolating, because it can be checked directly from both definitions without the full strength of theorem 5.3.4, and doing so makes the agreement concrete.

Example 5.3.6: Both Definitions Give $K_1(R) = \mathrm{GL}(R)/E(R)$

The plus construction and the Q-construction assign the same group to $K_1(R)$, namely the Whitehead group $\mathrm{GL}(R)/E(R)$.

The plus-construction side. By definition 5.1.1, $K_1(R) = \pi_1(\mathrm{BGL}(R)^+)$. The construction $\mathrm{BGL}(R) \to \mathrm{BGL}(R)^+$ is performed relative to the perfect normal subgroup $E(R) \leqslant \mathrm{GL}(R) = \pi_1(\mathrm{BGL}(R))$, perfect by lemma 4.1.3. The defining property in definition 5.1.1(1) and theorem 5.1.7 make $\pi_1(\mathrm{BGL}(R)) \to \pi_1(\mathrm{BGL}(R)^+)$ surjective with kernel exactly $E(R)$, so
$$\pi_1(\mathrm{BGL}(R)^+) \;\cong\; \mathrm{GL}(R)/E(R)$$
By theorem 4.1.4 this equals $\mathrm{GL}(R)/[\mathrm{GL}(R), \mathrm{GL}(R)]$, the K_1 of definition 4.1.1.

The Q-construction side. By definition 5.2.3, $K_1(R) = \pi_2(BQ\mathcal{P}(R))$. By theorem 5.3.4, $\Omega BQ\mathcal{P}(R) \simeq K_0(R) \times \mathrm{BGL}(R)^+$, so looping and reading π_1 of the connected factor,
$$\pi_2(BQ\mathcal{P}(R)) \;\cong\; \pi_1(\Omega BQ\mathcal{P}(R)) \;\cong\; \pi_1(\mathrm{BGL}(R)^+) \;\cong\; \mathrm{GL}(R)/E(R)$$

the discrete factor $K_0(R)$ contributing nothing to π_1.

Agreement. Both routes terminate at $\mathrm{GL}(R)/E(R)$. For a concrete instance take $R = \mathbb{Z}$: then $E(\mathbb{Z}) = \mathrm{SL}(\mathbb{Z})$ and $\mathrm{GL}(\mathbb{Z})/\mathrm{SL}(\mathbb{Z}) \cong \{\pm 1\} = \mathbb{Z}^\times$ via the determinant, so both definitions give $K_1(\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, matching example 4.1.5.

The example isolates the mechanism: on the plus side K_1 is the abelianization of $\mathrm{GL}(R)$ killing $E(R)$, and on the Q side the loop-space equivalence of theorem 5.3.4 transports the same computation up one degree. The two are forced to agree because the equivalence is a homotopy equivalence of spaces, not just an isomorphism of one chosen homotopy group. The same transport, applied in degree n, yields the agreement of all higher groups, which is the content the example cannot reach by hand but theorem 5.3.4 gives.

Exercise 5.3.1

1. Let R be a ring. Using theorem 5.3.4, prove that $\pi_n(\Omega BQ\mathcal{P}(R)) \cong K_n(R)$ for all $n \geq 1$ and that $\pi_0(\Omega BQ\mathcal{P}(R)) \cong K_0(R)$. Explain in one sentence why the loop Ω produces the index shift between the two definitions of K_n.
2. The plus-equals-Q theorem asserts a homotopy equivalence $\Omega BQ\mathcal{P}(R) \simeq K_0(R) \times \mathrm{BGL}(R)^+$, not just an isomorphism of homotopy groups. Explain why the stronger statement (an equivalence of spaces) is needed to conclude that the two definitions of $K_n(R)$ agree *naturally* in R for all n simultaneously, whereas a degree-by-degree isomorphism would not suffice.
3. In the $S^{-1}S$ construction of definition 5.3.2, show that $\pi_0(B(S^{-1}S))$ is the group completion of the monoid $\pi_0(BS)$. For $S = \mathrm{iso}\,\mathcal{F}(\mathbb{Z})$ the category of finitely generated free abelian groups under $\oplus$, identify both $\pi_0(BS)$ and $\pi_0(B(S^{-1}S))$ explicitly.
4. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor of small categories such that $\mathcal{D}$ has a terminal object t and F/t has an initial object. Use Theorem A of theorem 5.3.1 to decide whether BF must be a homotopy equivalence, justifying each hypothesis you invoke. (Hint: a comma category with an initial object has contractible classifying space.)
5. Compute $K_1(R)$ from both definitions when $R = k$ is a field with $k \notin \{\mathbb{F}_2, \mathbb{F}_3\}$, and confirm the two answers agree. State which result from the K_1 chapter you use to evaluate $\mathrm{GL}(k)/E(k)$.
6. Quillen's Theorem B (theorem 5.3.1) requires that for each morphism $d \to d'$ of $\mathcal{D}$ the functor $F/d \to F/d'$ be a homotopy equivalence. Explain, in the language of Step 3 of the proof of theorem 5.3.4, where this hypothesis is used and why it is what makes the homotopy fibre of $* \to BQ\mathcal{P}(R)$ deserve the name "$\Omega BQ\mathcal{P}(R)$".

5.4 Fundamental Theorems and Computations

With the Q-construction in hand and the plus-equals-Q theorem identifying $\pi_{n+1}(BQ\mathbb{C}\mathrm{P}(R))$ with $K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$, we now have a definition of higher K-theory that is flexible enough to admit genuine computation. The $+$-construction is tied to one specific category, the finitely generated projective modules over a single ring, and offers little leverage for relating the K-groups of one situation to another. The Q-construction takes an arbitrary exact category as input, and this is precisely what is needed to state the three structural theorems that drive almost every computation in the subject: the *resolution theorem*, which lets one replace a category of modules by a smaller subcategory adapted to a homological dimension bound; *dévissage*, which strips an abelian category down to its simple objects; and the *localization sequence*, which fits the K-theory of a Serre subcategory, an abelian category, and its quotient into a single long exact sequence. We state all three relative to the Q-construction K-groups $K_n(\mathcal{C})$ of definition 5.2.3, prove what is in scope, and cite [3] for the homotopy-theoretic core of those proofs that rests on Quillen's Theorem A and Theorem B. We then put the machinery to work: the K-theory of a field, recovered in low degrees by hand, and Quillen's computation of $K_*(\mathbb{F}_q)$, the first infinite family of higher K-groups ever determined.

Throughout, $\mathcal{C}$ denotes a (small) exact category in the sense of definition 3.2.1, and $K_n(\mathcal{C}) = \pi_{n+1}(BQ\mathcal{C})$ are its Q-construction K-groups (definition 5.2.3). When $\mathcal{C} = \mathbb{C}\mathrm{P}(R)$ is the category of finitely generated projective R-modules with the split exact sequences, $K_n(\mathbb{C}\mathrm{P}(R)) = K_n(R)$ recovers the K-groups of the ring (definition 5.1.1, definition 5.1.2), by the plus-equals-Q theorem. The notation $G_n(R)$ is reserved for the K-theory of the abelian category $\mathrm{Mod}_{fg}(R)$ of all finitely generated R-modules (defined when R is Noetherian, so that this category is abelian), the so-called *G-theory* of R.

The first theorem answers a question that already arose for K_0: when can the projective modules be traded for a larger or smaller class without changing the K-groups? The prototype is the comparison between projective modules and modules of finite projective dimension. Over a regular ring every finitely generated module has a finite projective resolution, so one expects the K-theory built from projectives to agree with the K-theory built from all finitely generated modules. The resolution theorem is the general mechanism behind such comparisons.

Theorem 5.4.1: Resolution Theorem

Let $\mathcal{A}$ be an abelian category and let $\mathcal{C} \subseteq \mathcal{A}$ be a full additive subcategory *closed under extensions in* $\mathcal{A}$, so that declaring the admissible sequences to be those exact in $\mathcal{A}$ makes $\mathcal{C}$ an exact category in the sense of definition 3.2.1. Let $\mathbb{C}\mathrm{P} \subseteq \mathcal{C}$ be a full subcategory, also closed under extensions, with the inherited structure. Suppose:

1. $\mathbb{C}\mathrm{P}$ is closed under kernels of admissible epimorphisms in $\mathcal{C}$; that is, if $0 \to C' \to P \to P'' \to 0$ is an admissible exact sequence in $\mathcal{C}$ with $P, P'' \in \mathbb{C}\mathrm{P}$, then $C' \in \mathbb{C}\mathrm{P}$.

(1′) $\mathcal{C}$ is closed under kernels of surjections in $\mathcal{A}$: if $f\colon B \to C$ is a map in $\mathcal{C}$ that is surjective in $\mathcal{A}$, then $\ker f \in \mathcal{C}$.

2. Every object $C \in \mathcal{C}$ admits a finite $\mathbb{C}\mathrm{P}$-*resolution*: an admissible exact sequence

$$0 \to P_n \to P_{n-1} \to \cdots \to P_1 \to P_0 \to C \to 0$$

with each $P_i \in \mathbb{C}\mathrm{P}$.

Then the inclusion $\mathbb{C}\mathrm{P} \hookrightarrow \mathcal{C}$ induces a homotopy equivalence $BQ\mathbb{C}\mathrm{P} \xrightarrow{\sim} BQ\mathcal{C}$, and hence isomorphisms

$$K_n(\mathbb{C}\mathrm{P}) \xrightarrow{\cong} K_n(\mathcal{C})$$

for all $n \geq 0$.

The content of the theorem is that a homological dimension bound is invisible to K-theory: enlarging $\mathbb{C}\mathrm{P}$ to the category $\mathcal{C}$ of all objects of finite $\mathbb{C}\mathrm{P}$-dimension changes nothing.

A word on the ambient $\mathcal{A}$, since definition 3.2.1 was deliberately set up *without* one. The K_0-level argument below computes homology of complexes and forms mapping cones, and hypothesis $(1')$ is what keeps the resulting syzygies inside $\mathcal{C}$; neither is available from the axioms alone. Requiring an ambient abelian category is no real loss, since every small exact category embeds in one with the admissible sequences exactly the exact ones (the Gabriel-Quillen embedding, [2]), but hypothesis $(1')$ is a genuine extra assumption and is stated rather than assumed silently. All three ambient conditions hold in every application made here: $\mathrm{Mod}_{fg}(R) \subseteq \mathrm{Mod}(R)$ and $\mathbf{Coh}(X) \subseteq \mathbf{QCoh}(X)$ sit in abelian categories, are closed under kernels of surjections because the ambient rings and schemes are Noetherian, and are closed under extensions. The remaining hypotheses encode what a resolution argument needs. Closure under kernels of admissible epimorphisms is what makes a bounded exact complex of $\mathbb{C}$P-objects have vanishing Euler characteristic in $K_0(\mathbb{C}\mathrm{P})$, which is the engine of the whole argument; note it does *not* force the syzygies of a resolution into $\mathbb{C}$P, as Step 1 of the proof spells out. The existence of finite resolutions guarantees that the enlargement is no bigger than $\mathbb{C}$P allows the homological algebra to reach. The principal application is to a regular ring: every finitely generated module over a regular Noetherian ring has a finite resolution by finitely generated projectives, so $K_n(R) \cong G_n(R)$ for R regular, the identification that makes G-theory computable through the localization and dévissage theorems below.

The homotopy equivalence $BQ\mathbb{C}\mathrm{P} \xrightarrow{\sim} BQ\mathcal{C}$ is proved by analysing the Q-construction with Quillen's Theorem A, filtering $\mathcal{C}$ by the length of the shortest $\mathbb{C}$P-resolution and showing each filtration step induces an equivalence. The argument is given in full in [3, Ch. V, Thm. 3.1]. What is fully in scope, and what already carries the whole geometric idea, is the statement on K_0, where the maps and relations can be written down explicitly.

***Proof*:**

The homotopy equivalence $BQ\mathbb{C}\mathrm{P} \xrightarrow{\sim} BQ\mathcal{C}$ requires Quillen's Theorem A applied to the inclusion functor, filtering by resolution length. This is carried out in [3, Ch. V, Thm. 3.1], and we do not reproduce the simplicial homotopy theory here. The argument on $\pi_1(BQ\mathcal{C}) = K_0$ is given below, with one comparison lemma cited. It exhibits the mechanism and suffices for all of the K_0-level applications.

Step 1: The induced map on K_0 is surjective. Recall from theorem 5.2.6 that $K_0(\mathcal{C})$ is the free abelian group on isomorphism classes $[C]$ of objects of $\mathcal{C}$, modulo the relations $[C] = [C'] + [C'']$ for every admissible exact sequence $0 \to C' \to C \to C'' \to 0$, and similarly for $K_0(\mathbb{C}\mathrm{P})$ using the admissible sequences with terms in $\mathbb{C}$P. Given $C \in \mathcal{C}$, choose a finite $\mathbb{C}$P-resolution

$$0 \to P_n \to \cdots \to P_0 \to C \to 0$$

Break it into short exact sequences $0 \to Z_i \to P_i \to Z_{i-1} \to 0$, where $Z_{-1} = C$ and $Z_i = \ker(P_i \to Z_{i-1})$. Each is admissible in $\mathcal{C}$, which is all that is needed here. The syzygies Z_i themselves need *not* lie in $\mathbb{C}$P, and hypothesis (1) does not put them there: its input is an admissible epimorphism between two objects *of* $\mathbb{C}$P, whereas $P_i \to Z_{i-1}$ has its target outside $\mathbb{C}$P in general. Over $R = k[x, y]$ with $\mathbb{C}\mathrm{P} = \mathbb{C}\mathrm{P}(R)$, for instance, the Koszul resolution of k has first syzygy the ideal (x, y), which is not projective. The defining relation of $K_0(\mathcal{C})$ gives, after telescoping,

$$[C] = \sum_{i=0}^{n} (-1)^i [P_i] \tag{5.1}$$

in $K_0(\mathcal{C})$. The right-hand side lies in the image of $K_0(\mathbb{C}\mathrm{P}) \to K_0(\mathcal{C})$, so the map is surjective.

Step 2: A well-defined inverse on classes. Define $\chi\colon \{\text{objects of } \mathcal{C}\} \to K_0(\mathbb{C}\mathrm{P})$ by the same formula, $\chi(C) = \sum_i (-1)^i [P_i]$ computed from any chosen finite $\mathbb{C}\mathrm{P}$-resolution. Independence of the choice rests on a comparison lemma of Grothendieck's: given a map $f\colon C' \to C$ in $\mathcal{C}$ and a finite $\mathbb{C}\mathrm{P}$-resolution $P_\bullet \to C$, there is a finite $\mathbb{C}\mathrm{P}$-resolution $P'_\bullet \to C'$ and a chain map $P'_\bullet \to P_\bullet$ lifting f ([3, Ch. II, Lem. 7.6.1]). One application produces a chain map to *one* prescribed resolution, so to compare two resolutions $P_\bullet$ and $P'_\bullet$ of C one applies it to the *diagonal* $C \to C \oplus C$ with the resolution $P_\bullet \oplus P'_\bullet \to C \oplus C$. That yields a single $P''_\bullet \to C$ mapping to $P_\bullet \oplus P'_\bullet$, hence to each of $P_\bullet$ and $P'_\bullet$ through the two projections. Each composite $P''_\bullet \to P_\bullet$ lifts id_C, so it induces the identity on H_0 and an isomorphism between zero groups elsewhere: both are quasi-isomorphisms.

That the alternating sums are then equal is the one place *hypothesis* (1) is used, and it is worth isolating. If $0 \to Q_N \to \cdots \to Q_0 \to 0$ is a bounded exact complex with every $Q_i \in \mathbb{C}\mathrm{P}$, then $\sum_i (-1)^i [Q_i] = 0$ in $K_0(\mathbb{C}\mathrm{P})$: break it at Q_0, so $Q_1 \twoheadrightarrow Q_0$ is surjective by exactness at Q_0, its kernel B_1 lies in $\mathcal{C}$ by hypothesis $(1')$ so that the sequence is admissible, and then $B_1 \in \mathbb{C}\mathrm{P}$ by hypothesis (1). Repeat with $Q_2 \twoheadrightarrow B_1$ and telescope; exactness at Q_N makes $Q_N \to B_{N-1}$ an isomorphism, which terminates the process. (For $N = 0$ exactness forces $Q_0 = 0$ and there is nothing to prove.) Applying this to the mapping cone of a quasi-isomorphism between finite $\mathbb{C}\mathrm{P}$-resolutions gives the equality, so χ depends only on the isomorphism class of C.

One consequence is recorded for use below: if C already lies in $\mathbb{C}\mathrm{P}$, the one-term resolution gives $\chi(C) = [C]$.

Step 3: χ respects the relations and inverts the inclusion. Let $0 \to C' \to C \to C'' \to 0$ be admissible in $\mathcal{C}$. The claim is $\chi(C) = \chi(C') + \chi(C'')$. This is the substantial step, and it is *not* the horseshoe lemma: that lemma needs the terms of the resolution of C'' to be projective objects, so that $P''_0 \to C''$ lifts through $C \twoheadrightarrow C''$, and the objects of $\mathbb{C}\mathrm{P}$ need not be projective in $\mathcal{C}$. Splicing arbitrary resolutions of the two ends does not produce one of the middle: on $\mathbb{P}^1$ with $\mathbb{C}\mathrm{P}$ the vector bundles, the Euler sequence $0 \to \mathcal{O}(-2) \to \mathcal{O}(-1)^2 \to \mathcal{O} \to 0$ and the length-zero resolutions of its ends would give $\mathcal{O}(-2) \oplus \mathcal{O} \cong \mathcal{O}(-1)^2$, which is false on global sections.

What replaces it is the mapping cone. Hypothesis $(1')$ is what keeps the construction inside $\mathcal{C}$; hypothesis (1) was used in Step 2.

Choose a finite $\mathbb{C}\mathrm{P}$-resolution $P_\bullet \to C$. Applying [3, Ch. II, Lem. 7.6.1] to the map $C' \to C$ produces a finite $\mathbb{C}\mathrm{P}$-resolution $P'_\bullet \to C'$ and a chain map $f\colon P'_\bullet \to P_\bullet$ over it. Form the mapping cone, with $P_n \oplus P'_{n-1}$ in degree n, so that by construction

$$\chi\big(\mathrm{cone}(f)\big) = \chi(P_\bullet) - \chi(P'_\bullet)$$

The long exact homology sequence $H_i(P') \to H_i(P) \to H_i(\mathrm{cone}\, f) \to H_{i-1}(P') \to H_{i-1}(P)$ has $H_i(P') = H_i(P) = 0$ for $i \neq 0$, with $H_0(P') = C'$ and $H_0(P) = C$. For $i \geq 2$ this gives $H_i(\mathrm{cone}\, f) = 0$ at once. Degree 1 needs one more word: that segment reads $0 = H_1(P) \to H_1(\mathrm{cone}\, f) \to H_0(P') = C' \to H_0(P) = C$, so $H_1(\mathrm{cone}\, f) = \ker(C' \to C)$, which vanishes because $C' \rightarrowtail C$ is the admissible monomorphism of the given sequence. In degree 0, $H_0(\mathrm{cone}\, f) = \mathrm{coker}(C' \to C) = C''$. The terms of the cone lie in $\mathbb{C}\mathrm{P}$ and its syzygies lie in $\mathcal{C}$ by hypothesis $(1')$, so it is admissible. Hence $\mathrm{cone}(f) \to C''$ is itself a finite $\mathbb{C}\mathrm{P}$-resolution, and

$$\chi(C'') = \chi\big(\mathrm{cone}(f)\big) = \chi(P_\bullet) - \chi(P'_\bullet) = \chi(C) - \chi(C')$$

which is the additivity wanted. Note that no induction on resolution length is needed, and in particular no claim that syzygies shorten: the cone does the bookkeeping in one step. The full argument in this form is [3, Ch. II, Thm. 7.6]. Hence χ descends to a homomorphism $K_0(\mathcal{C}) \to K_0(\mathbb{C}\mathrm{P})$.

On an object $P \in \mathbb{C}\mathrm{P}$ the one-term resolution $0 \to P \to P \to 0$ gives $\chi(P) = [P]$, so $\chi \circ (\text{inclusion}) = \mathrm{id}$ on $K_0(\mathbb{C}\mathrm{P})$; and (5.1) shows $(\text{inclusion}) \circ \chi = \mathrm{id}$ on $K_0(\mathcal{C})$. The inclusion therefore induces an isomorphism on K_0, completing the proof in the case $n = 0$, with the higher-degree statement given by [3, Ch. V, Thm. 3.1].

The alternating sum (5.1) is an Euler characteristic: the class of an object equals the Euler characteristic of any finite resolution by the preferred subcategory. This is the same formula by which a coherent sheaf is identified with the alternating sum of a locally free resolution in the K-theory of a variety, and the resolution theorem is the precise statement that the identification is an isomorphism of K-groups, not only a surjection.

Example 5.4.2: K-theory of a Regular Ring

Let R be a regular Noetherian ring, so every finitely generated R-module has finite projective dimension and a finite resolution by finitely generated projectives. Take $\mathcal{C} = \mathrm{Mod}_{fg}(R)$, the abelian category of finitely generated R-modules with its usual exact structure, and $\mathbb{C}\mathrm{P} = \mathbb{C}\mathrm{P}(R)$, the finitely generated projectives, inside the ambient $\mathcal{A} = \mathrm{Mod}(R)$. Hypothesis $(1')$ holds because R is Noetherian: a submodule of a finitely generated module is finitely generated, so $\mathrm{Mod}_{fg}(R)$ is closed under kernels of surjections in $\mathrm{Mod}(R)$. It is closed under extensions for a different and weaker reason, needing no Noetherian hypothesis at all: lift generators of M'' and adjoin generators of M'. Extension-closure of $\mathbb{C}\mathrm{P}(R)$ is likewise immediate, an extension of projectives splitting because the quotient is projective. Hypothesis (1) of theorem 5.4.1 holds unconditionally: a sequence $0 \to C' \to P \to P'' \to 0$ with P, P'' both projective splits, since P'' is projective, so C' is a direct summand of P and hence again projective. Hypothesis (2) is exactly regularity: finite projective dimension of every finitely generated module is equivalent to regularity for a Noetherian ring, and gives the required finite projective resolutions. The resolution theorem gives

$$K_n(R) = K_n(\mathbb{C}\mathrm{P}(R)) \xrightarrow{\cong} K_n(\mathrm{Mod}_{fg}(R)) = G_n(R)$$

for all $n \geq 0$. For $R = \mathbb{Z}$ this reads $K_n(\mathbb{Z}) \cong G_n(\mathbb{Z})$: the K-theory of free abelian groups agrees with the G-theory of all finitely generated abelian groups. The equality $K_0(\mathbb{Z}) = G_0(\mathbb{Z}) = \mathbb{Z}$ is visible directly, since the class of a finite abelian group is 0 in $G_0(\mathbb{Z})$ (it has a finite resolution $0 \to \mathbb{Z}^k \to \mathbb{Z}^k \to M \to 0$ with vanishing Euler characteristic), leaving only the rank.

Dévissage moves in the opposite direction from resolution. Where resolution trades a category for a smaller one inside it via resolutions, dévissage trades an abelian category for a subcategory of its *simple* layers, reached not by resolving but by filtering. The motivating computation is the G-theory of a ring modulo a nilpotent ideal, or of the torsion modules over a Dedekind domain: every such module is built from the residue-field modules by a finite filtration, and dévissage says the K-theory sees only those residue-field layers.

Theorem 5.4.3: Dévissage

Let $\mathcal{A}$ be a small abelian category and $\mathcal{B} \subseteq \mathcal{A}$ a nonempty full abelian subcategory that is closed under subobjects, quotient objects, and finite products in $\mathcal{A}$. Suppose every object $A \in \mathcal{A}$ admits a finite filtration

$$0 = A_0 \subseteq A_1 \subseteq \cdots \subseteq A_m = A$$

by subobjects with each successive quotient A_j/A_{j-1} lying in $\mathcal{B}$. Then the inclusion $\mathcal{B} \hookrightarrow \mathcal{A}$ induces a homotopy equivalence $BQ\mathcal{B} \xrightarrow{\sim} BQ\mathcal{A}$, hence isomorphisms

$$K_n(\mathcal{B}) \xrightarrow{\cong} K_n(\mathcal{A})$$

for all $n \geq 0$.

The closure conditions are what make a filtration usable: closure under subobjects and quotients guarantees that the layers of any filtration, and the layers of any comparison between filtrations, stay inside $\mathcal{B}$, so the combinatorics of the proof never leaves the subcategory. The conclusion is that an abelian category is K-theoretically determined by its simple objects together with the way arbitrary objects filter through them. The canonical instance takes $\mathcal{A}$ to be the finite-length modules over a commutative ring supported at a single maximal ideal $\mathfrak{m}$ and $\mathcal{B}$ the modules killed by $\mathfrak{m}$, that is, the $R/\mathfrak{m}$-vector spaces. Every finite-length module at $\mathfrak{m}$ has a composition series with $R/\mathfrak{m}$-quotients, so dévissage computes the local torsion G-theory as the K-theory of the residue field.

The homotopy equivalence is Quillen's Theorem 4 of the foundational paper. The proof analyses the map of Q-construction nerves using Theorem A, and is reproduced in [3, Ch. V, Thm. 4.1]. As with resolution, the K_0 statement is fully in scope and exhibits the mechanism: the filtration produces the surjectivity and the relation $[A] = \sum_j [A_j/A_{j-1}]$.

***Proof*:**
The homotopy equivalence on the full Q-construction spaces is [3, Ch. V, Thm. 4.1]. It rests on Quillen's Theorem A applied to the inclusion of $Q\mathcal{B}$ into $Q\mathcal{A}$, and we do not reproduce that simplicial argument. We prove the statement on $K_0(\mathcal{A}) = \pi_1(BQ\mathcal{A})$, which is fully in scope.

Step 1: Surjectivity on K_0. Let $A \in \mathcal{A}$ with a finite filtration $0 = A_0 \subseteq \cdots \subseteq A_m = A$ and quotients $S_j = A_j/A_{j-1} \in \mathcal{B}$. Each inclusion $A_{j-1} \hookrightarrow A_j$ is part of an admissible exact sequence $0 \to A_{j-1} \to A_j \to S_j \to 0$ in the abelian category $\mathcal{A}$. The defining relation of $K_0(\mathcal{A})$ (theorem 5.2.6) gives $[A_j] = [A_{j-1}] + [S_j]$, and summing,

$$[A] = \sum_{j=1}^{m} [S_j] \tag{5.2}$$

with every $S_j \in \mathcal{B}$. Hence $[A]$ lies in the image of $K_0(\mathcal{B}) \to K_0(\mathcal{A})$, and the inclusion is surjective on K_0.

Step 2: A left inverse. Define $\theta\colon \{\text{objects of } \mathcal{A}\} \to K_0(\mathcal{B})$ by $\theta(A) = \sum_j [S_j]$, computed from a chosen filtration. To see this is independent of the filtration, note that any two filtrations of A admit a common refinement (the Jordan-Hölder / Schreier refinement argument, valid in any abelian category with the given closure), and refining a filtration replaces a layer S by a two-step filtration $0 \to S' \to S \to S'' \to 0$ with $S', S'' \in \mathcal{B}$ (closure under subobjects and quotients), which leaves the sum $\sum[S_j]$ unchanged because $[S] = [S'] + [S'']$ in $K_0(\mathcal{B})$. Thus θ is well-defined on isomorphism classes.

Step 3: θ is additive and inverts the inclusion. Given an admissible $0 \to A' \to A \to A'' \to 0$ in $\mathcal{A}$, a filtration of A' followed by the preimage of a filtration of A'' is a filtration of A whose layers are those of A' together with those of A''; hence $\theta(A) = \theta(A') + \theta(A'')$, so θ descends to a homomorphism $K_0(\mathcal{A}) \to K_0(\mathcal{B})$. On $B \in \mathcal{B}$ the one-step filtration gives $\theta(B) = [B]$, so θ restricts to the identity on $K_0(\mathcal{B})$; and (5.2) shows the composite $K_0(\mathcal{A}) \to K_0(\mathcal{B}) \to K_0(\mathcal{A})$ is the identity. The inclusion is therefore an isomorphism on K_0, with the higher-degree statement furnished by [3, Ch. V, Thm. 4.1], completing the proof.

Equation (5.2) is again an Euler characteristic, now read off a filtration rather than a resolution: the class of an object is the sum of the classes of its filtration layers. The two theorems are genuinely dual, resolution descending through projective resolutions and dévissage ascending through composition series, and together they reduce the G-theory of a Noetherian ring to the K-theory of its residue fields once the localization sequence is available to organize the supports.

Example 5.4.4: Torsion Modules over a Dedekind Domain

Let R be a Dedekind domain and let $\mathcal{A} = \mathrm{Mod}_{fg,\mathrm{tors}}(R)$ be the abelian category of finitely generated torsion R-modules. Every such module is a finite direct sum of modules $R/\mathfrak{p}^k$ over nonzero primes $\mathfrak{p}$, and $R/\mathfrak{p}^k$ carries the filtration

$$0 \subseteq \mathfrak{p}^{k-1}/\mathfrak{p}^k \subseteq \cdots \subseteq \mathfrak{p}/\mathfrak{p}^k \subseteq R/\mathfrak{p}^k$$

whose successive quotients are all isomorphic to the residue field $R/\mathfrak{p} = \kappa(\mathfrak{p})$. Let $\mathcal{B} \subseteq \mathcal{A}$ be the subcategory of modules annihilated by some nonzero ideal that are semisimple, equivalently the finite products of residue-field modules $\kappa(\mathfrak{p})$. It is closed under subobjects, quotients, and finite products, and the filtrations just exhibited show every object of $\mathcal{A}$ filters with layers in $\mathcal{B}$. As $\mathcal{B}$ decomposes over the primes, $\mathcal{B} \cong \coprod_{\mathfrak{p}} \mathrm{Mod}_{fg}(\kappa(\mathfrak{p}))$, so dévissage gives

$$G_n(\mathcal{A}) \cong \bigoplus_{\mathfrak{p} \neq 0} K_n(\kappa(\mathfrak{p}))$$

the sum over all nonzero primes. For $n = 0$ this recovers the divisor group: G_0 of the torsion modules is $\bigoplus_{\mathfrak{p}} \mathbb{Z}$, the free abelian group on the nonzero primes, which is exactly the group of divisors of R.

The third theorem is the engine that connects the previous two. Resolution and dévissage each compare two categories by an inclusion. The localization sequence instead relates a category, a subcategory, and the quotient, producing a long exact sequence rather than an isomorphism. It is the K-theoretic form of the fact that the Q-construction sends a short exact sequence of categories to a fibration of spaces.

Theorem 5.4.5: Localization Exact Sequence for Abelian Categories

Let $\mathcal{A}$ be a small abelian category and $\mathcal{B} \subseteq \mathcal{A}$ a Serre subcategory ([13, Serre Subcategory]), and let $\mathcal{A}/\mathcal{B}$ be the Gabriel quotient with its exact quotient functor $\mathcal{A} \to \mathcal{A}/\mathcal{B}$ ([13, The Quotient Abelian Category]). Then there is a long exact sequence of abelian groups

$$\cdots \to K_n(\mathcal{B}) \to K_n(\mathcal{A}) \to K_n(\mathcal{A}/\mathcal{B}) \xrightarrow{\partial} K_{n-1}(\mathcal{B}) \to \cdots \to K_0(\mathcal{B}) \to K_0(\mathcal{A}) \to K_0(\mathcal{A}/\mathcal{B}) \to 0$$

in which the map $K_n(\mathcal{B}) \to K_n(\mathcal{A})$ is induced by the inclusion $\mathcal{B} \hookrightarrow \mathcal{A}$, the map $K_n(\mathcal{A}) \to K_n(\mathcal{A}/\mathcal{B})$ is induced by the quotient functor, and ∂ is the connecting homomorphism.

The exactness encodes a conservation principle: a class in $K_n(\mathcal{A})$ dies in the quotient $\mathcal{A}/\mathcal{B}$ exactly when it comes from the subcategory $\mathcal{B}$, and the failure of the quotient map to be surjective in degree n is measured by the image of ∂ in $K_{n-1}(\mathcal{B})$. The Serre condition is precisely what guarantees that the quotient $\mathcal{A}/\mathcal{B}$ is again abelian and that the quotient functor is exact, so that the K-theory of all three categories is defined and the sequence makes sense. The defining geometric input is Quillen's Theorem B: the sequence of Q-construction spaces $BQ\mathcal{B} \to BQ\mathcal{A} \to BQ(\mathcal{A}/\mathcal{B})$ is a homotopy fibration, and the long exact sequence of the theorem is the long exact homotopy sequence of that fibration, reindexed by $K_n(\mathcal{C}) = \pi_{n+1}(BQ\mathcal{C})$. The proof that this sequence of spaces is a fibration is the most demanding part of Quillen's foundational paper. It is given in full in [3, Ch. V, Thm. 5.1] and cited rather than reproduce the Theorem B argument.

***Proof Key ideas*:**
The proof is beyond the homotopy theory developed here: it requires Quillen's Theorem B to identify $BQ\mathcal{B}$ with the homotopy fibre of $BQ\mathcal{A} \to BQ(\mathcal{A}/\mathcal{B})$ over the basepoint, after first checking that the quotient $\mathcal{A}/\mathcal{B}$ is abelian and the quotient functor exact (which is where the Serre condition enters). The long exact sequence is then the homotopy long exact sequence of the fibration $BQ\mathcal{B} \to BQ\mathcal{A} \to BQ(\mathcal{A}/\mathcal{B})$ under the identification $\pi_{n+1}(BQ\mathcal{C}) = K_n(\mathcal{C})$. The connecting map ∂ is the fibration boundary. The full argument is [3, Ch. V, Thm. 5.1]. The right-hand end of the sequence, exactness at $K_0(\mathcal{A}/\mathcal{B})$ and the surjectivity $K_0(\mathcal{A}) \to K_0(\mathcal{A}/\mathcal{B}) \to 0$, is elementary from theorem 5.2.6: the quotient functor sends generators to generators and the relations of $K_0(\mathcal{A}/\mathcal{B})$ are images of relations in $K_0(\mathcal{A})$, so every class in $K_0(\mathcal{A}/\mathcal{B})$ lifts. This is as far as the elementary methods reach, completing the discussion.

The reason this sequence is so productive is that, combined with resolution and dévissage, it computes G-theory along a stratification. For a Noetherian ring R and a nonzerodivisor s, the modules supported on $\{s = 0\}$ form a Serre subcategory of $\mathrm{Mod}_{fg}(R)$ whose quotient is $\mathrm{Mod}_{fg}(R[1/s])$. The localization sequence then reads

$$\cdots \to G_n(R/s) \to G_n(R) \to G_n(R[1/s]) \to G_{n-1}(R/s) \to \cdots$$

after dévissage identifies the K-theory of the supported modules with $G_n(R/s)$. This is the algebraic K-theory analogue of the Gysin sequence relating a space, an open subspace, and a closed complement, and it is what inductive computations of G_n use over Dedekind domains and beyond.

Structure now gives way to computation, beginning with the simplest input of all, a field. Here the higher machinery is not yet needed for the bottom two groups, which can be extracted by hand from the definitions of K_0 and K_1 already established. This anchors the abstract theory to a concrete answer and serves as the base case against which the finite-field computation is checked.

Example 5.4.6: K_0 and K_1 of a Field

Let F be a field. Both of the lowest K-groups were computed in earlier chapters, and the point here is only to have them side by side before Quillen's general formula.

$K_0(F) = \mathbb{Z}$, by example 2.2.3(1): every finitely generated F-module is a finite-dimensional vector space, so the monoid of isomorphism classes under $\oplus$ is $(\mathbb{N}, +)$ and its Grothendieck completion is $\mathbb{Z}$, generated by $[F]$, with $[V] = \dim_F V$. The Q-construction gives the same answer, since theorem 5.2.6 presents $K_0(\mathbb{C}\mathrm{P}(F))$ on classes modulo $[V] = [V'] + [V'']$ and over a field every short exact sequence splits.

$K_1(F) = F^\times$, by example 4.1.5(2) together with theorem 4.1.4: the determinant induces $\mathrm{GL}(F)/E(F) \xrightarrow{\sim} F^\times$, injectivity being the statement $\mathrm{SL}(F) = E(F)$, which example 4.1.5(2) proves by the explicit commutator computation in GL_n for $n \geq 3$.

For $F = \mathbb{F}_q$ this gives $K_1(\mathbb{F}_q) \cong \mathbb{F}_q^\times$, cyclic of order $q-1$. That is the value that must reappear at $i = 1$ in theorem 5.4.7, and it is the only input from this example that the computation below consumes.

The computation of $K_1(F) = F^\times$ used that $\mathrm{SL}(F) = E(F)$ over a field, which is the field case of the general fact that SK_1 measures the failure of elementary generation. For a field it vanishes. With K_0 and K_1 of $\mathbb{F}_q$ in hand, the natural question is what happens above degree 1. The answer, due to Quillen, was the first complete determination of an infinite family of higher K-groups, and it set the template for the subject: identify the relevant classifying space with a space built from topological K-theory, then read off homotopy groups.

Theorem 5.4.7: Quillen's Computation of $K_*(\mathbb{F}_q)$

Let $\mathbb{F}_q$ be the finite field with q elements. Then

$$K_0(\mathbb{F}_q) = \mathbb{Z}, \qquad K_{2i-1}(\mathbb{F}_q) \cong \mathbb{Z}/(q^i - 1) \quad (i \geq 1), \qquad K_{2i}(\mathbb{F}_q) = 0 \quad (i \geq 1)$$

In particular the higher K-groups are finite cyclic in odd degrees and vanish in positive even degrees.

The pattern is concrete. In degree 1 the formula reads $K_1(\mathbb{F}_q) = \mathbb{Z}/(q-1) = \mathbb{F}_q^\times$, agreeing with example 5.4.6. In degree 3, $K_3(\mathbb{F}_q) = \mathbb{Z}/(q^2-1)$. In degree 5, $K_5(\mathbb{F}_q) = \mathbb{Z}/(q^3-1)$; and every positive even group vanishes. The method is the second of the two great ideas of the paper (the Q-construction being the first), and rests on comparing the algebraic classifying space with a space governed by the Adams operations of topological K-theory.

***Proof Key ideas*:**

The full proof determines the mod-ℓ homology of $\mathrm{GL}(\mathbb{F}_q)$ for all primes ℓ and assembles it into the homotopy type of $\mathrm{BGL}(\mathbb{F}_q)^+$. It is long and uses the representation theory of $\mathrm{GL}_n(\mathbb{F}_q)$ together with the homology of the topological space BU. We describe the architecture and defer the computation to [3, Ch. IV, Thm. 1.12], whose read-off is [3, Ch. IV, Cor. 1.13].

Step 1: The Brauer lift. The complex representations of $\mathrm{GL}_n(\mathbb{F}_q)$ are studied through the Brauer character, which lifts the eigenvalues of an $\mathbb{F}_q$-linear map (roots of unity in $\overline{\mathbb{F}_q}$) to roots of unity in $\mathbb{C}$ via a chosen embedding of the multiplicative group of $\overline{\mathbb{F}_q}$ into $\mathbb{C}^\times$. Because this is the one input to the computation that the series does not own, it is worth stating rather than gesturing at.

Definition 5.4.8: The Brauer Lift

Fix an embedding $\overline{\mathbb{F}_q}^{\times} \hookrightarrow \mathbb{C}^{\times}$ of the multiplicative group into the roots of unity. For a finite group G and a representation of G over $\overline{\mathbb{F}_q}$, say $\rho\colon G \to \mathrm{GL}(V)$, the *Brauer lift* is the class function whose value at g is the sum, *with multiplicity*, of the images under that embedding of the roots in $\overline{\mathbb{F}_q}$ of the characteristic polynomial of $\rho(g)$. Reading the eigenvalues off the characteristic polynomial is what makes this well defined at every g, including those of order divisible by $p = \operatorname{char}\mathbb{F}_q$, where $\rho(g)$ need not be diagonalisable. Three facts about it are used below and none is proved here. They are the section's whole external ledger. First, this class function is a *virtual* character, that is a $\mathbb{Z}$-combination of ordinary characters, which is a theorem of Green's. Second, the resulting assignment is a homomorphism of representation rings $R_{\mathbb{F}_q}(G) \to R_{\mathbb{C}}(G)$, and indeed of λ-rings ([3, Ch. IV, Ex. 5.8], where the character calculation behind the λ-ring half is itself omitted). Third, the assembly of proposition 5.4.9 below.
Virtuality is not a formality here, because *effectivity fails.* Take $q = 2$ and $n = 2$, so that $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$. The standard module has $\theta_2 = (2, 2, -1)$ on the classes of the identity, a transposition and a 3-cycle: a transvection is unipotent, so contributes $1 + 1$, while an element of order 3 has eigenvalues ω, ω^2 and contributes $\zeta_3 + \zeta_3^2 = -1$. Against the character table of S_3 this is $\mathbf{1} - \varepsilon + \chi_2$, with ε the sign character. The coefficient -1 means θ_2 is virtual and *not* effective, so no homomorphism $\mathrm{GL}_2(\mathbb{F}_2) \to U$ realises it.

The topological realisation is then one application of a universal property, with no colimit bookkeeping and no plus-construction step of our own.

Proposition 5.4.9: Realising the Brauer Lift

There is a map $\rho\colon \mathrm{BGL}(\mathbb{F}_q)^+ \to BU$, unique up to homotopy, through which the Brauer lift factors.

***Proof*:**
The natural transformation $R_A(\pi_1 X) \to [X, \mathrm{BGL}(A)^+]$ is universal for maps into representable functors: for any connected H-space H and any natural transformation $R_A(\pi_1 X) \to [X, H]$ there is a map $\mathrm{BGL}(A)^+ \to H$, unique up to homotopy, through which it factors ([3, Ch. IV, Prop. 5.7]). Apply this with $A = \mathbb{F}_q$ and $H = BU$, which is a connected H-space. The required natural transformation is the second fact recorded in definition 5.4.8, the ring map $R_{\mathbb{F}_q}(G) \to R_{\mathbb{C}}(G)$, followed by the assignment of a complex representation to its induced bundle, which is exactly $R_{\mathbb{C}}(\pi_1 X) \to [X, BU]$. The universal property returns ρ and its uniqueness at once, as we sought to show.

Uniqueness comes from the universal property directly, so nothing here rests on the vanishing of a $\lim^1$ term, and no compatible family over the tower $\mathrm{BGL}_n(\mathbb{F}_q)$ has to be assembled by hand. The target is the classifying space of the infinite unitary group, the representing space for complex topological K-theory: for a connected compact Hausdorff X one has $K^0(X) \cong [X, \mathbb{Z} \times BU]$ ([22, Representability of K^0]). That theorem is quoted to identify what BU *is*, and is not applied to $\mathrm{BGL}(\mathbb{F}_q)^+$, which is not compact.

The lift itself is Brauer's and the virtual-character theorem is Green's. What Quillen introduced, for exactly this computation, is the topological realisation ρ. The construction of ρ is set out in

[3, Ch. IV, §1, p. 294], though the prose there asserts that the lifts of the trivial and standard representations are themselves honest representations given by homomorphisms $\mathrm{GL}_n(\mathbb{F}_q) \to U$, which the $\mathrm{SL}_2(\mathbb{F}_4)$ example in definition 5.4.8 contradicts. The route through the representation ring, at [3, Ch. IV, Ex. 5.8 and Prop. 5.7], uses only virtuality and is the one followed above. Weibel's own definition of Brauer lifting is a different and coarser one, at [3, Ch. IV, Ex. 5.8]: a homomorphism $\mathbb{F}_q^\times \to \mathbb{C}^\times$ inducing $R_{\mathbb{F}_q}(G) \to R_{\mathbb{C}}(G)$, with no eigenvalue formula. The construction is natural enough to interact with the Frobenius.

Step 2: Frobenius and the Adams operation ψ^q. The Frobenius $x \mapsto x^q$ on $\overline{\mathbb{F}_q}$ induces an operation on the representation ring that the Brauer lift carries to the q-th Adams operation ψ^q, the self-map of BU representing that operation on $\widetilde{KU}^0(-) = [-, BU]$. What carries Frobenius across is the second fact recorded in definition 5.4.8: the lift is a map of λ-rings, not merely of groups. The fixed points of Frobenius are exactly $\mathbb{F}_q$, so the algebraic data over $\mathbb{F}_q$ are matched with the data on BU fixed by ψ^q. Concretely, Quillen identifies $\mathrm{BGL}(\mathbb{F}_q)^+$ with the homotopy fibre of the self-map

$$\psi^q - 1\colon BU \longrightarrow BU$$

an identification proved by comparing mod-ℓ homology on both sides for every prime ℓ, including $\ell = p$ where the source has the homology of a point in the relevant range.

Step 3: Reading off homotopy groups. The homotopy groups of BU are $\pi_{2i}(BU) = \mathbb{Z}$ for $i \geq 1$ and $\pi_{2i-1}(BU) = 0$ ([22, Bott Periodicity], which gives $\pi_k(U)$; the shift to BU is $\pi_k(BU) \cong \pi_{k-1}(U)$, from the fibration $U \to EU \to BU$ with EU contractible). The Adams operation ψ^q acts on $\pi_{2i}(BU) = \mathbb{Z}$ by multiplication by q^i. The relevant stretch of the long exact homotopy sequence of the fibration $\mathrm{Fib}(\psi^q - 1) \to BU \xrightarrow{\psi^q - 1} BU$ around degree $2i$ is

$$\pi_{2i+1}(BU) \to \pi_{2i}(\mathrm{Fib}) \to \pi_{2i}(BU) \xrightarrow{q^i - 1} \pi_{2i}(BU) \to \pi_{2i-1}(\mathrm{Fib}) \to \pi_{2i-1}(BU)$$

in which the two outer terms $\pi_{2i+1}(BU)$ and $\pi_{2i-1}(BU)$ vanish, so the sequence reduces to

$$0 \to \pi_{2i}(\mathrm{Fib}) \to \mathbb{Z} \xrightarrow{q^i - 1} \mathbb{Z} \to \pi_{2i-1}(\mathrm{Fib}) \to 0$$

Since $q^i - 1 \neq 0$, multiplication by $q^i - 1$ on $\mathbb{Z}$ is injective with cokernel $\mathbb{Z}/(q^i - 1)$. Hence $\pi_{2i}(\mathrm{Fib}) = \ker(q^i - 1) = 0$ and $\pi_{2i-1}(\mathrm{Fib}) = \mathrm{coker}(q^i - 1) = \mathbb{Z}/(q^i - 1)$. Under the identification of Step 2, $K_{2i}(\mathbb{F}_q) = \pi_{2i}(\mathrm{BGL}(\mathbb{F}_q)^+) = 0$ and $K_{2i-1}(\mathbb{F}_q) = \pi_{2i-1}(\mathrm{BGL}(\mathbb{F}_q)^+) = \mathbb{Z}/(q^i - 1)$ for $i \geq 1$, with $K_0(\mathbb{F}_q) = \mathbb{Z}$ from example 5.4.6. The verification that the Brauer lift induces the claimed homology isomorphism is [3, Ch. IV, Thm. 1.12], which Weibel states and attributes to Quillen rather than proving. The passage from it to the displayed groups is [3, Ch. IV, Cor. 1.13], and is the computation just carried out, completing the discussion.

The computation is the prototype for every later K-theory calculation by "homotopy fibre of an Adams operation": the same shape recurs in the étale and motivic descriptions of the K-theory of more general fields. The vanishing of $K_{2i}(\mathbb{F}_q)$ for $i \geq 1$ reflects that BU has no odd homotopy and that $\psi^q - 1$ is injective on the even homotopy. The odd groups $\mathbb{Z}/(q^i - 1)$ record exactly the cokernel of $q^i - 1$, the order of the cyclic group of q^i-power roots of unity caught by the i-th Bott class.

Example 5.4.10: $K_*(\mathbb{F}_q)$ in Low Degrees

Tabulating theorem 5.4.7 for the first several degrees:

n	$K_n(\mathbb{F}_q)$
0	$\mathbb{Z}$
1	$\mathbb{Z}/(q-1)$
2	0
3	$\mathbb{Z}/(q^2-1)$
4	0
5	$\mathbb{Z}/(q^3-1)$
6	0

The degree-1 entry is the consistency check. By example 5.4.6, $K_1(\mathbb{F}_q) = \mathbb{F}_q^\times$, the multiplicative group of the field, which is cyclic of order $q-1$. The formula gives $K_1(\mathbb{F}_q) = K_{2\cdot 1-1}(\mathbb{F}_q) = \mathbb{Z}/(q^1-1) = \mathbb{Z}/(q-1)$, and a cyclic group of order $q-1$ is exactly $\mathbb{Z}/(q-1)$. The two computations agree.

As a numerical instance, take $q=2$. Then $K_1(\mathbb{F}_2) = \mathbb{Z}/1 = 0$ (the field $\mathbb{F}_2$ has trivial unit group), $K_3(\mathbb{F}_2) = \mathbb{Z}/3$, $K_5(\mathbb{F}_2) = \mathbb{Z}/7$, $K_7(\mathbb{F}_2) = \mathbb{Z}/15$, and all even higher groups vanish. For $q=3$: $K_1(\mathbb{F}_3) = \mathbb{Z}/2$, $K_3(\mathbb{F}_3) = \mathbb{Z}/8$, $K_5(\mathbb{F}_3) = \mathbb{Z}/26$. The orders q^i-1 grow geometrically, so unlike a field of characteristic 0, all higher K-groups of a finite field are finite.

Exercise 5.4.1

1. Let $\mathcal{A}$, $\mathcal{C}$ and $\mathbb{C}\mathrm{P}$ satisfy the hypotheses of theorem 5.4.1. Suppose $C \in \mathcal{C}$ has two finite $\mathbb{C}\mathrm{P}$-resolutions of lengths n and n'. Using only the K_0-level argument of the resolution theorem, show that the two Euler characteristics $\sum_i(-1)^i[P_i]$ and $\sum_i(-1)^i[P_i']$ agree in $K_0(\mathbb{C}\mathrm{P})$, and deduce that the class $[C] \in K_0(\mathcal{C})$ is independent of the chosen resolution.

2. Verify Quillen's formula in degree 1 against the elementary computation. That is, show directly from definition 4.1.1 that $K_1(\mathbb{F}_q) \cong \mathbb{F}_q^\times$ is cyclic of order $q-1$, and confirm this matches $K_{2i-1}(\mathbb{F}_q) = \mathbb{Z}/(q^i-1)$ at $i=1$. Then compute $K_3(\mathbb{F}_4)$ and $K_5(\mathbb{F}_4)$ explicitly as cyclic groups, giving their orders.

3. Let R be a Dedekind domain with fraction field F. Applying the localization sequence (theorem 5.4.5) to the Serre subcategory of finitely generated torsion modules inside $\mathrm{Mod}_{fg}(R)$, together with dévissage (theorem 5.4.3), write down the resulting long exact sequence in terms of $G_n(R)$, $K_n(F)$, and the residue fields $\kappa(\mathfrak{p})$. Identify the degree-0 portion and explain how it recovers the exact sequence $0 \to \mathrm{Cl}(R) \to G_0(R) \to \mathbb{Z} \to 0$ relating $G_0(R)$ to the class group, given that $G_0(R) \cong K_0(R)$ for R regular.

4. In the proof of theorem 5.4.7, the Adams operation ψ^q acts on $\pi_{2i}(BU) = \mathbb{Z}$ by multiplication by q^i. Using only this fact and the long exact homotopy sequence of the fibration $\mathrm{Fib}(\psi^q - 1) \to BU \to BU$, rederive that $\pi_{2i}(\mathrm{Fib}) = 0$ and $\pi_{2i-1}(\mathrm{Fib}) = \mathbb{Z}/(q^i-1)$. Where does the hypothesis $q \geq 2$ (equivalently $q^i - 1 \neq 0$) enter, and what would go wrong at the formal value $q=1$?

5. Let R be a regular Noetherian ring. Combining example 5.4.2 with the resolution theorem, explain why $K_n(R) \cong G_n(R)$ for all n, and why this isomorphism can fail when R is not regular. Give an explicit non-regular example where the Cartan map $K_0(R) \to G_0(R)$ fails to be an isomorphism, computing both sides and identifying the map. (Note the two groups may

still be abstractly isomorphic. What fails is the comparison map.)

6

K-Theory of Schemes

Everything so far has taken a ring as input. A scheme is glued from rings, and the invariants glue with it, but the gluing is not formal: the two natural categories attached to a scheme, its vector bundles and its coherent sheaves, sit on opposite sides of the divide chapter 3 was built to straddle. Vector bundles form an exact category and not an abelian one; coherent sheaves form an abelian category in which almost nothing is locally free. Each has a K_0, the two are related but not equal, and the whole of intersection theory turns on when the comparison map between them is an isomorphism.

This chapter develops that comparison. The present section constructs both groups, equips them with the structure they carry, and builds the two functorialities that make them useful: pullback along an arbitrary morphism and pushforward along a proper one.

6.1 K_0 of a Scheme

Throughout this chapter every scheme is Noetherian of *finite Krull dimension*, not X alone but the source and target of every morphism that appears. The finiteness is not automatic, a Noetherian ring being able to have infinite dimension, and it is what makes the alternating sums below terminate; stating it for all schemes at once is what lets theorem 6.1.4 compose its pushforwards, since g_* needs the bound on $\dim Y$ that f_* needed on $\dim X$. Two categories present themselves. The locally free sheaves of finite rank, which over a base field are the vector bundles by [4, Vector Bundle / Locally Free Sheaf Correspondence] and which we work with sheaf-theoretically in general, are closed under $\oplus$ and $\otimes$ but not under kernels: where a map of bundles drops rank, its kernel *gains* generators and stops being locally free. On $\mathbb{A}^3$ the map $\mathcal{O}^3 \xrightarrow{(x,y,z)} \mathcal{O}$ has kernel of generic rank 2 needing three generators at the origin, the fibre dimension of a coherent sheaf being upper semicontinuous. The coherent sheaves ([4, Coherent Sheaf]) are closed under kernels and cokernels, so they form an abelian category, but they contain torsion objects that no bundle sees.

The first is an exact category in the sense of definition 3.2.1, with the sequences that happen to be exact as sheaves declared admissible. The second is abelian. So definition 3.2.4 and definition 3.1.1 apply respectively, and give two different groups.

Definition 6.1.1: K_0 of a Scheme

Let X be a Noetherian scheme. Write $\mathrm{Vect}(X)$ for the category of locally free $\mathcal{O}_X$-modules of finite rank, with a sequence declared admissible when it is exact as a sequence of $\mathcal{O}_X$-modules, and $\mathbf{Coh}(X)$ for the category of coherent $\mathcal{O}_X$-modules. Then

$$K^\circ(X) := K_0\big(\mathrm{Vect}(X)\big), \qquad K_\circ(X) := K_0\big(\mathbf{Coh}(X)\big)$$

The first is the K-theory of X, the second its G-theory.

Two checks are owed. That $\mathbf{Coh}(X)$ is abelian uses the Noetherian hypothesis in exactly the way definition 3.1.3 did: affine-locally a map of coherent sheaves is a map of finitely generated modules over a Noetherian ring. Cokernels and images are quotients of finitely generated modules, hence automatic. The Noetherian hypothesis is what makes the *kernel* finitely generated. Coherence being local, these glue, and the coimage-image comparison is an isomorphism because it is one on each affine. That $\mathrm{Vect}(X)$ is exact is the observation that an exact sequence of $\mathcal{O}_X$-modules whose three terms are locally free is locally split, since the quotient is locally free hence locally projective. The four clauses of definition 3.2.1 then hold, though not quite for the reason they held in $\mathscr{P}(R)$: there every admissible sequence split globally, whereas here splitting is only local. The pushouts and pullbacks of clauses (2) to (4) are global sheaf constructions, and what is checked locally is only that they land back in $\mathrm{Vect}(X)$. For clause (4), the pullback of an admissible mono $\mathscr{E}' \rightarrowtail \mathscr{E}$ along an admissible epi $p\colon \mathscr{F} \twoheadrightarrow \mathscr{E}$ is the subsheaf $p^{-1}(\mathscr{E}') \subseteq \mathscr{F}$. It sits in $0 \to \ker p \to p^{-1}(\mathscr{E}') \to \mathscr{E}' \to 0$ with both ends locally free, so it is locally free by the splitting just noted; and $\mathscr{F}/p^{-1}(\mathscr{E}') \cong \mathscr{E}/\mathscr{E}'$ is locally free as well. All three terms of $0 \to p^{-1}(\mathscr{E}') \to \mathscr{F} \to \mathscr{E}/\mathscr{E}' \to 0$ are therefore locally free, which is what admissibility asks.

For $X = \operatorname{Spec} R$ affine the two groups are the ones already met: $\mathrm{Vect}(\operatorname{Spec} R)$ is $\mathscr{P}(R)$ and $\mathbf{Coh}(\operatorname{Spec} R)$ is $\mathrm{Mod}_{fg}(R)$, so $K^\circ(\operatorname{Spec} R) \cong K_0(R)$ by proposition 3.2.6 and $K_\circ(\operatorname{Spec} R) = G_0(R)$ by definition 3.1.3.

Proposition 6.1.2: Ring and Module Structures

Let X be a Noetherian scheme. Then $\otimes_{\mathcal{O}_X}$ makes $K^\circ(X)$ a commutative ring with identity $[\mathcal{O}_X]$, and makes $K_\circ(X)$ a module over it.

***Proof*:**
A tensor product of locally free sheaves of finite rank is locally free of finite rank, and a locally free sheaf is flat, so $\mathscr{E} \otimes -$ is exact. Hence for fixed $\mathscr{E} \in \mathrm{Vect}(X)$ the assignment $\mathscr{F} \mapsto [\mathscr{E} \otimes \mathscr{F}]$ is additive on admissible sequences, and proposition 3.2.5 makes it a homomorphism $K^\circ(X) \to K^\circ(X)$. Varying $\mathscr{E}$ and using additivity in the first variable the same way gives a biadditive pairing, hence a multiplication on $K^\circ(X)$; associativity, commutativity and the unit are inherited from $\otimes$ on generators.

For the module structure, $\mathscr{E} \otimes \mathscr{F}$ is coherent when $\mathscr{E}$ is locally free of finite rank and $\mathscr{F}$ is coherent, and the pairing needs additivity in *each* variable separately, so both universal properties are used rather than one replacing the other. Fixing $\mathscr{E} \in \mathrm{Vect}(X)$, the functor $\mathscr{E} \otimes -$ is exact by flatness and proposition 3.1.2 applies to the coherent variable. Fixing $\mathscr{F} \in \mathbf{Coh}(X)$, flatness is unavailable (a coherent $\mathscr{F}$ need not be flat), and the reason $- \otimes \mathscr{F}$ carries admissible sequences of $\mathrm{Vect}(X)$ to exact sequences is instead that the *quotient* in such a sequence is locally free, so the sequence is locally split (as noted above) and stays exact after any tensor; then proposition 3.2.5 applies to the locally free variable. Together these give a pairing $K^{\circ}(X) \otimes K_{\circ}(X) \to K_{\circ}(X)$ satisfying the module axioms on generators, completing the proof.

Note where flatness was used, because it is what fails on the other side: $\mathscr{F} \otimes -$ for a general coherent $\mathscr{F}$ is only right exact, so $\otimes$ does not descend to a well-defined pairing on $K_{\circ}(X)$. That is a statement about this construction, not about $K_{\circ}(X)$: when X satisfies the hypothesis of theorem 6.1.6 the two groups are identified and $K_{\circ}(X)$ inherits a ring structure after all. What fails in general is the naive product, and exercise set 6.1.1 makes the failure explicit.

Proposition 6.1.3: Pullback

Let $f\colon X \to Y$ be a morphism of Noetherian schemes. Then $\mathscr{E} \mapsto f^*\mathscr{E}$ induces a ring homomorphism

$$f^*\colon K^{\circ}(Y) \longrightarrow K^{\circ}(X)$$

and $(g \circ f)^* = f^* \circ g^*$. If f is flat, the same formula induces $f^*\colon K_{\circ}(Y) \to K_{\circ}(X)$.

***Proof*:**
Pullback carries a locally free sheaf of finite rank to one of the same rank, and it carries an admissible sequence of $\mathrm{Vect}(Y)$ to an admissible sequence of $\mathrm{Vect}(X)$: such a sequence is locally split, and f^* preserves local splittings, being additive. So f^* is an exact functor and proposition 3.2.7 applies. It respects $\otimes$ because $f^*(\mathscr{E} \otimes \mathscr{E}') \cong f^*\mathscr{E} \otimes f^*\mathscr{E}'$ and sends $\mathcal{O}_Y$ to $\mathcal{O}_X$, so it is a ring map, and functoriality is the corresponding statement for f^* on sheaves.

On coherent sheaves f^* is only right exact in general, which is why the second claim needs flatness: for flat f the functor f^* is exact and preserves coherence, so proposition 3.1.2 applies, completing the proof.

The pushforward is the harder of the two, and it is what the properness hypothesis is for. It cannot be defined by $\mathscr{F} \mapsto [f_*\mathscr{F}]$: pushforward is only left exact, so that assignment is not additive. The repair is to take the whole derived sequence in an alternating sum, which is exactly what makes it additive.

Theorem 6.1.4: Proper Pushforward

Let $f\colon X \to Y$ be a proper morphism of Noetherian schemes of finite Krull dimension. Then

$$f_*[\mathscr{F}] := \sum_{i \geq 0} (-1)^i \left[R^i f_* \mathscr{F}\right]$$

is a well-defined homomorphism $f_*\colon K_{\circ}(X) \to K_{\circ}(Y)$, and $(g \circ f)_* = g_* \circ f_*$ for proper $f\colon X \to Y$ and $g\colon Y \to Z$ with Z likewise Noetherian of finite dimension.

***Proof*:**
The sum makes sense. Each $R^if_*\mathscr{F}$ is coherent by [4, Finiteness of Higher Direct Images, Proper Case](1), whose hypotheses are exactly those in force, and the same result's part (2) makes $R^if_*\mathscr{F}=0$ for $i>\dim X$. So the sum is finite and lands in $K_\circ(Y)$.

Additivity. Let $0\to\mathscr{F}'\to\mathscr{F}\to\mathscr{F}''\to 0$ be exact in $\mathbf{Coh}(X)$. The long exact sequence of higher direct images reads

$$0\to f_*\mathscr{F}'\to f_*\mathscr{F}\to f_*\mathscr{F}''\to R^1f_*\mathscr{F}'\to\cdots$$

and terminates, by the vanishing above. Every term is coherent. A finite exact sequence in an abelian category has alternating sum zero in K_0: applying definition 3.1.1 to the short exact sequences it breaks into, and summing with signs, the images cancel in pairs. Grouping the terms of the long exact sequence by which of $\mathscr{F}',\mathscr{F},\mathscr{F}''$ they come from gives

$$\sum_i(-1)^i[R^if_*\mathscr{F}]=\sum_i(-1)^i[R^if_*\mathscr{F}']+\sum_i(-1)^i[R^if_*\mathscr{F}'']$$

so $\mathscr{F}\mapsto f_*[\mathscr{F}]$ is additive, and proposition 3.1.2 makes it a homomorphism.

Functoriality. For proper f and g the composite $g\circ f$ is proper, and [4, Higher Direct Images Along a Composite] gives a spectral sequence $E_2^{p,q}=R^pg_*(R^qf_*\mathscr{F})\Rightarrow R^{p+q}(g\circ f)_*\mathscr{F}$. All terms are coherent and the sequence is bounded, so the alternating sum of the E_2-page equals that of the abutment, each page having alternating sum equal to the next by the same cancellation argument. That identity is exactly $g_*(f_*[\mathscr{F}])=(g\circ f)_*[\mathscr{F}]$, completing the proof.

The two functorialities interact through a single formula, which is what lets a computation be pushed from X to Y and back.

Proposition 6.1.5: Projection Formula

Let $f\colon X\to Y$ be proper, $\alpha\in K^\circ(Y)$ and $\beta\in K_\circ(X)$. Then

$$f_*\big(f^*\alpha\cdot\beta\big)=\alpha\cdot f_*\beta$$

in $K_\circ(Y)$.

***Proof*:**
Both sides are additive in α and in β, so it suffices to check the case $\alpha=[\mathscr{E}]$ with $\mathscr{E}$ locally free of finite rank and $\beta=[\mathscr{F}]$ with $\mathscr{F}$ coherent. The sheaf-level projection formula gives a natural isomorphism

$$R^if_*\big(f^*\mathscr{E}\otimes\mathscr{F}\big)\cong\mathscr{E}\otimes R^if_*\mathscr{F}$$

for every i. Locality alone does not prove this, two sheaves that are locally isomorphic need not be isomorphic, so one first builds the comparison map globally: adjunction gives $\mathscr{E}\to f_*f^*\mathscr{E}$, and cup product with it gives a morphism $\mathscr{E}\otimes R^if_*\mathscr{F}\to R^if_*(f^*\mathscr{E}\otimes\mathscr{F})$ natural in $\mathscr{E}$ and $\mathscr{F}$. That it is an isomorphism is then checked locally on Y, where $\mathscr{E}$ is free of finite rank: for $\mathscr{E}\cong\mathcal{O}_Y^n$ both sides are $(R^if_*\mathscr{F})^n$, since R^if_* commutes with finite direct sums. Taking alternating sums and using that $\mathscr{E}\otimes-$ is exact, hence commutes with the alternating sum in $K_\circ(Y)$, gives the stated identity, completing the proof.

Finally, the comparison between the two groups. Every locally free sheaf is coherent, so there is a map $K^\circ(X) \to K_\circ(X)$. The content is when it is an isomorphism.

Theorem 6.1.6: The Duality Map

Let X be a Noetherian scheme and suppose every coherent sheaf on X admits a finite resolution by locally free sheaves of finite rank. Then the map

$$K^\circ(X) \longrightarrow K_\circ(X), \qquad [\mathscr{E}] \mapsto [\mathscr{E}]$$

is an isomorphism, with inverse $[\mathscr{F}] \mapsto \sum_i (-1)^i [\mathscr{E}_i]$ for any finite locally free resolution $\mathscr{E}_\bullet \to \mathscr{F}$.

The hypothesis holds whenever X is regular ([4, Regular Scheme]) and *separated*. This is [3, Ch. II, Thm. 8.2].

***Proof*:**

This is theorem 5.4.1 applied to $\mathcal{C} = \mathbf{Coh}(X)$ and $\mathbb{CP} = \mathrm{Vect}(X)$, inside the ambient abelian category $\mathcal{A} = \mathbf{QCoh}(X)$, so the work is checking that theorem's hypotheses. There are five, not two: besides the numbered clauses (1) and (2), the theorem asks that *both* $\mathcal{C}$ and $\mathbb{CP}$ be closed under extensions, and, as $(1')$, that $\mathcal{C}$ be closed under kernels of surjections in $\mathcal{A}$.

Hypothesis $(1')$, *and extension-closure of* $\mathcal{C}$. Both hold for the same reason. The kernel of a surjection of coherent sheaves is coherent, and an extension of one coherent sheaf by another is coherent, since affine-locally these are statements about finitely generated modules over a Noetherian ring. This is the same Noetherian input that made $\mathbf{Coh}(X)$ abelian above.

Closure under extensions. If $0 \to \mathscr{E}' \to \mathscr{G} \to \mathscr{E}'' \to 0$ is exact with $\mathscr{E}'$ and $\mathscr{E}''$ locally free of finite rank, then it is locally split because $\mathscr{E}''$ is, so $\mathscr{G}$ is locally $\mathscr{E}' \oplus \mathscr{E}''$ and therefore locally free of finite rank. This is not the same statement as the local splitting used for hypothesis (1), where all *three* terms were assumed locally free. Here the middle term is what is being concluded.

Hypothesis (1), that $\mathbb{CP}$ is closed under kernels of admissible epimorphisms, is the local-splitting observation made after definition 6.1.1: if $0 \to \mathscr{K} \to \mathscr{E} \to \mathscr{E}'' \to 0$ is exact with $\mathscr{E}, \mathscr{E}''$ locally free of finite rank, then the sequence is locally split because $\mathscr{E}''$ is locally free, so $\mathscr{K}$ is locally a direct summand of $\mathscr{E}$ and hence locally free of finite rank.

Hypothesis (2), that every object of $\mathcal{C}$ has a finite $\mathbb{CP}$-resolution, is precisely the standing assumption of this theorem.

Theorem 5.4.1 then gives that the inclusion induces an isomorphism on K_0, with inverse the Euler characteristic $[\mathscr{F}] \mapsto \sum_i (-1)^i [\mathscr{E}_i]$ of any finite resolution, as claimed.

It remains to justify the sufficient condition. On a regular scheme every local ring has finite global dimension by [13, Auslander-Buchsbaum-Serre Theorem], so every coherent sheaf has finite projective dimension at each point. Promoting those local resolutions to a single global one of finite length is the substantial step. When X carries an ample invertible sheaf it can be taken, and the resolution property then holds outright, by [4, Finite Locally Free Resolutions on a Regular Scheme]. That covers every quasi-projective case. For a general separated regular scheme the promotion is a different argument, not developed in this series, which is why the resolution property is stated as a hypothesis of the theorem rather than derived.

Separatedness is not a convenience of the proof. Affine n-space with a doubled origin $(n \geq 2)$ is regular and Noetherian but not separated, and there $K^\circ(X) \cong \mathbb{Z}$ while $K_\circ(X) \cong \mathbb{Z} \oplus \mathbb{Z}$, so the map of this theorem is not surjective ([3, Ch. II, Ex. 8.2.4]). That completes the proof.

One scheme has to be computed explicitly before the theory can be used, because it is the one every projective computation is reduced to. Projective space is that scheme, and its K-theory is free of rank $m+1$ over the base, on the twists of the tautological bundle.

Theorem 6.1.7: K_0 of a Relative Projective Space

Let Y be a regular Noetherian separated scheme of finite Krull dimension. The hypotheses do different jobs, and all of them serve the *freeness* half: regular, Noetherian and separated feed theorem 6.1.6, and finite Krull dimension feeds theorem 6.1.4. Let $p\colon \mathbb{P}^m_Y = Y \times \mathbb{P}^m \to Y$ and $\rho\colon \mathbb{P}^m_Y \to \mathbb{P}^m$ be the projections. Write $t = [\rho^*\mathcal{O}(-1)]$. Then $K^\circ(\mathbb{P}^m_Y)$ is a free $K^\circ(Y)$-module with basis

$$1,\ t,\ t^2,\ \ldots,\ t^m$$

so that the classes $p^*\beta \cdot \rho^*[\mathcal{O}(n)]$, for $\beta \in K^\circ(Y)$ and $n \in \mathbb{Z}$, generate $K^\circ(\mathbb{P}^m_Y)$ as a group. The relation $(1-t)^{m+1} = 0$ holds, and as a $K^\circ(Y)$-algebra

$$K^\circ(\mathbb{P}^m_Y) \ \cong\ K^\circ(Y)[t]/\big((1-t)^{m+1}\big)$$

The case $Y = \operatorname{Spec} k$ is $K^\circ(\mathbb{P}^m) \cong \mathbb{Z}[t]/((1-t)^{m+1})$, free of rank $m+1$.

***Proof*:**
The relation. Write $E = \mathcal{O}(-1)^{\oplus(m+1)}$ for the middle term of the twisted Euler sequence $0 \to \Omega^1_{\mathbb{P}^m} \to \mathcal{O}(-1)^{\oplus(m+1)} \to \mathcal{O} \to 0$ on $\mathbb{P}^m$. The Koszul complex $\wedge^\bullet E$ of the surjection $E \twoheadrightarrow \mathcal{O}$ is exact, and $\wedge^k E = \mathcal{O}(-k)^{\oplus\binom{m+1}{k}}$. Pulling back along the flat ρ gives the exact sequence of bundles on $\mathbb{P}^m_Y$

$$0 \to \rho^*\mathcal{O}(-m-1) \to \cdots \to \rho^*\mathcal{O}(-i)^{\oplus\binom{m+1}{i}} \to \cdots \to \rho^*\mathcal{O}(-1)^{\oplus\binom{m+1}{1}} \to \mathcal{O} \to 0$$

whose alternating sum in K° is $\sum_{i=0}^{m+1}(-1)^i\binom{m+1}{i}t^i = (1-t)^{m+1} = 0$. In particular t is a unit, so every $[\rho^*\mathcal{O}(n)]$ with $n \in \mathbb{Z}$ lies in the $K^\circ(Y)$-span of $1, t, \ldots, t^m$.

Generation. What is needed is that every class in $K^\circ(\mathbb{P}^m_Y)$ is a $K^\circ(Y)$-combination of $1, t, \ldots, t^m$, and the natural first attempt does not give it. That attempt resolves a coherent $\mathscr{F}$ by repeatedly surjecting $p^*\mathscr{E} \otimes \rho^*\mathcal{O}(-n) \twoheadrightarrow \mathscr{F}$ and iterating on the kernel, hoping the syzygies terminate because Y is regular of finite Krull dimension. Two things go wrong. Finite global dimension makes the terminal kernel *locally free*, not a sum of twists of pullbacks, and a locally free sheaf need not be such a sum: on $\mathbb{P}^2$ the tangent bundle has rank 2, $c_1 = 3$ and $c_2 = 3$, while $\mathcal{O}(a) \oplus \mathcal{O}(b)$ would need $a + b = 3$ and $ab = 3$, which no pair of integers satisfies. So the alternating sum retains a term of exactly the kind being expressed, and the argument is circular. Second, the intermediate kernels are coherent rather than locally free, so the identity obtained lives in $K_\circ(\mathbb{P}^m_Y)$ and returning it to $K^\circ(\mathbb{P}^m_Y)$ needs theorem 6.1.6 for $\mathbb{P}^m_Y$ itself, which is a further step.

Quillen's resolution gives what is wanted directly, and stays inside K° throughout. Call a vector bundle $\mathscr{F}$ on $\mathbb{P}^m_Y$ *Mumford-regular* when $R^q p_*\mathscr{F}(-q) = 0$ for all $q > 0$. Every class in $K^\circ(\mathbb{P}^m_Y)$ is represented by a Mumford-regular bundle, since a large enough twist of any bundle is one and twisting is invertible in K° by the relation above ([3, Ch. II, Prop. 8.7.10]). For such an $\mathscr{F}$

there is an exact sequence of bundles

$$0 \to p^*T_m\mathscr{F} \otimes \rho^*\mathcal{O}(-m) \to \cdots \to p^*T_i\mathscr{F} \otimes \rho^*\mathcal{O}(-i) \to \cdots \to p^*T_0\mathscr{F} \to \mathscr{F} \to 0$$

with each $T_i\mathscr{F}$ locally free of finite rank on Y ([3, Ch. II, Thm. 8.7.8 and Cor. 8.7.9]). Note where this is better: it terminates at $i = m$ for *cohomological* reasons, the fibres of p having dimension m, rather than because of any global dimension of Y, and every term already has the form $p^*[\mathscr{E}] \cdot t^i$. Taking the alternating sum, $[\mathscr{F}] = \sum_{i=0}^m (-1)^i p^*[T_i\mathscr{F}] \cdot t^i$, which is a $K^\circ(Y)$-combination of $1, t, \ldots, t^m$ as required. This is the one step of the theorem we cite rather than prove. The construction is Quillen's.

Freeness. Suppose $\sum_{i=0}^m \beta_i t^i = 0$ with $\beta_i \in K^\circ(Y)$. Apply p_* (theorem 6.1.4) after multiplying by t^{-j} for $0 \le j \le m$. The projection formula (proposition 6.1.5) gives $p_*\big(p^*\beta \cdot t^{i-j}\big) = \beta \cdot p_*\big(t^{i-j}\big)$, and $p_*[\rho^*\mathcal{O}(l)] = \binom{m+l}{m} [\mathcal{O}_Y]$ for $l \ge -m$ by flat base change and the cohomology of twists on $\mathbb{P}^m$ ([4, Twisted Sheaf Cohomology]), vanishing for $-m \le l \le -1$ and equal to $[\mathcal{O}_Y]$ at $l = 0$. The coefficients here are integers, so what the relations say is that the *integer* matrix $\big(\binom{m+j-i}{m}\big)_{0\le i,j\le m}$, which is unitriangular and hence invertible over $\mathbb{Z}$, annihilates the vector $\big(\iota(\beta_0), \ldots, \iota(\beta_m)\big)$. Here $\iota\colon K^\circ(Y) \to K_\circ(Y)$ is the duality map, and it appears because p_* and the projection formula both land in $K_\circ(Y)$. So every $\iota(\beta_i)$ vanishes, and ι is injective by theorem 6.1.6, whose hypotheses Y was assumed to satisfy. Hence every β_i vanishes.

The hypotheses on Y are what *this* argument needs, not what the theorem needs. The projective bundle theorem holds over an arbitrary quasi-compact scheme ([3, Ch. II, Thm. 8.5], originally Berthelot), by a proof that never leaves K° and so never needs ι to be injective. The basis is therefore free, and the algebra presentation follows, as we sought to show.

Exercise 6.1.1

1. Let $X = \operatorname{Spec} R$ be affine and Noetherian. Identify the map of theorem 6.1.6 with the comparison $K_0(R) \to G_0(R)$ of definition 3.1.3, and say what the hypothesis of that theorem becomes for R.

2. Let $f\colon X \to Y$ be a closed immersion of Noetherian schemes. Show that f is proper and that f_* of theorem 6.1.4 reduces to $[\mathscr{F}] \mapsto [f_*\mathscr{F}]$ with no higher terms.

3. Show that $\otimes$ does not in general descend to a well-defined pairing on $K_\circ(X)$, by exhibiting a Noetherian scheme on which two coherent sheaves with the same class have tensor products with different classes. (A double point will do.) Why does this not contradict theorem 6.1.6?

4. Let X be a Noetherian scheme and $Z \subseteq X$ a closed subscheme with open complement U. Show that $[\mathscr{F}] \mapsto [\mathscr{F}|_U]$ defines a surjection $K_\circ(X) \to K_\circ(U)$, and identify a set of classes in its kernel. (For surjectivity you may assume that every coherent sheaf on U extends to a coherent sheaf on X. This is the one input the series does not prove.)

5. Show that the projection formula of proposition 6.1.5 makes f_* a homomorphism of $K^\circ(Y)$-modules, where $K_\circ(X)$ is a $K^\circ(Y)$-module via f^*.

6. For $f\colon X \to Y$ proper and $\mathscr{E}$ locally free on X, is $\sum_i (-1)^i [R^i f_*\mathscr{E}]$ in the image of $K^\circ(Y) \to K_\circ(Y)$? Decide, and either prove it or give the obstruction.

Index

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